真经 GRE

张巍GRE

数学机经900题

（2026年）

（难度分级版）

关 注 公 众 号
「张巍GRE」

答案及解析获取方式

1. 关注公众号“张巍GRE”（扫右边二维码关注微信公众号）

2.在对话框中输入“GRE最新数学机经900题答案”，系统会自动弹出答案和解析。

张巍 GRE 数学机经 900 题

(难度分级版）

资料说明：本资料在最受欢迎的《张巍 GRE 数学机经 800 题》的基础上增加了精选的 2025 年和 2026 年最新机经 100 题，根据难度分为了 easy，medium，hard 三种模式，再另外汇总了图表题方便考生在备考过程中更有针对性的使用。本资料由真经 GRE 数学名师教研团老师共同编写，并含全套答案和解析。

使用说明：
1，easy 部分的题目建议大家在刚开始基础阶段的时候使用，medium 和 hard 部分可以在刷题冲击高分时使用，图表题建议平时不擅长该类型或者该类型错误率比较高的同学重点使用。
2，此资料和考试难度一致，若做此资料错误率很高，则建议把基础知识和考点先夯实一遍，推荐用《GRE 数学满分宝典》，是教材级别的备考材料，涵盖了所有的考试知识点。

答案和解析获取方式：
1. **关注微信公众号“张巍 GRE”**
2. **在对话框中输入“GRE 最新数学机经 900 题”，系统会自动弹出答案和解析。**

张巍老师介绍：真经 GRE 创始人，GRE 行业大牛老师，多年 GRE 授课经验，累计培训学员超过 90000 人，帮大量学员考出 GRE 高分，进入名校。深入研究 GRE 真题和教学方法，著有《GRE 填空机经 2000 题》和解析，《GRE 阅读机经 440 篇》和解析，《数学 170 难题》，《数学回忆机经 360》，《GRE 数学满分宝典》，《最新数学机经 900 题》和解析，《GRE 镇考机经词》，《GRE 等价词汇总》《GRE 镇考 3000 词》等学生喜爱的资料。

真经GRE 资料一览图

GRE Verbal资料

《张巍GRE阅读机经440篇》(2026年)(题型分类版)

推荐程度: ★★★★★

推荐理由: 按题型分类，针对性训练，市面唯一。

《张巍GRE填空机经2000题》(2026年)(难度分级版)

推荐程度: ★★★★★

推荐理由: 最新机经难度分级，配套最新最全解析。

GRE模考资料

「内部学员专享」

《张巍GREVerbal模考套题50套》(改革后)

推荐程度: ★★★★★

推荐理由: 难度系数标注，真实考试高度还原verbal题目和出题顺序，配套详细解析。备考必刷!

「内部学员专享」

《真经GRE新版模考套题》(3套)(改革后)

推荐程度: ★★★★★

推荐理由: 改革后的考试套题，自适应仿真结构。高度还原真实考试。

GRE数学资料

《张巍GRE最新数学机经900题》(2026年)(难度分级版)

推荐程度: ★★★★★

推荐理由: 最新真题，难度分级，针对性训练。

《GRE数学170难题4.0》(2026年版)

推荐程度: ★★★★★

推荐理由: 高分必备，难题合集，冲击170。

《GRE数学回忆版机经360》

推荐程度: ★★★★

推荐理由: 考场回忆版真题，贴近考试难度和思路。

《GRE数学满分宝典3.0》(2026年版)

推荐程度: ★★★★★

推荐理由: 涵盖所有数学知识点考点，最完整的数学知识体系教材。

「内部学员专享」

《真经GRE| 数学考点分类真题练习册》

推荐程度: ★★★★★

推荐理由: 分最新真题收录、考点的模块化训练，针对性训练。

GRE写作资料

《2026 issue官方题库》

推荐程度: ★★★★

推荐理由: 官方最新题库汇总。

《GRE issue 范文精析》

推荐程度: ★★★★★

推荐理由: 高分范文的分析和梳理，掌握写作逻辑。

「内部学员专享」

《Issue万能素材》

推荐程度: ★★★★★

推荐理由: 扩充素材库，深挖素材每个解析角度，丰富论证过程!

「内部学员专享」

《改革后Issue题库公式匹配》

推荐程度: ★★★★★

推荐理由: Issue题库中所有题目的公式匹配和题目思路，秒出思路!

GRE单词资料

《GRE镇考3000词》(2026年版)

推荐程度: ★★★★★

推荐理由: 紧跟考试趋势，最新选词，配备视频讲解。

《GRE镇考机经词7.0》(2026年版)

推荐程度: ★★★★★

推荐理由: GRE核心高频词，真题机经答案词，配套视频讲解。

《真经GRE等价词汇总》(2026年版)

推荐程度: ★★★★★

推荐理由: 六选二等价词汇必背词汇。

《巍哥GRE数学词汇汇总》

推荐程度: ★★★★

推荐理由: 按照知识点分类整理GRE全部数学专有名词。

《GRE阅读机经核心词汇》

推荐程度: ★★★★

推荐理由: GRE阅读中词汇题扩展，阅读必备词汇。

《张巍GRE熟词僻义汇总》(2026年版)

推荐程度: ★★★★

推荐理由: GRE常考的160多个熟词僻义。

真经GRE 课程一览图

真经 GRE

从入门到出分，不同课程满足不同类型学员的需求

01 王牌冲分班

性价比之王 COST PERFORMANCE

高质量课程设置 QUALITY
- GRE业内**名师亲授**
- 各科考点**无死角覆盖**
- 搭建解题**方法和逻辑**
- **核心高频题目**精讲
- **科学课业比** 提高效率

+

全阶一站课程服务 SERVICE
- **一站式**学习规划
- **终身答疑** 陪你出分
- **班级群打卡&交流**
- 1年**超长回放期**

+

最新最全备考资料 RESOURCE
- 词汇&机经**全阶段**
- 课后作业**专项训练**
- **模考练习**查缺补漏

适合下类考生杀G

考点体系薄弱
还未建立GRE考点和方法体系的小白同学

快速进阶出分
自学了一段时间想要快速进阶的同学

备考遭遇瓶颈
备考中遇见了瓶颈的同学

02 极速提分小班

极速杀G EXTREME SPEED

闭环课程设置 COURSE
- **打基础 为上课做准备**
  全面诊断 课前规划
- **建立考点及解题体系**
  课中系统+针对授课
- **检测课程吸收情况**
  定制刷题方案
  科学规划，不走弯路

+

极致监督服务 SUPERVISE
- **针对更强 服务更高效**
  3-6人极致小班
- **课前+课中+课后刷题**
  +考前冲刺计划
  定制各阶段学习计划
- **细致答疑，陪你出分**
  名师1对1终身答疑
- **落实每日学习计划**
  专属班主任督促学习

+

专属资料解析 RESOURCE
- **业内最新最全机经**
- **各科名师内部资料**
- **课后作业专项练习**
  高质量刷题

适合下类考生杀G

追求极速出分
时间紧任务重的同学

追求小班授课
能动性较弱想要严格督促的同学

03 个性化一对一

专属定制&专治各种"疑难杂症" VVVVIP

报名后**全面诊断**，找出"病根"
量身定制**针对性的授课内容**
课上实时互动，**监控学习效果**
一对一**定制刷题和冲刺方案**

更强针对性
更高的效率
更稳的提升

顶级名师亲授 更懂"对症下药
名师**终身答疑，负责到底**
专属班主任一对一监督，拒绝拖延
1年超长回放，可反复听课消化

业内**最新最全机经真题**
课后作业**专项练习** 高质量刷题

各科名师的**核心内部资料**
模考训练查缺补漏 适应考试节奏

适合下类考生杀G

GRE钉子户
上过课做了很多题、但没出分，严重瓶颈的学员

GRE小白
想极速提分，帮你快准狠杀G

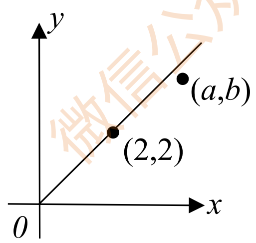
Section 1 - Easy

1. $x^{-2} = \frac{1}{y+1}$【微信公众号：张巍 GRE】

   Quantity A: x^2
   Quantity B: y

2. The lengths of the three sides of a triangle are 6, 8, and S.
   Quantity A: s
   Quantity B:10

3. Quantity A: a
   Quantity B: b

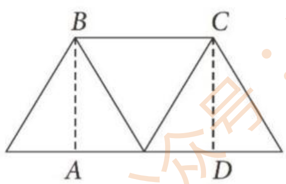


4. $\frac{m^3}{n^6} = \frac{1}{27}$

   Quantity A: 3m
   Quantity B: n^2

5. x > 0 and x + y = x
   Quantity A: x
   Quantity B: y

6. If the trapezoid above is composed of three equilateral triangles, each having area $\sqrt{3}$, what is the area of rectangular region ABCD?



A. $\sqrt{3}$
B. $2\sqrt{3}$
C. $3\sqrt{3}$
D. $4\sqrt{3}$
E. 3

7. If the average (arithmetic mean) of x+2 and y+4 is 11, what is the average of x and y?

8. $-3\leq x\leq 4$
Quantity A: x
Quantity B: 0.3

9. $-6<r<-3$, $-7<m<-5$
Quantity A: r^2
Quantity B: m^3

10. j, k, and e are three positive consecutive integers, and j < k < e.
Quantity A: jke
Quantity B: k^3

11. In the xy-plane, the x-intercept and the y-intercept of line j are 3 and -2, respectively, and line k is parallel to j. If the y-intercept of k is 6, what is the x-intercept of k?

12. $x^2 = (x-1)^2$
Quantity A: x
Quantity B: 1/4

13. x= -0.5
Quantity A: $4x^3-x^2+6x+3$
Quantity B: $4x^3-x^2-6x-3$

14. Quantity A: The least value of y for which $-5\leq 1-2y\leq 9$
Quantity B: -2

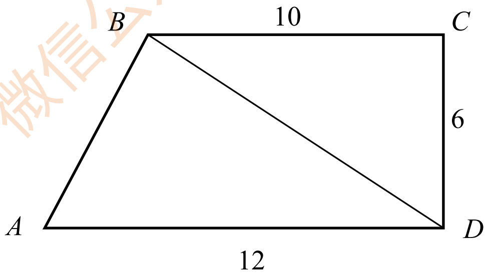
15. If a train traveled 3,200 yards in 80 seconds, approximately what was the average speed of the train in miles per hour? (1 mile = 1,760 yards)

A. 60
B. 70
C. 80
D. 90
E. 100

16. In the xy-plane, the distance from point (3, 1) to which of the following points is greatest?

A. (−2,−1)
B. (−2,1)
C. (3,−5)
D. (3, 5)
E. (7, 1)

17. In the figure above, if angles BCD and ADC are both right angles, which of the following is true?【微信公众号：张巍 GRE】



A. AB= 6.
B. BD= 16.
C. The area of triangular region ABD is 36.
D. The area of triangular region BCD is 60.
E. The area of quadrilateral region ABCD is 72.

18. If a number is to be chosen at random from the integers 1 through 30, what is the probability that the number chosen will be divisible by 3 or by 5?

A. $\frac{1}{15}$
B. $\frac{7}{15}$
C. $\frac{1}{2}$
D. $\frac{8}{15}$

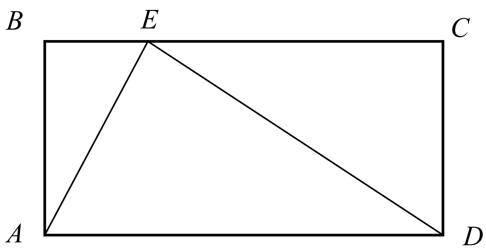
E. $\frac{17}{30}$

19. The table shows the frequency distribution of the random variable Y. What is the median of the distribution of the values of Y?

Y	Frequency
0	9
10	12
20	15
30	22
40	17

A. 18
B. 20
C. 22
D. 24
E. 30

20. ABCD is a rectangle【微信公众号：张巍 GRE】



Quantity A: The combined areas of the triangular regions ABE and ECD
Quantity B: The area of triangular region AED

Section 2 - Easy

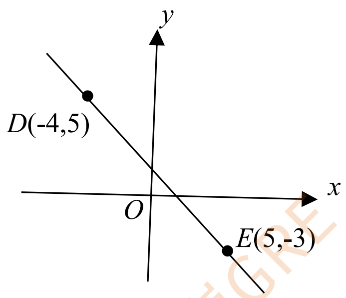
21. x>0, y>0【微信公众号：张巍 GRE】

    Quantity A: $(\sqrt{x})(\sqrt{y})$

    Quantity B: $\sqrt{x+y}$

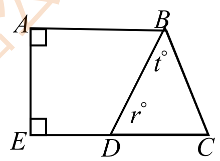
22. Three variables, x, y, and z, in a scientific experiment are related by the equation $z=x^2y$. In the experiment, if the value of x decreases by 40 percent while the value of y increases by 50 percent, what is the percent decrease in the value of z?
    A. 10%
    B. 18%
    C. 24%
    D. 36%
    E. 46%

23. Quantity A: The slope of line DE
    Quantity B: -4/3



24. xy= -6
    Quantity A: $(x-y)^2$
    Quantity B: $(x+y)^2$

25. Quantity A: r
    Quantity B:t



BC=ED

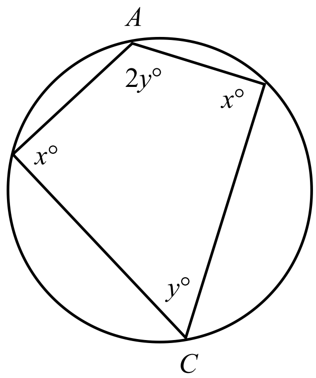
26. $x<0$ and $y>0$
    Quantity A: $(x^{-1})((y^2)$
    Quantity B: $(x)((y^{-2})$

27. A company has 420 employees, of whom 240 received a raise, 115 received a promotion, and 60 received both a raise and a promotion. How many of the employees received neither a

raise nor a promotion?

28. Which of the following is true about the x-intercept(s) of the graph of the equation $y=x^2+2x-15$ in the xy-plane? 【微信公众号：张巍 GRE】
A. The graph has no x-intercept.
B. The graph has one x-intercept: -15.
C. The graph has two x-intercepts: -3 and 5.
D. The graph has two x-intercepts: -5 and 3.
E. The graph has two x-intercepts: 3 and 5.

29. In the Figure above, a quadrilateral is inscribed in a circle. Line segment AC (not shown) is a diameter of the circle. What is the value of x + y?



30. It costs d dollars to buy s sandwiches. At this rate, what is the cost of s+2,850 sandwiches, in dollars?

A. $\frac{sd+s}{2{,}850d}$

B. $\frac{sd+d}{2{,}850d}$

C. $\frac{sd+d}{2{,}850s}$

D. $\frac{2{,}850d+sd}{s}$

E. $\frac{2{,}850s+d}{d}$

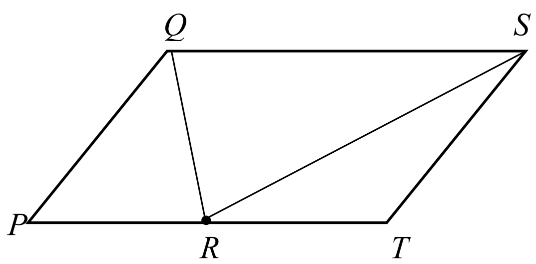
31. Quantity A: 3^{120}
Quantity B: 6^{60}

32. The area of square region *S* is equal to the area of circular region *C*.
Quantity A: The length of a diagonal of S
Quantity B:The length of a diameter of C

33. S is the set of all integers x such that 2x＜15.
T is the set of all integers y such that 3y＞10.
Quantity A: The number of integers that are in both set S and set T
Quantity B: 4

34. x=y-3
x= -x
Quantity A: y
Quantity B: 2

35. *PQST* is a parallelogram and *R* is the midpoint of side *PT*.
Quantity A: The area of triangular region PQR
Quantity B: The area of triangular region RST



36. -4<x<2 and -2<x<4
Quantity A: |x|
Quantity B: 2

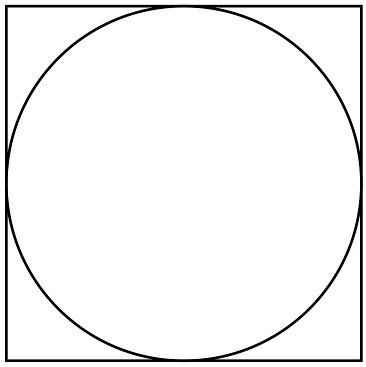
37. $\frac{2}{x} = \frac{y}{2}$
Quantity A: x
Quantity B: y

38. x and y are positive even integers.

Quantity A: $(-1)^{\frac{x+y}{2}}$

Quantity B: $(-1)^{x+y}$

39. As shown in the figure above, a circular flower bed lies in a square garden plot that is 60 meters on each side. What fraction of the garden plot area is not part of the flower bed?



A. $\frac{4-\pi}{2}$

B. $\frac{\pi-2}{2}$

C. $\frac{4-\pi}{3}$

D. $\frac{4-\pi}{4}$

E. $\frac{\pi-2}{8}$

40. For x=97, which of the following fractions has the least value?【微信公众号：张巍 GRE】

A. $\frac{97}{x}$

B. $\frac{x}{x+1}$

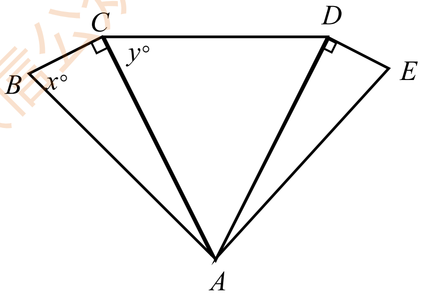
C. $\frac{x}{x-1}$

D. $\frac{x+1}{x}$

E. $\frac{x-1}{x}$

Section 3 - Easy

41. Triangles *ABC* and *AED* are congruent, where *BC=DE*, $\angle BAE=90°$. If y=63, what is the value of x? 【微信公众号：张巍 GRE】



42. s =-t
Quantity A: $-s^3$
Quantity B: ts^2

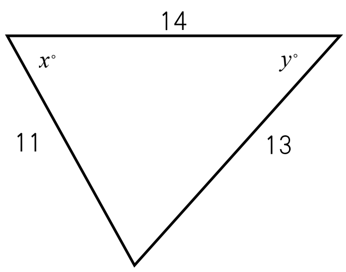
43. $\sqrt{x^2} = 9$
Quantity A: 2^x
Quantity A: 2^{-x}

44. -3<x<7
Quantity A: x-2
Quantity B: 6

45. Quantity A: The product of the positive prime factors of 30
Quantity B: The sum of the positive prime factors of 30

46. $(3^x)(3^y) = 1$
Quantity A: x+y
Quantity B: 1

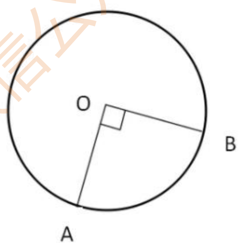
47. Quantity A: x
Quantity B: y



48. s and t are positive integers such that $st = 30$
Quantity A: The least possible value of s+t
Quantity B: 13

49. By draining 40 gallons of water from a tank, the amount of water in the tank was decreased from 1/5 of the tank 's full capacity to 2/11 of the tanks full capacity. Water was then added to the tank until the tank was full. How many gallons of water were added to the tank?
A 1,800
B 2,000
C 2,200
D 2,400
E 2,600

50. In the figure above. O is the center of the circle. If minor arc AB has length 2π, what is the area of triangular region AOB?



A 8
B 16
C 4π
D 8π
E 16π

51. At a certain florist shop, each rose costs r dollars and each tulip costs t dollars. If 2 roses and 4 tulips cost a total of $11.00, and if 3 roses and 3 tulips cost a total of $12.00, how much does 1 rose cost at the shop?【微信公众号：张巍 GRE】
A $2,50
B $2.75
C $3.00
D $3.25
E $3.50

52. In the xy-plane, a circle with radius 4 has its center at the point (-1. 2) Which of the following is an equation of the circle?

A $(x-1)^2+(y+2)^2=2$

B $(x-1)^2+(y+2)^2=4$

C $(x-1)^2+(y+2)^2=16$

D $(x+1)^2+(y-2)^2=4$

E $(x+1)^2+(y-2)^2=16$

53. If x and y are positive numbers and the ratio of x to y is 5 to 4, which of the following ratios must be equal to 6 to 5? 【微信公众号：张巍 GRE】
A 2x to 3y
B 12x to 10y
C 24x to 25y
D x+1 to y+1
E x+6 to y+5

54. $x<0$

Quantity A：$(x^{-1})^{-3}$

Quantity B：x^{-4}

55. At a certain farm, there are horses, goats, and sheep. The ratio of the number of horses to the number of goats is 4 to 3. The ratio of the number of horses to the number of sheep is 3 to 2. If there are 24 sheep at the farm, how many goats are at the farm?
A 18
B 24
C 27
D 36
E 48

56. Metal rod M measures between 8.65 meters and 8.75 meters in length; metal rod R measures between 8.70 meters and 8.80 meters in length.
Quantity A：The length of metal rod M
Quantity B：The length of metal rod R

57. Quantity A：$(-7)^{52}$
Quantity B：$(-7)^{81}$

58. Three of the sides of quadrilateral ABCD have lengths 7, 8, and 9, respectively.
Quantity A: The perimeter of ABCD
Quantity B: 32

59. If x + y = 5, x – y = 4, and z =$\frac{x^2-y^2}{x^2-2xy+y^2}$, what is the value of z ?
Give your answer as a fraction. 【微信公众号：张巍 GRE】

60. $\frac{51!-50!}{50!-49!}$ =

A. $\frac{1}{49}$

B. 1

C. $\frac{50}{49}$

D. $\frac{(50)(50)}{49}$

E. $\frac{(50)(50)(50)}{(49)}$

Section 4 - Medium

1. A collection of solid-colored marble, 60 percent of which are red, 30 percent blue, and 10 percent white, is enlarged by adding 12 marbles, 6 of which are red.
   Quantity A: The percent of marbles in the enlarged collection that are red
   Quantity B: 60 percent

2. In a normal distribution, the amount of the distribution that lies within 1 standard deviation of the mean is 68 percent, rounded to the nearest whole percent, A certain set of values is normally distributed with a mean of 81 and a standard deviation of 5, A value x will be randomly selected from the set.【微信公众号：张巍 GRE】
   Quantity A: The probability that x will be less than 76 or greater than 86
   Quantity B: The probability that x will be between 81 and 86

3. The function f is defined for all numbers x by $f(x) = 57x^2 – kx + 925$, where k is a constant, and $f(x) = f(-x)$ for all x.
   Quantity A: k
   Quantity B: 0

4. Last week Tom worked x hours and earned \$256, and Frank worked y hours and earned \$144. If Tom and Frank worked a combined total of 50 hours last week and they were paid the same constant hourly rate, how many more hours did Tom work than Frank?
   A. 12
   B. 14
   C. 17
   D. 18
   E. 32

5. If $0 < |x| - 2x < 3$, which of the following is true?
   A. $x < -1$
   B. $x = -1$
   C. $-1 < x < 0$
   D. $x = 0$
   E. $0 < x < 1$

6. Last month, each salesperson at a certain company sold one of three products J, K, and L. For the 3 salespeople who sold product J, the average (arithmetic mean) amount of sales per salesperson was $14,000. For the 7 salespeople who sold product K, the average amount of sales per salesperson was $12,000. For the n salespeople who sold product L, the average amount of sales per salesperson was $21,000. If the average amount of sales per salesperson for all the salespeople at the company was $15,400, what is the value of n?

7. If d is the standard deviation of the three numbers w, p, and u, what is the standard deviation the three numbers w+6, p+6, and u+6?【微信公众号：张巍 GRE】

A) d
B) d+6
C) d+18
D) 6d
E) 18d

8. The greatest common divisor of two positive integers k and n is 5, and the least common multiple of k and n is 30. If k=10, what is the value of n?

A) 5
B) 10
C) 15
D) 20
E) 30

9. An appliance dealer purchased refrigerators of a certain model for $850 each. The dealer sold each refrigerator at a 25 percent discount of a regular price of p dollars. If the dealer's profit on each refrigerator was 20 percent of the amount for which the dealer purchased each refrigerator, what is the value of p? (<u>Note</u>: Profit equals revenue minus cost.)

A)1,063
B)1,150
C)1,233
D)1,328
E)1,360

10. When the positive integer n is divided by 4, the remainder is 3; when n is divided by 3, the remainder is 2.

Quantity A: The least possible value of n
Quantity B: 12

11. Last semester at School X, there was a total of 6 biology classes, and the numbers of students enrolled in these classes were 33, 34, 32, 32, 34, and 35. Of the total number of students

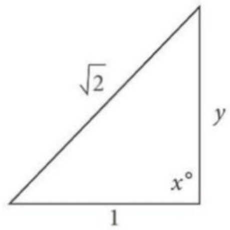
enrolled in the biology classes, 20 percent were biology majors. Of the students enrolled in the biology classes who were not biology majors, 10 percent were enrolled in a chemistry class.

Quantity A: The number of students enrolled in the biology classes who were not biology majors and were enrolled in a chemistry class

Quantity B: 18

12. $1 \le |x| \le 2$
$4 \le |y| \le 5$
$6 \le |z| \le 7$
Quantity A: The range of the possible values of $|z - x|$
Quantity B: The range of the possible values of $|z - y|$

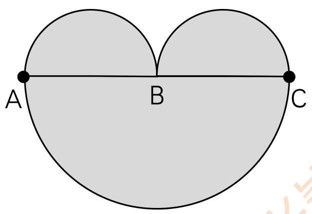
13. Quantity A: y
Quantity B: 1



$x > 90^0$

14. Bucket Q currently contains 5 times as much water as bucket R contains. If 10 liters of water will be transferred from Q to R, then Q will contain 2 times as much water as R will contain. How many liters of water does Q currently contain?【微信公众号：张巍 GRE】
A) 10
B) 20
C) 30
D) 40
E) 50

15. The figure shown consists of three semicircles with diameters AB, BC, and AC, where B is the midpoint of AC. If the area of the shaded region is 48π, what is the length of AC?



A) 4
B) 8
C) 16
D) 32

E) 64

16. A water trough is in the shape of a rectangular solid. The length of the trough is 4 times its width, and its depth is 5 feet. A new trough is constructed with length 4 feet greater than the length of the original trough, a width 1 foot less than the width of the original trough and a depth equal to that of the original trough. If the capacity of the new trough is 300 cubic feet, what is the capacity, in cubic feet, of the original trough? 【微信公众号：张巍 GRE】
A) 280
B) 300
C) 320
D) 340
E) 360

17. In a bucket containing y golf balls, 3/16 of the balls are white and the rest are yellow. After t additional balls are placed in the bucket, where half of the additional balls are white and half are yellow, 3/8 of the balls in the bucket will be white. What is the ratio of t to y?
Give your answer as a fraction.

18. List L consists of 9 different numbers, all of which are positive integers. The median of the numbers in L is 11. What is the least possible average (arithmetic mean) of the numbers in L?
A. 5
B. $5\frac{2}{9}$
C. $6\frac{5}{9}$
D. $7\frac{2}{3}$
E. $8\frac{1}{3}$

19. If $-3 < \sqrt[3]{a}-1 < 3$, which of the following statements must be true?
Indicate <u>all</u> such statements.
A) $a<64$
B) $a>0$
C) $a>-8$

20. P, Q, and T are three distinct points in a plane.
Quantity A: The number of lines in the plane that pass through P, Q and T

Quantity B: 1

Section 5 - Medium

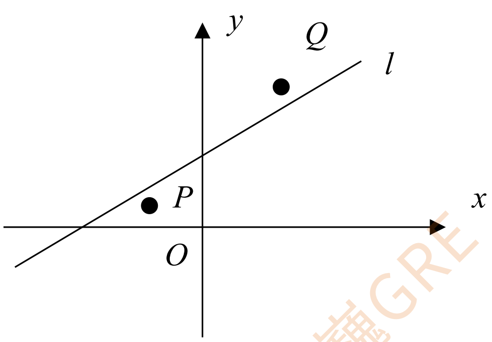
21. List R and list S each consist of 100 numbers, where 50 of the numbers occur on both lists. The average (arithmetic mean) of the numbers in list R is less than the average of the numbers in list S. 【微信公众号：张巍 GRE】
Quantity A: The median of the numbers in list R
Quantity B: The median of the numbers in list S

22. x>1
Quantity A: x^{-3}
Quantity B: $\sqrt{x^{-5}}$

23. Let *x, y*, and z be positive numbers. If *x* is 12.5 percent of *y* and *x* is 3.125 percent of *z*, then *x* is what percent of *y*+*z*?

24. After a bird feeder was filled with *x* pounds of birdseed, 12 ounces of birdseed were consumed each day for the next 8 days.（1 pound = 16 ounces）
Quantity A: At the endof the 8 days, the weight of the birdseed in the bird feeder that was not consumed
Quantity B: *x*-5 pounds

25. In the rectangular coordinate system, point *P* has coordinates (-2, 1) and point *Q* has coordinates(3, 6).



Quantity A: The slope of line *l*
Quantity B: 1

26. The units digit of 3^n is 1, where n is an integer greater than 50.
Quantity A: The units digit of 3^{n+3}
Quantity B: 7

27. For all integers k>1. Pk = $\frac{k+1}{k}$. For al integers k>2, $b_k = P_k - P_{k-1}$. The integer n is greater than 2
Quantity A: b_n
Quantity B: $\frac{-1}{n^2-n}$

28. On his first bitrthday, John weighed 3 times his birth weight. On his second birthday, he weighed 50 percent more than he had on his first birthday. On each of his third, fourth, and fifth birthdays, he weighed 10 percent more, respectively, than he had on each of the immediately preceding birthdays. If Johm weighed 18 kilograms on his fifth birthday, which of the following represents his birth weight, in kilograms?【微信公众号：张巍 GRE】

A. $\frac{18}{(1.1)^3(1.5)(3)}$

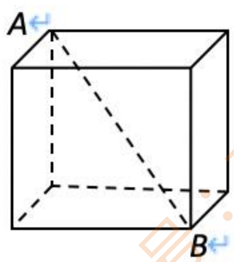
B. $\frac{18}{(1.1)^3(0.5)(3)}$

C. $\frac{18}{(1.1)^2(1.5)(3)}$

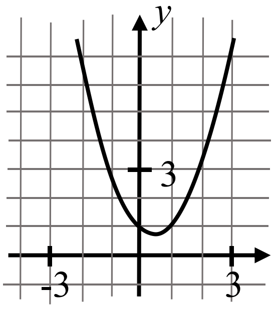
D. $\frac{18}{(1.1)^2(0.5)(3)}$

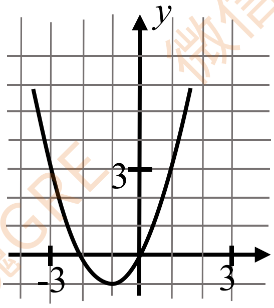
E. $\frac{18}{(0.1)^3(1.5)(3)}$

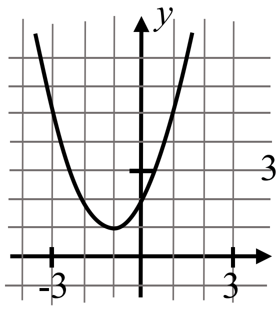
29. In the cube shown above, the length of diagonal AB is 6. What is the surface area of the cube?
A. 24
B. 36
C. 54
D. 72
E. 108

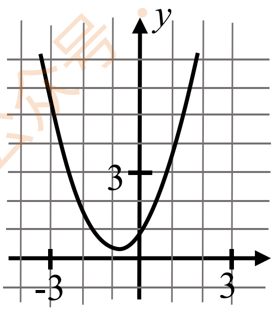


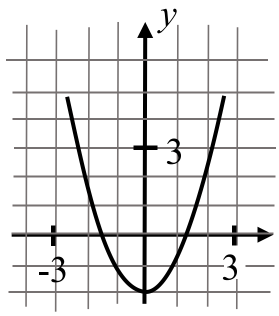
30. The functions f and g are defined for all numbers x by $f(x) = x^2$ and $g(x) = bx+1$, where b is a constant and $0< b< 2$. Which of the following could be the graph of $y = f(x) + g(x)$ in the xy -plane? 【微信公众号：张巍 GRE】











31. The positive integers w, x, y, and z are consecutive multiples of 6; and w＜x＜y＜z.

Quantity A: $\frac{w+z}{2}$

Quantity B: $x+\frac{z-y}{2}$

32. One integer will be randomly chosen from a set of 6 consecutive positive integers. If p is the probability that the chosen integer will be a multiple of 5, what is the greatest possible value of p?

A. 1/6
B. 1/5
C. 1/4
D. 1/3
E. 1/2

33. If n is a positive integer, which of the following CANNOT be the units digit of 2^n-1?
    A. 1
    B. 3
    C. 5
    D. 7
    E. 9

34. Today a certain machine is worth 20 percent less than is was worth a year ago, and it is worth x percent less than it was worth two years ago. A year ago the machine was worth 20 percent less than it was worth two years ago.
    Quantity A: x
    Quantity B: 40

35. The sum of the lengths of two sides of isosceles triangle T is 10, and T has a side of length 5.
    Quantity A: The perimeter of T
    Quantity B: 15

36. *n* is an integer.【微信公众号：张巍 GRE】
    Quantity A: $(-1)^n(-1)^{n+2}$
    Quantity B: 1

37. Terry deposits a total of $400 in two different accounts. One account pays 3 percent annual interest and the other pays 5 percent annual interest. The total yearly interest on these investments is $l6.
    Quantity A: The amount of money that is invested in the account that pays 3 percent annual interest
    Quantity B: The amount of money that is invested in the account that pays 5 percent annual interest

38. A certain experiment has exactly two mutually exclusive outcomes with probabilities p and 2p, respectively.
    Quantity A: p
    Quantity B: 1-p

39. In the xy-plane, line k passes through the points (-3, 2) and (0, 0). If line m is perpendicular to line k, what is the slope of line m?
    A. 2/3
    B. 3/2
    C. 1/3
    D. -2/3
    E. -3/2

40. Row A: 15, 16, 17, 18, 19
    Row B: 8, 9, 10, 11, 12, 13, 14
    One number is to be selected from row A and another from row B. The sum of the two numbers will have how many possible values? 【微信公众号：张巍 GRE】
    A.9
    B.10
    C.11
    D.12
    E.13

Section 6 - Medium

41. A market analyst estimated that 1.9 million smart watches were sold in 2013 and 6.8 million smart watches were sold in 2014. In 2015 the analyst predicted that the percent increase in the number sold from 2014 to 2015 would be half of the percent increase in the number sold from 2013 to 2014. Approximately how many smart watches were predicted by the analyst to be sold in 2015? 【微信公众号：张巍 GRE】
    A. 9.3 million
    B. 12.2 million
    C. 15.6 million
    D. 19.0 million
    E. 24.3 million

42. Which of the following inequalities is consistent with the statement that the time it took train K to travel r miles at a constant rate of s miles per hour is less than the time it took train M to travel y miles at a constant rate of z mile per hour?
    A. $yz - rs > 0$
    B. $rs - yz > 0$
    C. $rz - sy > 0$
    D. $sy - rz > 0$
    E. $ry - sz > 0$

43. Set *S* consists of all three-digit positive integers that contain only the digits 1, 2, 3, 4, or 5.
    Quantity A: The number of integers in *S* in which all three digits are different
    Quantity B: 60

44. If the average (arithmetic mean) of three consecutive odd integers is 11, then 7 more than the least of the three integers is
    A. 16
    B. 17
    C. 18
    D. 19
    E. 20

45. If $z=\frac{x}{y}$, $w=\frac{x}{x+y}$,and $0<x<y$, what is w in terms of z?

A. z^2

B. z+1

C. $\frac{1}{z+1}$

D. $\frac{z}{z+1}$

E. $\frac{z}{z-1}$

46. If two different numbers are selected at random from the numbers 1, 2, 3, 4 and 5, what is the probability that their product will be odd?【微信公众号：张巍 GRE】

A. $\frac{1}{2}$

B. $\frac{1}{12}$

C. $\frac{1}{20}$

D. $\frac{3}{5}$

E. $\frac{3}{10}$

47. The first term of an infinite sequence is 4, and each term after the first term is 7 greater than the preceding term. What is the 64th term of the sequence?

A. 188

B. 431

C. 445

D. 448

E. 452

48. A wall in a museum is covered by 60 flat rectangular panels. If the dimensions of each panel are 4 feet by 7.5 feet, what is the total area, in <u>square yards</u>, of the panels?(<u>Note</u>: 1 yard=3 feet.)

A. 200

B. 600

C. 1,800

D. 5,400

E. 16,200

49. Last year and this year, there were both men and women in a certain choir. This year there are x fewer men and x fewer women in the choir than there were last year, where $x>0$, and there are fewer men than women in the choir.

Quantity A: The percent decrease from last year to this year in the number of men in the choir

Quantity B: The percent decrease from last year to this year in the number of women in the choir

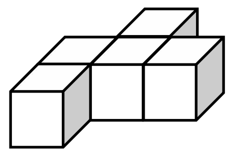
50. Quantity A: The standard deviation of the numbers 45, 64, 83, and 53

Quantity B: The standard deviation of the numbers 55, 81, 47, and 62

51. The solid shown consists of five cubes, each with a volume of 27, that are joined so that four pairs of faces coincide.

Quantity A: The surface area of the solid

Quantity B: 198



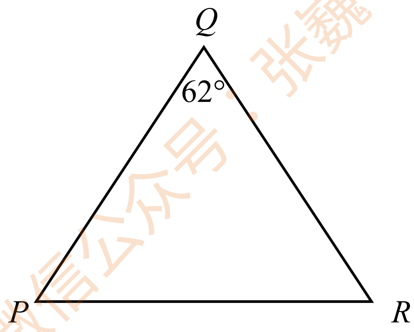
52. A total of $24,000 was invested for one month in a new money market account that paid simple annual interest at the rate of r percent. If the investment earned $180 in interest for the month, what is the value of r?【微信公众号：张巍 GRE】

A. 7.5
B. 8.0
C. 8.5
D. 9.0
E. 10.0

53. A pianist agreed to perform one concert at a fee $12\frac{1}{2}$ percent less than her usual fee and a second concert at a fee 20 percent greater than the first fee. The fee for the second concert was what percent greater than her usual fee?

A. 5%
B. $7\frac{1}{2}\%$
C. 15%
D. $16\frac{1}{4}\%$
E. $32\frac{1}{2}\%$

54. $\triangle PQR$ is isosceles.
Quantity A: The measure of angle P
Quantity B: The measure of angle R



55. A certain list has 5 entries and each entry is an integer between 55 and 70, inclusive. The median of the 5 entries is 60. If m is the average (arithmetic mean) of the 5 entries, which of the following must be true?【微信公众号：张巍 GRE】
A. $54 \le m \le 60$
B. $55 \le m \le 61$
C. $56 \le m \le 62$
D. $57 \le m \le 63$
E. $58 \le m \le 64$

56. List L consists of 11 different positive numbers. The sum of the 6 smallest numbers in L is 35, and the sum of the 6 greatest numbers in L. is 125. If the sum of all of the numbers in L is 142, what is the median of the numbers in L?

57. At a certain sale, the price of a jacket was reduced by 5 percent and the price of a pair of slacks was reduced by 20 percent.
Quantity A: The dollar reduction in the price of the jacket
Quantity B: The dollar reduction in the price of the pair of slacks

58. Last month Paul and Jack each sold newspapers and Paul sold 5,000 more newspapers than Jack. If Paul and Jack sold a combined total of 35,000 newspapers last month, what fraction of this combined total did Jack sell?
A. 3/4
B. 4/7
C. 3/7
D. 4/10
E. 3/10

59. In sequence T, each term after the first term is d more than the preceding term. The sum of the first 10 terms of T is 210. The sum of the first 20 terms of T is 820. What is the value of d?
【微信公众号：张巍 GRE】
A. 4
B. 4.5
C. 5
D. 5.5
E. 6

60. The function f is defined for all integers n greater than 2 by f(n)=(1)(2)(3)...(n-2). That is, f(n) is the product of the first n-2 integers. What is the value of f(9)/f(8)?
A 9
B 8
C 7
D 8/7
E 7/6

Section 7 - Medium

61. In the sophomore class of a certain university, 85 percent of the students were taking English composition, 70 percent were taking biology, and 60 percent were taking both English composition and biology. What percent of the students in the class were taking neither English composition nor biology? 【微信公众号：张巍 GRE】
A 0%
B 5%
C 10%
D 15%
E 25%

62. Evelyn owns two different properties with areas of 1.5 acres and 0.5 acre, respectively. What is the total area, in square feet of the two properties combined? (1 acre = 4,840 square yards, 1 yard = 3 feet)

63. John bought a suit for a selling price of s dollars, where s is an integer greater than 50. He paid a tax on the suit that was r percent of the selling price of the suit. where r is a positive integer less than 10.
Quantity A： The tax John paid on the suit
Quantity B： sr/100

64. A bag contains 6 balls of which 2 are red and 4 are green. If 2 balls are to be chosen at random one after the other, without replacement. what is the probability that the second ball chosen will be red?
A 1/6
B 1/5
C 1/3
D 2/5
E 2/3

65. If x=2y+1, and y=2w, where w, x, and y are integers, which of the following must be an odd integer?
A xy+w
B xy+w+1
C (x+y)w
D wy+x
E wx+y

66. In a university, a certain committee consists of 6 faculty members, 4 administrators and 3 students. A subcommittee of 5 members will be selected from the committee. Professor Smith, who is one of the 6 faculty members, and Ms. Wilson, who is one of the 4 administrators must be on the subcommittee. The other 3 subcommittee members will be selected at random from the rest of the committee, How many different 5-member subcommittees can be selected? 【微信公众号：张巍 GRE】

A 45
B 72
C 165
D 720
E 1.287

67. Based on the table shown, the braking distance of car C for a speed of 50 miles per hour is what percent greater than the braking distance of car C for a speed of 40 miles per hour?

Braking Distance of Car C

Speed (miles per hour)	Braking Distance (feet)
10	20
20	45
30	78
40	125
50	188
60	272

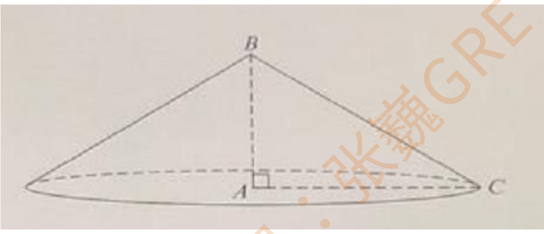
68. At a convention, 30 percent of the attendees are vegetarians, and of those vegetarians 55 percent are men.

Quantity A： The percent of the attendees who are not men and not vegetarians

Quantity B： 25%

69. The figure shows a right circular cone, where A is the center of the circular base. If the measure of angle ACB is 30°and the length of line segment BC is 2, what is the volume of the cone? (The volume of a right circular cone is one-third of the product of the height of the cone and the area of the circular base.)

Give your answer to the nearest whole number.



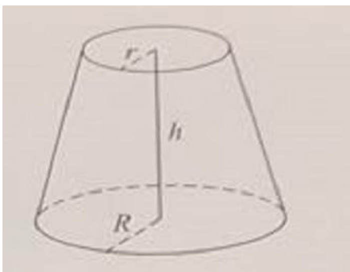
70. R：x，y，z，2.7，3.8，5.5
S：x，y，z，-2.7，-3.8，-5.5
If the average (arithmetic mean) of the 6 numbers in list R is r and the average of the 6 numbers in list S is s， then r-s is the average of which of the following lists?
A x，y，z
B 2.7，3.8，5.5
C –x，-y，-z
D -2.7，-3.8，-5.5
E 2.7，3.8，5.5，-2.7，-3.8，-5.5

71. $(3.8\times 6^{25}) - (0.24\times 6^{26}) =$
A 2.36×6^{25}
B 2.36×6^{26}
C 3.56×6^{25}
D 3.56×6^{26}
E 3.76×6^{25}

72. In plane P, point Q is on line l.
Quantity A： The number of points in plane P whose distance from line l is 2 and whose distance from point Q is 3
Quantity B： 4

73. The function f is defined by $f(x)=\sqrt{x}-2$ for all numbers $x\geq 0$. If $f(f(v))=0$ and $v\geq 4$, what is the value of v? 【微信公众号：张巍 GRE】
A 8
B $8\sqrt{2}$
C 16
D $16\sqrt{2}$
E 36

74. The volume V of the frustum of a right circular cone is given by $V=(\frac{\pi h}{3})(R^2+Rr+r^2)$, where h is the height of the frustum, R is the radius of its lower base, and r is the radius of its upper base, as shown. If the volume of the frustum of a right circular cone is 148π, its height is 12, and the radius of its upper base is 3, what is the radius of the lower base of the frustum?
A 4
B 5
C 6
D 7
E 8

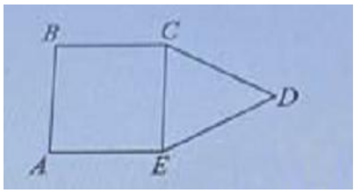


75. The perimeter of parallelogram ABCD is 360 feet.

Quantity A: The length of diagonal AC
Quantity B: $90\sqrt{2}$ feet

76. List S consists of 10 positive numbers. The range of the numbers in S is 124. Each of the numbers in S will be increased by x percent, where x>0. List T consists of the 10 resulting numbers.
Quantity A：The range of the numbers in T
Quantity B：124

77. Quadrilateral ABCE is a square, and triangle CDE is equilateral. If the perimeter of pentagon ABCDE is 30, what is the area of pentagon ABCDE?
A $25+25\sqrt{3}$
B $25+50\sqrt{3}$
C $36+6\sqrt{3}$
D $36+9\sqrt{3}$
E $6+18\sqrt{3}$



78. An apple falls straight to the ground from a tree branch 8 meters above the ground. The apple's height above the ground, t seconds after it starts falling, is H meters, where $H=8-4.9t^2$ and $t\geq 0$. How many seconds after the apple starts falling does it hit the ground? Give your answer to the nearest 0.1 second.【微信公众号：张巍 GRE】

79. N copies of a certain health magazine cost a total of \$64.
R copies of a certain news magazine cost a total of \$80.
Quantity A：The ratio of the cost of 1 of the health magazines to the cost of 1 of the news magazines
Quantity B：4/5

80. Quantity A：The number of odd integers between $\sqrt{12}$ and 12^2
Quantity B：70

Section 8 - Medium

81. -1, 2, -3, 4, ... , a_n, ... , -99, 100

In the finite sequence, $a_n = (-1)^n \times n$ for all integers n between 1 and 100, inclusive.

Quantity A：The sum of the terms in the sequence

Quantity B：51

82. When the integer n is divided by 33, the remainder is 24. Which of the following must be a divisor of n? 【微信公众号：张巍 GRE】

A 11

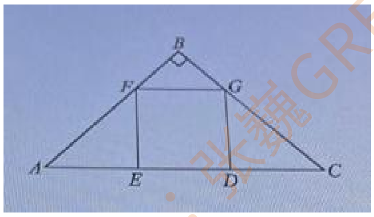
B 9

C 7

D 4

E 3

83. Square DEPG is inscribed in isosceles right triangle ABC. If the area of triangular region ABC is r, what is the area of triangular region AFE?



A $\frac{1}{6}$ r

B $\frac{2}{9}$ r

C $\frac{1}{4}$ r

D $\frac{1}{3}$ r

E $\frac{4}{9}$ r

84. A bag contains 13 blue pens and 3 red pens. Two pens are to be randomly selected from the bag, one at a time and without replacement. What is the probability that both of the pens selected will be blue?

85. In the xy-plane, 8 data points lie on the line with equation y=40+5x. If the standard deviation of the x-coordinates of the 8 points is 4.6, what is the standard deviation of the y-coordinates of the 8 points?

86. How many different positive three-digit integers are there that have an odd hundreds digit?
A 400
B 405
C 495
D 500
E 1,000

87. The reduced price of a certain television was x percent less than the television's original price.
The ratio of the reduced price of the television to the original price was 5 to 6.
Quantity A： x
Quantity B： 20

88. $a_k = \left(\frac{1}{k} - \frac{1}{k+1}\right)$ for any positive Integer k.【微信公众号：张巍 GRE】

Quantity A： $a_3 + a_4 + a_5 + a_6 + a_7$

Quantity B： 1/8

89. List G consists of positive integers, each of which is less than or equal to 300. The average (arithmetic mean) of the integers in G is greater than 25c, where c is a positive number.
Quantity A： c
Quantity B： 13

90. In a survey of educators. 40 percent reported that they used the Internet for research. Also in the survey, 65 percent reported that they used the Internet for E-mail.
Quantity A： The percent of the educators surveyed who reported that they used the Internet for both research and E-mail
Quantity B： 26%

91. Set K consists of 9 positive integers, 5 of which are prime numbers. How many subsets of K consist of 3 integers such that 2 integers are prime numbers and 1 integer is not a prime number?

92. If x and y are positive integers in the equation shown, what is the least possible value of x+y?

$$\frac{x+3y}{-2}=\frac{2x+y}{-3}$$

93. a and b are distinct odd prime numbers. 【微信公众号：张巍 GRE】

Quantity A：The number of factors of $2ab^2$

Quantity B：The number of factors of a^2b^3

94. x and y are integers such that 0<y<x.
H=100x+10x+4
G=100y+10y+2
M=(H-G)(H+G)
Quantity A：The units digit of M
Quantity B：2

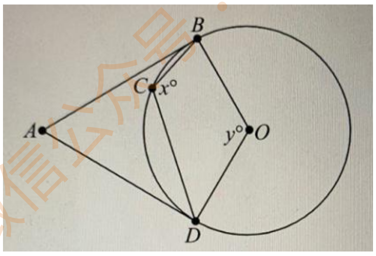
95. x>0, y<0
Quantity A：x+y-1
Quantity B：x-y+1

96. An apartment building contains 60 apartments. Of the apartments in the building, 3/5 have one bedroom and the rest have 2 or 3 bedrooms. If the 60 apartments have a total of 94 bedrooms, how many of the apartments have 3 bedrooms?
A 10
B 12
C 16
D 24
E 36

97. Pet drove a car 400 miles in 8 hours. For the first 6 hours, Pat drove on highway X; and the rest of the time, Pet drove on Highway Y. If the average speed for the car on highway X was 20 miles per hour more than the average speed on Highway Y, what was the car's average speed, in miles per hour, on highway Y?
A 35
B 45
C 55
D 60
E 65

98. In the figure above the circle has center O, and AB and AD are tangent to the circle. If the degree measures of angles ABC and ADC are 20 and 40, respectively, what is the value of x+y?

A 140
B 160
C 190
D 210
E 240



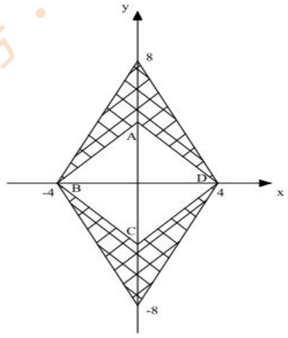
99. How many integers between 100 and 1, 000 have a hundreds digit that is even, a tens digit that is odd, and a non-zero units digit that is divisible by 3? 【微信公众号：张巍 GRE】

A 15
B 30
C 45
D 60
E 75

100. a is a positive integer.
x is the remainder when 15a is divided by 6.
Quantity A： x
Quantity B： 2

Section 9 - Medium

101. ABCD is a square, what is the area of the shaded region?
【微信公众号：张巍 GRE】

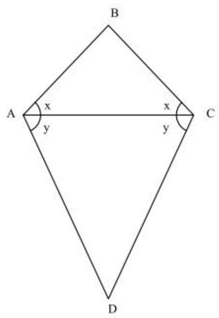


102. $k+\frac{1}{k}=3$

Quantity A： $k+\frac{1}{k^2}$

Quantity B： $k^2+\frac{1}{k^3}$

103. AB<AD
Quantity A： x
Quantity B： y



104. B is a positive integer, and A is 125 greater than B. Which of the following statements must be true?
Indicate all such statements.

A. A is odd.
B. B is even.
C. A+B is odd.
D. A-B is even.
E. AB is even.

105. On the number line, the distance between points P and Q is 18, point M is the midpoint of line segment PQ, and point N is the midpoint of line segment MQ. If M is to be the left of N and the coordinate of N is 0, what is the coordinate of P?
A. -13.5
B. -4.5
C. 4.5
D. 13.5
E. 22.5

106. When positive integer m is divided by 6, the remainder is 4. When positive integer p is divided by 6, the remainder is 5. What is the remainder when the product mp is divided by 6?

A. 1
B. 2
C. 3
D. 4
E. 5

107. How many ordered pairs (x,y), where both x and y are integers, are solutions of the equation $x^2+y^2=5$? 【微信公众号：张巍 GRE】

A. 2
B. 4
C. 8
D. 10
E. 12

108. Nancy invested $5,000 for x years at an annual interest rate of 6 percent, compounded annually. David invested $6,000 for y years at annual interest rate of 6 percent, compounded semiannually.

Quantity A: The total amount of interest that Nancy's investment earned
Quantity B: The total amount of interest that David's investment earned

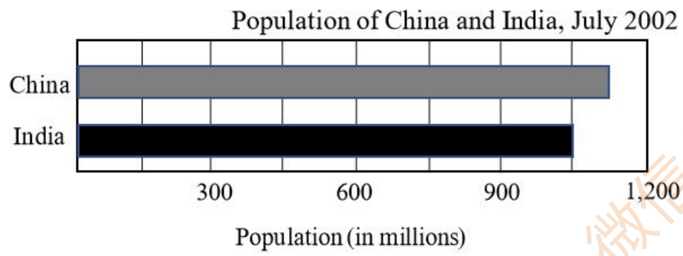
109. The dollar value of a certain carving in 1975 was 25 percent greater than its dollar value in 1970. The dollar value of the carving in 1980 was 80 percent greater than its dollar value in 1970.

Quantity A: The percent increase in the dollar value of the caring from 1975 to 1980.
Quantity B: 55%

110. x and y are integers.

Quantity A: $2^{(x+y)^2}$
Quantity B: $2^{x^2+y^2}$

111. According to the graph, in July 2002, the population of China was approximately what percent greater than the population of India?



A. 2%
B. 7%
C. 17%
D. 22%
E. 27%

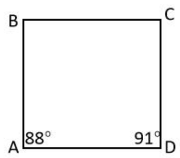
112. If $a<b$, which of the following must be true?【微信公众号：张巍 GRE】

A. $|a| < |b|$
B. $a<|ab|$
C. $a-b < |a-b|$
D. $b-a < |a+b|$
E. $a+b < |a| + |b|$

113. The standard deviation of the numbers a, b, c, d, e, f, and g is 3.6. What is the standard deviation of a+2.4, b+2.4, c+2.4, d+2.4, e+2.4, f+2.4, and g+2.4?

A. 1.2
B. 2.4
C. 3.6
D. 4.8
E. 6.0

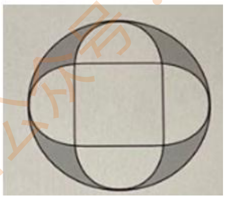
114. AD || BC



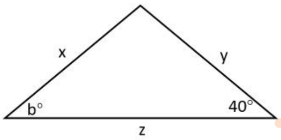
Quantity A: The length of line segment AB
Quantity B: The length of line segment DC

115. Each of the four semicircles shown in the figure has a side of the square as its diameter and has exactly one point in common with the larger circle. If each side of the square has length 4, What is the sum of the areas of the shaded region? 【微信公众号：张巍 GRE】

A. $8\pi-16$
B. $12\pi-16$
C. $14\pi-16$
D. $16\pi-16$
E. $18\pi-16$



116. In the triangle shown above, x<y<z. Which of the following could be the value of b?



Indicate all such values.
A. 35
B. 39
C. 48
D. 67
E. 71

117. The length of a rectangular rug is 3 times its width, and the area of the rug is between 24 square feet and 63 square feet. Which of the following values could be the perimeter of the rug, in feet?
Indicate all such values.
A. 10
B. 24
C. 32
D. 3

118. p, q, r, and s are different positive prime numbers
Quantity A: The greatest common factor of p^2qr and qr^2s
Quantity B: qr

119. In the xy-plane(not shown), a right triangle has its right angle at the origin and has its hypotenuse along the line y =7x-1. If none of the sides of the triangle are vertical, what is the

product of the slopes of the three sides of the triangle?

A. -7

B. -1

C. -1/7

D. 1/7

E. 1

120. $a<-4<b<c<-3<0<2<d<3$

Given the compound inequality above, which of the following statements is true?【微信公众号：张巍 GRE】

A. $a^2 < b^2 < c^2 < d^2$

B. $b^2 < c^2 < d^2 < a^2$

C. $c^2 < b^2 < d^2 < a^2$

D. $c^2 < d^2 < a^2 < b^2$

E. $d^2 < c^2 < b^2 < a^2$

Section 10 - Medium

121. In the xy-plane, line k has a slope of 2 and passes through the point(5,5). Which of the following points lie on line k? 【微信公众号：张巍 GRE】

Indicate all such points.

A. (-5,5)

B. (0,-5)

C. (10,15)

122. A ball is dropped from a height of 6 meters and bounces to no more than 90 percent of its original height. It falls back down and then repeatedly bounces to no more than 90 percent of its maximum height after the previous bounce. What is the maximum height, in meters, that the ball can reach after the 5th bounce?

A. $(6 \times 0.9)^4$

B. $(6 \times 0.9)^5$

C. $6(0.9)^4$

D. $6(0.9)^5$

E. $6(0.9)^6$

123. 15,000 is divisible by $25a^kb^2$, where a and b are prime numbers, a≠b, and k is a positive integer.

Quantity A: The greatest possible value of k

Quantity B: 3

124. The planet Mars travels around the Sun at an average speed of 29.8 miles per second.

Quantity A: The average speed at which Mar travels around the Sun, in miles per hour

Quantity B: 100,000 miles per hour

125. A list consists of 10 positive integers. The sum of the 10 integers is 101. If no integer appears more than twice in the list, what is the greatest possible integer that can appear in the list?

126. A rectangular floor will be completely covered with square tiles, each of which has sides of length 6 inches. If tiles are laid side by side with no space between them and no tiles are cut, then the number of tiles needed to cover the floor is 1,080. Which of the following could be the dimensions of the floor? (Note: 12 inches = 1 foot.)

A. 15 feet by 9 feet
B. 15 feet by 18 feet
C. 30 feet by 18 feet
D. 30 feet by 36 feet
E. 45 feet by 36 feet

127. Fred's suitcase contains 4 shirts, 3 pair of pants, and 2 pairs of shoes. A matching outfit consists of any shirts, any pair of pants, and a pair of shoes, except that one of the shirt does not match one of the pairs of pants. How many matching outfits can be selected from Fred's suitcase?【微信公众号：张巍 GRE】

128. What is the median of the number shown?

$$2\sqrt{2}-1,\quad \sqrt{2}-1,\quad 1+\sqrt{2},\quad 1-\sqrt{2},\quad \sqrt{2}$$

A. $2\sqrt{2}-1$
B. $\sqrt{2}-1$
C. $1+\sqrt{2}$
D. $1-\sqrt{2}$
E. $\sqrt{2}$

129. If q and r are prime numbers greater than 3 and $qr^2 < 450$, what is the greatest possible value of r?

A. 5
B. 7
C. 11
D. 13
E. 17

130. r is the remainder obtained dividing $7^{995} + 7^{50} - 4$ by 7

Quantity A: r
Quantity B: 4

131. 11,12,13,14,15

Which of the following changes to the list of numbers above would increase the average (arithmetic mean) of the numbers but not affect their standard deviation?

A. Adding the number "13" to the list

B. Adding the number “20” to the list
C. Multiplying each number in the list by 2
D. Adding 3 to each number in the list
E. Replacing the “15” with a “20”

132. A dealer’s gross profit on the sale of a car is defined as the selling price of the car minus its cost to the dealer. A car dealer sold n cars for s dollars each. If the dealer’s total gross profit was p dollars, what was the dealer’s average(arithmetic mean) cost per car, in dollars?

A. $\frac{s-p}{n}$

B. s-$\frac{p}{n}$

C. $\frac{s}{n}-$p

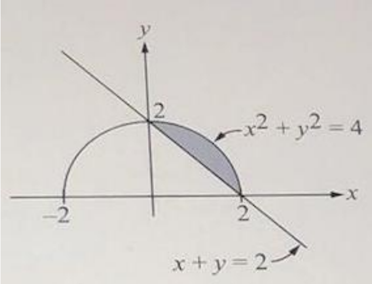
D. p-$\frac{s}{n}$

E. $\frac{sn}{p}$

133. If a pencil is to be chosen at random from the 20 colored pencils in a pencil holder, the probability that the pencil chosen will be yellow is 0.8. If 6 of the yellow pencils in the pencil holder don’t have an eraser, what is the probability that the pencil chosen will be a yellow pencil that has an eraser?【微信公众号：张巍 GRE】
A. 1/10
B. 1/5
C. 3/10
D. 2/5
E. 1/2

134. The number of children in a certain family is a prime number less than 10. The number of boys in the family is greater than the number of girls, and the number of boys is a prime number. If at least 1 of the children in the family is a girl, which of the following could be the number of boys in the family?
Indicate all such numbers.
A. 2
B. 3
C. 4
D. 5
E. 6
F. 7

135. The figure shows a semicircle intersected by a straight line in the rectangular coordinate system. What is the area of the shaded region? 【微信公众号：张巍 GRE】



A. π-2
B. π-4
C. 4π-2
D. 4π-4
E. 16π-2

136. If w, x, y, and z are consecutive positive integers, where w<x<y<z, which of the following statements must be true?
Indicate <u>all</u> such statements.
A. wxy is a multiple of 3.
B. wxz is a multiple of 3.
C. wxy + wyz is a multiple of 3

137. What is the least positive integer b such that the equation $x^2+bx+18=0$ has two integer solutions for x?

138. Three pipes P1, P2, and P3, working together at their respective constant rates, can fill a tank in 6 hours. P1 and P2, working together at their respective constant rates, can fill the same tank in 8 hours. How many hours will it take P3, working alone at its constant rate, to fill this tank?
A. 2
B. 12
C. 18
D. 24
E. 36

139. Solution A is 10 percent salt, by weight, and solution B is 20 percent salt, by weight. How many grams of solution B must be added to 200 grams of solution A to make a new solution that is 12 percent salt, by weight?
A. 2

B. 20
C. 50
D. 100
E. 200

140. The “deviation” of x from y is defined as y-x. Based on this definition, what is the sum of the deviations of 39 from 22, 34, and 46? 【微信公众号：张巍 GRE】

A. -15
B. -5
C. 5
D. 15
E. 29

Section 11 - Hard

1. Each of 12 books on a shelf has a single classification as Mathematics Science, or English. At least 1book is classified as Mathematics, at least 3 books are classified as Science, and at least 3 books are classified as English. There are an odd number of books classified as Science and an odd number of books classified as English. Which of the flowing could be the number of books classified as Mathematics? 【微信公众号：张巍 GRE】

   Indicate all such numbers.

   A) 2
   B) 4
   C) 5
   D) 6
   E) 7
   F) 8

2. Each day for 25 days, the number of people who used a certain library that day was recorded. The standard deviation of the 25 recorded numbers is 12.

   Which of the following statements individually provide(s) sufficient additional information to determine the mean of the 25 recorded numbers?

   Indicate all such statements.

   A. The median of the recorded number is 49.
   B. One of the recorded numbers, 45, is 1/3 standard deviation below the mean.
   C. More than 50 percent of the recorded numbers are between 37 and 61.

3. The odds against the occurrence of an event are defined to be the ratio of the probability that the event will not occur to the probability that the event will occur.

   An integer is to be selected at random from the set of integers from k to n, where k < n. The odds against the event that the integer selected will be an even integer are 1 to 0.95. Which of the following could be the integers k and n, respectively?

   A)13 and 33
   B)14 and 33
   C)15 and 33
   D)16 and 55
   E)17 and 55

4. The standard deviation of n numbers x_1, x_2, x_3, …, xn with mena $\overline{x}$ is equal to $\sqrt{\frac{s}{n}}$ where s is the sum of the squared differences $(x_i - \overline{x})^2$ for 1≤i≤n.

c, c, c, 2c, 4c, 4c, 5c, 6c

In the list of 8 numbers shown, C>0. Which of the following is closest to the standard deviation of the 8 numbers? 【微信公众号：张巍 GRE】

A) 1.4c
B) 1.9c
C) 2.2c
D) 3.0c
E) 3.5c

5.

1, 2, 3, … , m
1, 2, 3, … , n

The first sequence consists of m consecutive integers, and the second sequence consists of n consecutive integers, where m is even and n is odd.

Quantity A: The percent of integers in the first sequence that are odd
Quantity B: The percent of integers in the second sequence that are even

6. The acreage of Hopetown, a town in Lewis County, is equal to 20 percent of the acreage of the region that is in Lewis County but not in Hopetown.
Quantity A: The acreage of Hopetown as a percent of the entire acreage of Lewis County
Quantity B: 18%

7. Bruce, Carlos, Debbie, and Esther are to be seated in four adjacent chairs arranged in a row. If Debbie and Esther are to be seated next to each other, how many different seating arrangements are possible?

A. 6
B. 8
C. 9
D. 12
E. 16

8. An escalator installed at a new shopping mall operates at a speed of 90 feet per minute. The handrail of the escalator operates on a separate motor at a speed of 93 feet per minute. A person steps onto the escalator and, at the same time, grabs the handrail. Assuming that the person's hand and shoes do not move on the escalator in how many seconds will the person's hand have moved 0.25 foot more than the person's shoes?

9. Mr. Thomas gave a chemistry test to 25 students and assigned each student a score. Of the 25 students, 12 students received scores that were greater than 80.

Which of the following statements individually provide(s) sufficient additional information to determine the median of the 25 scores?

Indicate all such statements.

A. The average (arithmetic mean) of the 25 scores was 80.
B. One student received a score of 80.
C. Twelve students received scores that were less than or equal to 75.

10. List L: 84, 73,87, 88, 75, 90

The standard deviation of the numbers in list *L* is *s*. List *M* consists of 6 different numbers, each of which is obtained by adding 9 to one of the numbers in list *L*. The standard deviation of the numbers in list *M* is *t*.

Quantity A: s+3

Quantity B: t

11. For City *O*, the composite cost-of-living index (not shown) is the weighted mean of the four component indexes shown, using the percents as relative weights. Approximately what is the composite cost-of-living index for City *O* ?

Food (20%)	Housing (20%)	Transportation and Energy (25%)	Miscellancous Goods and Services (35%)	Composite (100%)
95	84	148	50	

A. 98
B. 94
C. 90
D. 86
E. 82

12. The units digit of 7^n is r, and the units digit of 9^n is t, where n, r, and t are positive integers. Which of the following could be the value of r+t?【微信公众号：张巍 GRE】

Indicate all such values

A. 10
B. 12
C. 14
D. 16

13. How many distinct solutions does the equation $\left|1-|x-215|\right|=1$ have?

A. None
B. One
C. Two
D. Three
E. Four

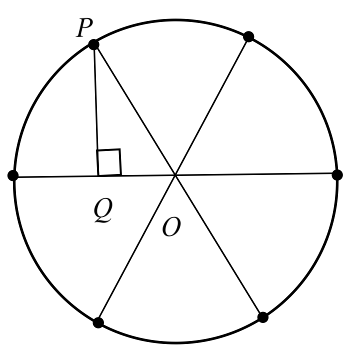
14. Of the 120 first-semester students entering a certain college, 3/5 registered for a statistics class and 1/2 registered for a calculus class. Which of the following could be the number of first-semester students entering the college who registered for a statistics class but did not register for a calculus class? 【微信公众号：张巍 GRE】

Indicate all such numbers.

A. 0
B. 6
C. 12
D. 30
E. 60
F. 72
G. 90

15. Five different families received tax bills at the beginning of the year. Later, the four lowest tax bills were each reduced by $200 while the highest tax bill remained the same.
Quantity A: The standard deviation of the original five tax bills
Quantity B: The standard deviation of the resulting five tax bills after the four lowest tax bills were reduced

16. The circle has center O and radius 3. The 6 points indicated on the circle divide the circle into 6 arcs of equal length.
Quantity A: The length of PQ
Quantity B: $\frac{3\sqrt{3}}{2}$



17. The tens digit of the product $(4^{100})(5^{99})$
The tens digit of the product $(4^{100})(5^{101})$

18. A restaurant has a total of 16 tables, each of which can seat a maximum of 4 people. If 50 people were sitting at the tables in the restaurant, with no tables empty, what is the greatest possible number of tables that could be occupied by just 1 person?
    A. One
    B. Two
    C. Three
    D. Four
    E. Five

19. When the positive integer w is divided by 13, the remainder is 1, and when w is divided by 15, the remainder is 14. Which of the following is a possible value for w?【微信公众号：张巍 GRE】
    A. 14
    B. 17
    C. 29
    D. 40
    E. 44

20. For the cars parked in a parking lot, the range of their odometer readings is 52,000 kilometers. Two of the odometer readings are 66,000 kilometers and 84,000 kilometers. Which of the following values could be the odometer reading, in kilometers, of one of the cars in the parking lot?

    Indicate all such values.
    A. 10,000
    B. 30,000
    C. 60,000
    D. 90,000
    E. 120,000
    F. 140,000

Section 12 - Hard

21. Let g be the function defined for all numbers x such that g(x) is the greatest integer that is less than or equal to x. If g(x) = -3, which of the following could be the value of x?
Indicate all such values. 【微信公众号：张巍 GRE】
A. -3.9
B. -3.1
C. -3.0
D. -2.9
E. -2.1

22. 2, 4, 14, 64
In the sequence shown, each term after the first term is c less than the product of d and the preceding term, where c and d are constants. What is the value of c + d?

23. For a positive integer *n*, the remainder is 3 when it is divided by 7, and the remainder is *R* when it is divided by 4.
Quantity A: R
Quantity B: 2

24. Last spring, a gardening store bought 100 trees for $6 each to resell to customers. By the end of the year, 20 of the trees were sold for $30 each, 30 of the trees were sold for $50 each, 40 of the trees were sold for $60 each, and 10 of the trees were damaged and not sold. For the 100 trees, what was the average (arithmetic mean) profit per tree?(Note: Profit equals revenue minus costs.)

25. A research report states that the average (arithmetic mean) of 120 measurements was 72.5, the greatest of the 120 measurements was 92.8, and the range of the 120 measurements was 51.6.

The information given above is sufficient to determine the value of which of the following statistics?

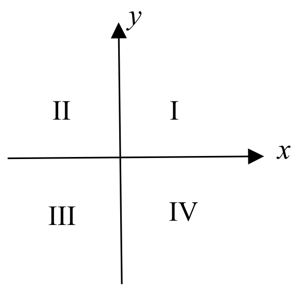
Indicate all such statistics.

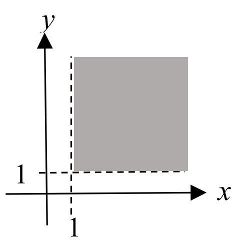
A. The least of the 120 measurements
B. The median of the 120 measurements
C. The standard deviation of the 120 measurements
D. The sum of the 120 measurements

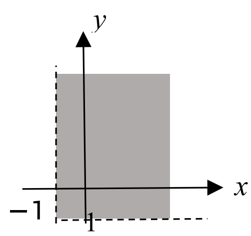
26. For all numbers x, the function $h(x)$ is defined as 1 more than the greatest integer less than or equal to x. 【微信公众号：张巍 GRE】
Quantity A: h(1.5)
Quantity B: h(1.75)

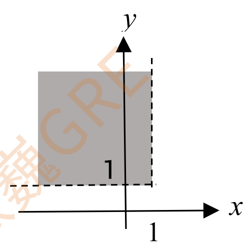
27. A set consists of all three-digit positive integers with the following properties. Each integer is of the form *JKL*, where *J,K,* and *L* are digits; all the digits are nonzero; and the two-digit integers *JK* and *KL* are each divisible by 9. How many integers are in the set?
A. 5
B. 9
C. 15
D. 27
E. 36

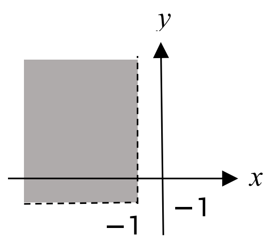
28. Which of the following shaded regions represents the set of all points (a, b) in the xy-plane above such that $(a+1, b+1)$ is in quadrant I? (Note that a point on an axis is not in any quadrant.)

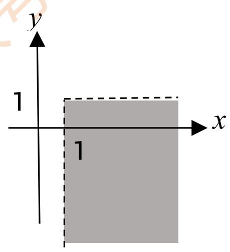












29. A certain museum contains a total of 1,500 works of art, of which 800 are paintings. Of the twentieth century works of art in the museum, 40 percent are paintings. Of the works of art that are <u>not</u> paintings, 490 are <u>not</u> twentieth-century works of art.

How many twentieth-century works of art does the museum contain?

30. For each integer greater than 1, the sum of the first positive integers is given by the formula shown.

If the average (arithmetic mean) of the first n positive integers is k, what is the sum of the first n positive integers in terms of k? 【微信公众号：张巍 GRE】

$$1+2+3+\ldots+n = \frac{n(n+1)}{2}$$

A. $k^2 - k$

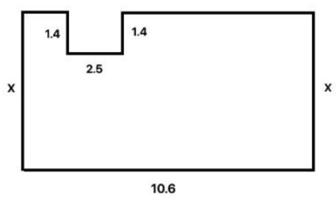
B. $\dfrac{k^2 - k}{2}$

C. $\dfrac{k^2 + k}{2}$

D. $2k^2 - k$

E. $4k^2 + 2k$

31. The figure above represents the floor plan of a basement in which all the walls meet at right angles. The measurements shown in the floor plan, including the two sides of length x, are in centimeters. The floor plan is drawn using a scale of 1 centimeter to 1.5 meters. If the perimeter of the floor plan is 39.4 centimeters, approximately what is the area, in square meters, of the basement floor?



A. 78
B. 117
C. 122
D. 176
E. 184

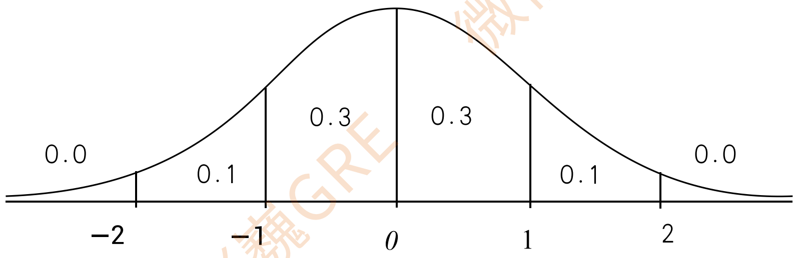
32. Let a be the greatest integer such that 5^a is a factor of 1,500, and let b be the greatest integer such that 3^b is a factor of 33,333,333. Which of the following statements are true?

Indicate all such statements.

A. $ab = 3$
B. $a > b$
C. $2a > 5b$

33. The figure shows the standard normal distribution, with mean 0 and standard deviation 1, including approximate probabilities corresponding to the six intervals shown.

The random variable X is normally distributed with mean 140 and standard deviation 12.



Quantity A: $P(125<x<131)$
Quantity B: $P(152<x<158)$

34. A shop sells only two types of beverages: coffees and teas. If 36 percent of the shop's beverages are teas that have caffeine and 60 percent of the shop's teas have caffeine, what percent of the shop's beverages are coffees? 【微信公众号：张巍 GRE】

A. 24%
B. 30%
C. 40%
D. 50%
E. 60%

35. List L consists of the integers from 1 to 99 and two integers c and d such that $c + d = 100$ and $cd<0$.

Which of the following statements must be true?

Indicate <u>all</u> such statements.

A. The average (arithmetic mean) of the numbers in L is equal to the median of the numbers in L.
B. The range of the numbers in L is greater than 100.
C. The range of the numbers in L is less than 200.

36. Among the incoming students at a certain college, an equal number of full-time and part-time

students received a welcome package. If $\frac{3}{16}$ of the incoming full-time students and $\frac{5}{12}$ of the incoming part-time, students received a welcome package, what fraction of the incoming students received a welcome package?

A. $\frac{15}{58}$

B. $\frac{2}{7}$

C. $\frac{29}{96}$

D. $\frac{9}{29}$

E. $\frac{19}{48}$

37. A circle in the xy-plane is given by the equation $(x-a)^2+(y-b)^2=c^2$, where a, b, and c are nonzero constants. If the point（$b+a$，$b-a$）lies on the circle, which of the following is an equation of the tangent line to the circle at this point?【微信公众号：张巍 GRE】

A. $y=\frac{a}{b}(x-a-b)+(a-b)$

B. $y=\frac{a}{b}(x-a+b)+(a+b)$

C. $y=\frac{a}{b}(x-b+a)+(a+b)$

D. $y=\frac{b}{a}(x-b+a)+(a+b)$

E. $y=\frac{b}{a}(x-b-a)+(b-a)$

38. In a game, the cards in a certain deck sare distributed to players one at a time until each player has the same number of cards and there are not enough cards left in the deck to distribute one more card to each player. The number of cards in the deck is between 60 and 100. If the cards in the deck are distributed o 5 players, 2 cards will be left in the deck. If the cards in the deck are distributed to 6 players, 4 cards will be left in the deck. If the cards in the deck are distributed to 7 players, how many cards will be left in the deck?

39. The standard deviation of n numbers $x_1, x_2, x_3, \ldots, x_n$ with mean $\bar{x}$ is equal to $\sqrt{\frac{S}{n}}$, where S is the sum of the squared differences $(x_i-\bar{x})^2$ for $1 \le i \le n$.

If the standard deviation of the 4 numbers $140-a$, 140, 160, and $160+a$ is 50, where $a>0$, what is the value of a?

40. Point P lies inside rectangle $ABCD$ so that $PA=2$, $PB=3$, and $PC=4$. What is the value of PD?

A. $\sqrt{11}$

B. $\sqrt{13}$

C. $\sqrt{20}$

D. $\sqrt{21}$

E. $\sqrt{29}$

Section 13 - Hard

41. In two equilateral parallelograms, *X* and *Y*, the sum of the lengths of the diagonals of *X* is equal to the sum of the lengths of the diagonals of *Y*. In *X*, the length of the longer diagonal is 20 more than the length of the shorter diagonal. In *Y*, the length of the longer diagonal is 8 more than the length of the shorter diagonal. 【微信公众号：张巍 GRE】
What is the area of *Y* minus the area of *X*?
A. 336
B. 168
C. 84
D. 42
E. 0

42. The probability distribution function f of a continuous random variable x is defined by f(x) = $(\frac{2}{13})*|x|$ for $-3\le x\le 2$.
Quantity A: The median of the distribution of X
Quantity B: -9/5

43. k is an integer greater than 2. The remainder when 3^k is divided by 13 is equal to r.
Quantity A：The remainder when 9 is divided by r
Quantity B：3

44. Several teams will play in a tournament. Each game in the tournament will be played by two teams, where one team will win or the teams will tie. Each team will earn 3 points for each win, 1 point for each tie, and 0 points for each loss. If a team plays only four games in the tournament, which of the following could be the total number of points that the team will earn in the tournament?
Indicate <u>all</u> such numbers.
A 0
B 1
C 5
D 6
E 11
F 12

45. Let x and y be numbers such that xy< 0.
Quantity A：||x| – |y||
Quantity B：|x – y|

46. Sets R, S and T are finite and T contains more elements than S. The number of elements in T minus the number of elements in S is equal to the number of elements in the set R∩T minus the number of elements in the set R∩S.

Quantity A： The number of elements in the set RUS

Quantity B： The number of elements in the set RUT

47. Carlene played in two chess tournaments, each consisting of a number of chess games, and she played more games in the second tournament than in the first tournament. In each tournament, she won at least one game and won twice as many games as she lost. She scored 1 point for each game she won, 0 points for each game she lost, and 1/2 point for each game she neither won nor lost.

Quantity A： Carlene's total score in the first tournament divided by the number of games she played in that tournament

Quantity B： Carlene's total score in the second tournament divided by the number of games she played in that tournament

48. When r percent is expressed as a fraction and the fraction is reduced to lowest terms, the result is n/20, where n is an integer. Which of the following could be the value of r?

A 25

B 28

C 30

D 32

E 35

49. The population of City X in 1999 was r percent greater than it was in 1990, and the average income per person in 1999 was i percent greater than it was in 1990. The percent increase in the total income of the entire population from 1990 to 1999 would have been greatest for which of the following values of r and i?【微信公众号：张巍 GRE】

A r=6 and i=9

B r=7 and i=8

C r=9 and i=5

D r=10 and i=7

E r=11 and i=5

50. In a certain state, 60 percent of the eligible voters are registered to vote and 48 percent of the eligible voters in that state voted in a recent election. If only registered voters are permitted to vote, what percent of the registered voters in the state voted in the election?

A 30%

B 50%

C 60%

D 70%
E 80%

51. Of the artists in the graphics department of a multimedia company, 60 percent are working on digital projects, and the remaining artists are working on hard-copy projects. Of the artists working on digital projects. 20 percent attended a fine arts school. Of the artists working on hard-copy projects, 10 percent attended a fine arts school. Of the artists in the graphics department who attended a fine arts school, what percent are working on digital projects?
【微信公众号：张巍 GRE】
A 12%
B 33.33%
C 66.67%
D 72%
E 75%

52. The table above summarizes customer satisfaction ratings for two banks, where each rating is an integer from 1 to 10. Which of the following statements are true?

Customer Satisfaction Ratings

Rating Information	Bank I	Bank II
Number of ratings	155	160
Mean rating	7.4	5.9
Standard deviation of ratings	1.6	1.8

Indicate all such statements.
A. For Bank I, if a rating is within 0.5 standard deviation of the mean rating, then the rating is 7.
B. For Bank II, if a rating is within 0.4 standard deviation of the mean rating, then the rating is 6.
C. The sum of all the ratings for Bank I is less than the sum of all the ratings for Bank II.

53. Quantity A： $(\sqrt{999999}+\sqrt{1000001})^2$
Quantity B： $(\sqrt[3]{999999}+\sqrt[3]{1000001})^3$

54. Let W be a continuous random variables such that $P\left(W>\frac{1}{2}\right)=\frac{9}{10}$ and $P\left(W>\frac{3}{4}\right)=\frac{7}{20}$.

What is the value of $P\left(\frac{1}{2}<W\leq\frac{3}{4}\right)$?

Give your answer as a decimal.

55. The figure represents a flat rectangular plot of land that consists of a rectangular parking lot and a surrounding sidewalk of uniform width. If the area of the parking lot is 3/5 of the area

of the entire plot of land, what is the width, in feet, of the sidewalk?

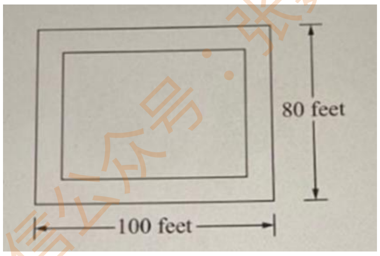
A 20

B 16

C 11

D 10

E 8



56. List L consists of n values, and the average (arithmetic mean) of the values in L is 10. List M consists of r values, and the average of the values in M is 15. The average of the values in List L and M combined is 11. If r>100, which of the following could be the value of n? Indicate all such values. 【微信公众号：张巍 GRE】

A 320

B 400

C 420

D 450

E 844

57. On her walk today, Emma walked along a straight path between the entrance to her house, H, and the entrance to library, L, passing a crosswalk sign, J, and then a crosswalk sign, K. First Emma walked from H to J and then turned around and walked back from J to H. Then she walked directly from H to L, stopping briefly at J and K on her way. The total distance that Emma walked from H to J and back from J to H was 2/9 of the distance that she walked directly from H to K and 1/6 of the distance that she walked directly from H to L. If Emma walked a total of 420 meters during her walk today. what was the distance, in meters, that she walked from K to L?

A 84

B 90

C 100

D 105

E 120

58. Each side of square S has length 2. Five different points lie inside S but not on the sides of S.

Quantity A： The distance between the closest two points among the five points

Quantity B： $\sqrt{2}$

59. If -x<y<0, which of the following statements is true about the three quantities $(x+y)^2$, x^2-y^2, and y^2-x^2 ?

A $x^2 - y^2 < y^2 - x^2 < (x+y)^2$
B $x^2 - y^2 < (x+y)^2 < y^2 - x^2$
C $(x+y)^2 < y^2 - x^2 < x^2 - y^2$
D $y^2 - x^2 < x^2 - y^2 < (x+y)^2$
E $y^2 - x^2 < (x+y)^2 < x^2 - y^2$

60. Water started running into an empty tank and continued running until it was filled to its capacity. Water ran at a constant rate of 200 kiloliters per minute for the first 20 minutes and then at a constant rate of 100 kiloliters per minute until the tank was filled. If it took a total to 80 minutes to fill the empty tank to its capacity. how many minutes did it take to fill the tank to 1/2 of its capacity? 【微信公众号：张巍 GRE】

Section 14 - Hard

61. The figure shows a window that is in the shape of a rectangle with a semicircle at the top. The radius of the semicircle is x feet, and the perimeter of the window is 26 feet. What is the total area, in square feet, of the window in terms of x? 【微信公众号：张巍 GRE】

A $13x-(\pi+2)x^2$

B $26x-2x^2$

C $26x-(\frac{\pi}{2}+2)x^2$

D $26x-(\frac{\pi}{2}+4)x^2$

E $26x-(\frac{3\pi}{2}+2)x^2$

62. List S consists of 19 numbers, and the median of the numbers in S is 8. List T consists of the numbers in S and the number 7, Which of the following numbers could be the median of T? Indicate all such numbers.

A 7
B 7.1
C 7.3
D 7.5
E 7.7
F 7.9
G 8

63. Which of the following is an odd integer?

A $\sqrt{\frac{1}{(3^{-4})(5^{-2})}}$

B $\sqrt{\frac{1}{(2^{-2})(3^{-3})}}$

C $\sqrt{\frac{1}{(3^{4})(5^{-2})}}$

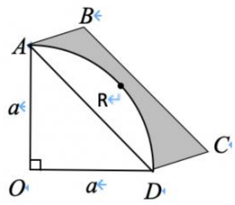
D $\frac{1}{(2^{-5})(3^{-3})}$

E $\frac{1}{3^{-1}+5^{-2}}$

64. For all positive even integers n. n* represents the product of all even integers from 2 to n, inclusive. For example, 12*=12×10×8×6×4×2. What is the greatest prime factor of 20*+22*?

A 41
B 23
C 19
D 17
E 11

65. In the figure, ARD is an arc of a circle centered at point O, ABCD is a rectangle, and CD=a/2. What is the area of the shaded region? 【微信公众号：张巍 GRE】



A $(\frac{\sqrt{2}}{2}-\frac{1}{2})a^2$

B $(\frac{\sqrt{2}}{2}-\frac{\pi}{4})a^2$

C $(\frac{\sqrt{2}}{2}-\frac{\pi}{4}-\frac{1}{2})a^2$

D $(\frac{\sqrt{2}}{2}+\frac{\pi}{4}-\frac{1}{2})a^2$

E $(\frac{\sqrt{2}}{2}-\frac{\pi}{4}+\frac{1}{2})a^2$

66. In the xy-plane, line m intersects the y-axis above the origin and intersects the x-axis to the right of the origin. The distance between the x- and y-intercepts of line m is 5. If the ratio of the distance between the origin and the x-intercept to the distance between the origin and the y-intercept is 4 to 3, which of the following is an equation of line m?

A $y=\frac{3}{4}x$

B $y=\frac{4}{3}x$

C $y=\frac{4}{3}x+3$

D $y=-\frac{4}{3}x+3$

E $y=-\frac{3}{4}x+3$

67. At Company X last year, 25 percent of all of the employee were promoted on June 1. Of

those employees who were promoted on June 1 last year, 80 percent had been with the company for 10 years or more.

Quantity A： The percent of all of the employees at Company X who were promoted on June 1 last year but were with the company for less than 10 years.

Quantity B： 5%

68. In the distribution of measurements of the variable x, the mean is 56 and the measurement r lies between the 65th and 70th percentiles. In the distribution of measurements of the variable y, the mean is 56 and the measurement t lies between the 75th and 80th percentiles【微信公众号：张巍 GRE】

Quantity A： r

Quantity B： t

69. Percent of households with Selected Appliances, 1980

Microwave Oven	14%
Central Air Conditioner	27%

According to the table, the percent of households in 1980 with neither a microwave oven nor a central air conditioner must be in which of the following inclusive intervals?

A 14% to 27%

B 27% to 41 %

C 41% to 59%

D 59% to 73%

E 73% to 86%

70. The sequence of the 50 numbers $a_1, a_2, \ldots, a_{50}$ is defined as follows, where n is an integer between 1 and 50, inclusive.

$$a_n = \begin{cases} \frac{n+1}{n} - 1 \; if \; n \; is \; odd \\ -a_{n-1} \; if \; n \; is \; even \end{cases}$$

What is the range of the 50 numbers in the sequence?

A 2/3

B 1

C 4/3

D 2

E 8/3

71.

$$d = \left(\frac{v_i + v_f}{2}\right)t$$

For an object that travels in a straight line with constant acceleration, the distance d that the object travels during time t is given by the formula shown, where v_i and v_f are the initial velocity and the final velocity of the object, respectively. A vehicle traveled distance of 32 kilometers along a straight route with constant acceleration. According to the formula, approximately how many minutes did it take the vehicle to travel the distance if the initial velocity and the final velocity of the vehicle were 14 meters per second and 20 meters per second, respectively?

A 9

B 17

C 23

D 31

E 40

72. The number of elements in set A is 60 percent greater than the number of elements in set B, and the number of elements in the set A ∩ B is 15. If set B contains at least one element that is not in set A, what is the least possible number of elements in the set A U B? 【微信公众号：张巍 GRE】

73. An empty water storage tank can be filled using two pumps, A and B. Pump A, working alone at its constant rate, takes x minutes to fill the tank. Pump B, working alone at its constant rate takes 5/4 times as long as pump A to fill the tank. If it takes y minutes for pumps A and B, working simultaneously at their respective constant rates, to fill the tank, what is the ratio of x to y?

Give your answer as a fraction.

74. In the xy-plane, if three of the vertices of a parallelogram have coordinates(2, -1), (1, 2), and(5, 1), which of the following could be the coordinates of the fourth vertex?

Indicate all such coordinates

A (6，-2)

B (-2，0)

C (4，4)

75. One thousand people voted for candidate X and Y, and X got 9% more of total votes, how many votes did X get?

76. The integers a, b, and c are positive, and a>b>c. When a, b, and c are divided by 3, the remainders are 0, 1, and 1, respectively.

Quantity A: The remainder when a+b is divided by 3

Quantity B: The remainder when a-c is divided by 3

77. Three 16-ounce bottles, M,R, and S, are filled with three different beverages. The amount of sugar contained in bottle M is 1/4 of the amount of sugar contained in bottle R and is 2/5 of the amount of sugar contained in bottle S. The amount of sugar contained in bottles R and S combined is what fraction of the amount of sugar contained in bottle R?

78. If x, y and z are consecutive positive integers and if x+y+z is even, how many of the four integers xy, yz, zx and xyz are even? 【微信公众号：张巍 GRE】

A. None
B. One
C. Two
D. Three
E. Four

79. When a certain stock exchanged closed on Tuesday, 1/5 of the stocks closed at the same price as on the previous day. Of the remaining stocks, 3 stocks closed at a higher price for every 4 stocks that closed at a lower price. What fraction of all the stocks on this exchange closed at a higher price than on the previous day?

A. 12/35
B. 3/7
C. 16/35
D. 4/7
E. 3/4

80. $0 < n < 10^7$

The integer n above is the square of an integer and the cube of an integer. If the ones digit of n is 5, what is the value of n?

Section 15 - Hard

81. Let n be a nonnegative integer such that when 6n is divided by 75, the remainder is 30. Which of the following is a list of all possible remainders when 7n is divided by 75?
A. 5, 35
B. 10, 55
C. 35, 60
D. 5, 30, 55
E. 10, 35, 60

82. Of the paperbacks in a private library, 5 percent are biographies. If 3 percent of all the books in the library are paperbacks that are biographies, what percent of all the books in the library are paperbacks? 【微信公众号：张巍 GRE】
A. 2%
B. 15%
C. 30%
D. 48%
E. 60%

83. According to a tax rate formula for a certain year, the amount of tax owed by an individual whose annual income was between $31,850 and $77,100 was equal to a base tax of $4,386 plus 24 percent of the annual income that exceeded $31,850. According to this formula, what was the amount of tax owed by an individual whose annual income that year was $42,000?

84. The probability that event A will occur is 0.8, and the probability that event B will occurs is 0.6. If x is the probability that events A and B will both occur, which of the following must be true?
A. $0.0 \le x \le 0.2$
B. $0.2 \le x \le 0.4$
C. $0.4 \le x \le 0.6$
D. $0.6 \le x \le 0.8$
E. $0.8 \le x \le 1.0$

85. The probability that each toss of a certain coin will result in heads is 0.5. If the probability that n tosses of the coin will each result in heads is greater than 0.0002, what is the greatest possible value of n?

86. A survey conducted by the department of motor vehicles on a sample of 200 car owners revealed that 15 percent of the sample had an expired driver's license, 10 percent had one or more outstanding parking tickets, and 78 percent had neither an expired driver's license nor any outstanding parking tickets. How many car owners in the sample had both an expired driver's license and on or more outstanding parking tickets?
A. 3
B. 6
C. 8
D. 9
E. 10

87. List P: 2, 11, 7, x The range of the four numbers in list P is 15 and for one of the numbers, n, in the list, |n-x| = 11. What is the value of x? 【微信公众号：张巍 GRE】
A. 21
B. 17
C. 1
D. -4
E. -8

88. List A: -2, 0, 2
List B: 3, 4, 5
List C: 9, 9.5, 10
Let a, b, and c be the standard deviations of the numbers in lists A, B, and C, respectively. Which of the following shows the order of a, b, and c?
A. a<b<c
B. a<c<b
C. b<a<c
D. c<a<b
E. c<b<a

89. The table above shows the annual revenues of a company for seven years. Based on the information given, which of the following statements are true for the seven years?

Year	Annual Revenue(millions of dollars)
2008	6
2009	6
2010	8
2011	13
2012	11
2013	16
2014	14

A. The average(arithmetic mean) of the revenues was greater than half of the greatest of the revenues.

B. The median of the revenues was greater than twice the least of the revenues.

C. The revenue for one of the years after 2008 was more than 60 percent greater than the revenue for the preceding year.

90. Larry and Tony work for different companies. Larry's salary is the 90th percentile of the salaries in his company, and Tony's salary is the 70th percentile of the salaries in his company. Which of the following statements individually provide(s) sufficient additional information to conclude that Larry's salary is higher than Tony's salary? Indicate all such statements. 【微信公众号：张巍 GRE】

A. The average(arithmetic mean) salary in Larry's company is higher than the average salary in Tony's company.

B. The median salary in Larry's company is equal to the median salary in Tony's company.

C. The 80th percentile salary in Larry's company is higher than the 70th percentile salary in Tony's company.

91. Of the 80 juniors at a certain school, 30 are taking biology and 40 are taking chemistry. Which of the following statement individually provide(s) sufficient additional information to determine the number of juniors who are taking both biology and chemistry?
Indicate all such statements.

A. Of the 80 juniors, 35 are taking chemistry but are not taking biology.

B. Of the 80 juniors, a total of 65 are taking biology or chemistry.

C. Of the 80 juniors, 25 are taking biology but are not taking chemistry.

92. The inside of a rectangular aquarium has an inner base that measures 8 inches by 10 inches, and has an inner height of 10 inches. The rectangular base of the aquarium is level, and the aquarium is filled with water so that the water level is 6 inches higher than the baes. After a solid piece of steel is completely submerged in the water, the water level is 7 inches higher than the base.

Quantity A: The volume of the piece of steel

Quantity B: 100 cubic inches

93. A right triangle has sides of length 2,5, and x. A second right triangle has sides of length 4,7, and y. A third right triangle has sides of length x, y, and n. Which of the following could be the value of n?

A. 3

B. 6

C. 8

D. 9

E. 10

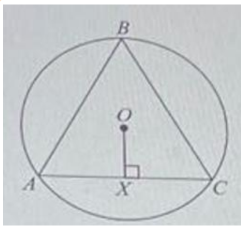
94. On each side of a parade route, spectators occupy a sidewalk that is 10 feet wide and 1.5 miles long. The parade organizers estimate that the average amount of sidewalk space occupied per spectator is 6 square feet. Based on this estimate, which of the following is closest to the total number of spectators occupying the two sidewalks?(1mile = 5,280 feet)

A. 9,000
B. 14,000
C. 18,000
D. 22,000
E. 26,000

95. A monument is constructed on level round in the shape of a pyramid with a square base. If each edge of the monument is 10 meters long, how many meters above the ground is the tip of the monument? 【微信公众号：张巍 GRE】

A. $5\sqrt{2}$
B. $10\sqrt{2}$
C. $2\sqrt{5}$
D. 5
E. 10

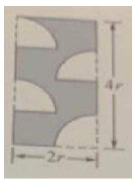
96. In the figure, equilateral triangle ABC is inscribed in the circle with center O. If OX is OX is perpendicular to AC and the area of circle O is 64π, what is the area of triangular region ABC?



A. $96\sqrt{3}$
B. $48\sqrt{3}$
C. $96\sqrt{3}$
D. 48
E. 144

97. What is the perimeter of the shaded region of the rectangle shown above if each of the unshaded region is a 90-degree sector of a circle with radius r?

A. (12+2π)r
B. (11+2π)r
C. (10+2π)r
D. (8+2π)r
E. (4+2π)r



98. All angles in the figure above are right angles. What is the straight-line distance from point P to point Q?

A. $\sqrt{5}$
B. $\sqrt{5}+\sqrt{2}+1$
C. $\sqrt{5}+2\sqrt{2}$
D. 5
E. 7

99. Two officers, stationed at opposite ends of a 1.5-mile zone where the spend limit is 40 miles per hour, use stopwatches and two-way radios to check speeds through the zone. A certain driver traveled through the entire zone in exactly 108 seconds. If the penalty for speeding in this zone is $50 plus $10 for each mile per hour that the driver exceed the speed limit, how much should the penalty be for this driver? 【微信公众号：张巍 GRE】
A. $60
B. $90
C. $100
D. $120
E. $150

100. A package contains between 75 and 100 marbles. When the marbles are divided into as many groups of 6 marbles as possible, the same number of marbles are left over as when the marble are divided into as many groups of 7 marbles as possible.
Quantity A: The number of marbles in the package
Quantity B: 80

Section 16 - Hard

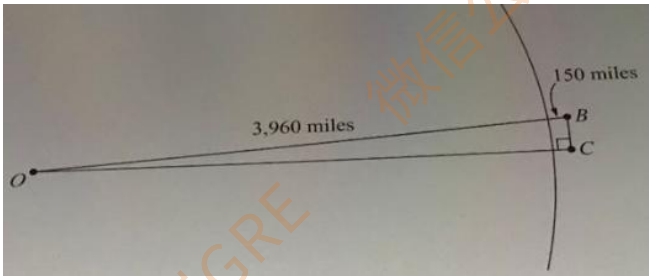
101. Two water faucets are used to fill a certain tank. Running individually at their respective constant rates, these faucets fill the empty tank in 12 minutes and 20 minutes, respectively. If no water leaves the tank, how many minutes will it take for both faucets running simultaneously at their respective rates to fill the empty tank?
   A. 16.0
   B. 10.5
   C. 8.0
   D. 7.5
   E. 6.0

102. It cost a certain company a total of $7,150.00 to make and sell 6,500 widgets. If the company sold each of the 6,500 widgets for $2.50, what was the company's profit per widget?(Profit is equal to the selling price minus the cost.)【微信公众号：张巍 GRE】

103. In the xy-plane, line m passes through the point (7,7) and is perpendicular to the line x+y =4. The point (a,b) is on line m and is halfway between the point (7,7) and the line x+y=4. What is the value of a+b?
   A. 8
   B. 9
   C. 10
   D. 11
   E. 12

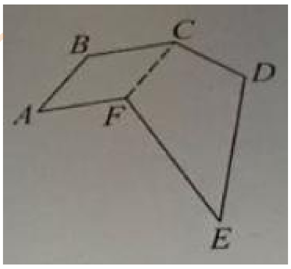
104. Stations A, M and B are located along a certain train route, and Station M is between A and B. At noon, a train engine passed Station A traveling at a constant speed of 80 kilometers per hour toward Station B. Also at noon, another train engine passed Station B traveling at a constant speed of 60 kilometers per hour toward Station A. Both train engines passed Station M at the same time. What is the ratio of the distance along the route between Station A and M to the distance along the route between Station A and B?
   A. 1/4
   B. 3/7
   C. 1/2
   D. 4/7
   E. 3/4

105. The figure shows the positions of two satellites, B and C, above the surface of Earth at a certain time. The area is part of a circle centered at O, which represents the center of Earth. The radius of Earth is approximately 3,960 miles. Satellite B is 150 miles above the surface of Earth. Satellite C is 300 miles from satellite B so that a right triangle with vertices at B, C, and O is formed. Which of the following is the best estimate of the distance, in miles, from satellite C to the surface of Earth?



A. 120
B. 125
C. 130
D. 135
E. 140

106. Hexagon ABCDEF shown is composed of parallelogram ABCF and quadrilateral CDEF. If CF=CD and DE = EF = AB + BC, what is the perimeter of the hexagon in terms of DE? 【微信公众号：张巍 GRE】



A. 2(DE)
B. 3(DE)
C. 4(DE)
D. 5(DE)
E. 6(DE)

107. If r, s, t, and u are positive integers such that (r+s)(t+u) is an odd integer and (r+s+t)(s+t+u) is an odd integer, which of the following statements must be true?

A. s and t are both even.
B. s and t are both odd.
C. s is even and t is odd.
D. s is odd and t is even.
E. s is even if t is odd, and s is odd if t is even.

108. The repeating decimal $1.\overline{ab}$, where a and b are different digits, is equivalent to the fraction n/d, where n and d are positive integers whose greatest common factor is 1. What is the greatest possible value of n+d?

A. 296
B. 297
C. 298
D. 299
E. 301

109. Set D consists of the numbers of the form $\frac{1}{2^n}$, where n is an integer from 1 to 10. What is the hundredths digit in the decimal representation of the product of 8,000 and the sum of the numbers in D? 【微信公众号：张巍 GRE】

A. 1
B. 2
C. 3
D. 5
E. 8

110. List K consists of 5 different positive integers, each of which is less than or equal to 20. The range of the integers in K is 10. List M consists of 4 of the integers in K and another integer that is 2 times the remaining integer in K. Which of the following values could be the range of the 5 integers in M?
Indicate all such values.

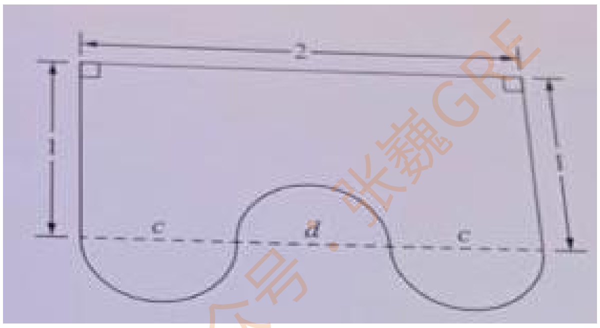
A. 8
B. 15
C. 30

111. A large pump and a small pump deliver water into a tank at their respective constant rates. Working simultaneously, the two pumps take 6 hours to fill the empty tank. If each pump worked alone, then the amount of time that it takes the large pump to fill the empty tank is 2/3 of the amount of time that it takes the small pump to fill the empty tank. How many hours does it take the large pump, working alone, to fill the empty tank?

A. 15
B. 12
C. 10
D. 8
E. 5

112. On level ground, a dog park is being built in the shape of an equilateral parallelogram whose diagonals have lengths 36.6 meters and 42.1 meters. A fence will be installed along the edges of the dog park. Approximately what is the total length, in meters, of the fence that will be installed?

A. 112
B. 123
C. 157
D. 212
E. 223



113. The figure above shows a design for the top surface of a desk, where all measurements are in meters. The curved edge of the desk consists of three semicircles with diameters c, d, and c. Which of the following statements are true?
Indicate all such statements.

A. If c=d, then the area of the surface is greater than 2 square meters.
B. If d=2c, then the area of the surface is equal to 2 square meters.
C. If c=3/4, then the area of the surface is less than 2 square meters.

114. Some standard Paper Sizes

Type	Width(millimeters)	Length(millimeters)
A0	841	1,189
A1	594	841
A2	420	594

Quantity A: The ratio of the area of a sheet of A0 paper to the area of a sheet of A1 paper
Quantity B: The ratio of the area of a sheet of A1 paper to the area of a sheet of A2 paper

115. Quantity A: For sample I, the average (arithmetic mean) minus the median
Quantity B: For sample II, the average (arithmetic mean) minus the median

Sample I	Sample II
400	400
500	500
700	800
900	900
1,000	1,000

116. A tank initially contains g gallons of water. Beginning at 1 o'clock in the afternoon, water flows into the top of the tank at a constant rate of x gallons per minute and out of the bottom of the tank at a constant rate of y gallons per minutes. If x<y, in how many minutes after 1 o'clock in the after noon is the volume of water in the tank equal to $\frac{1}{2}$g, in terms of g, x, and y? 【微信公众号：张巍 GRE】

A. $\frac{g}{y} + \frac{g}{2x}$

B. $\frac{g}{2y-2x}$

C. $\frac{g}{y} - \frac{g}{2x}$

D. $\frac{g}{2x} - \frac{g}{y}$

E. $\frac{2y-2x}{g}$

Method of Transportation	Percent of Workers	
	2000	2010
Private Vehicle	85.9%	88.0%
Public transportation	6.2%	5.1%
Walk	5.6%	3.9%
None(work at home)	2.3%	3.0%

117. The table above shows four categories of methods of transportation to work, including the category None for people who work at home. For each category and for the years 2000 and 2010, the table shows the percent of workers in a certain city who used that method most of the time during the year. Based on the information given, which of the following statements must be true?

Indicate all such statements.

A. In 2010 the ratio of the number of workers in the private vehicle category to the number of workers who were not in that category was 22 to 3.

B. The number of workers in the public transportation category in 2010 was less than that in 2000.

C. The number of workers in the category None in 2010 was greater than in 2000.

118. Kara went on a canoe trip. During this trip she paddled x kilometers downstream and then returned to her starting point by padding x kilometers upstream. Her average speed paddling downstream was 3 kilometers per hour and her average speed paddling upstream was 2 kilometers per hour. If the combined time she spent paddling downstream and upstream was 3 hours, how many hours did she spend paddling downstream?

A. $\frac{2}{3}$ hour

B. 1 hour

C. $1\frac{1}{5}$ hour

D. $1\frac{1}{3}$ hour

E. $1\frac{1}{2}$ hour

119. 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, …, 24

The list above consists of 300 entries that are integers in increasing order, and each integer n occurs n times for 1 <=n <=24. What is the least integer k in the list for which at least 15 percent of all the entries are less than k? 【微信公众号：张巍 GRE】

A. 8
B. 9
C. 10
D. 11
E. 12

120. In a certain building in a business district, a newspaper service delivers 104 papers every day to its customers. Twice as many customers have 3 papers delivered each day as have one paper delivered, and three times as many customers have 2 papers delivered each day as have one paper delivered. If no customer has more than 3 papers delivered, then the number of customers who have 2 papers delivered each day is

A. 16
B. 18
C. 20
D. 22
E. 24

Section 17 – Hard

Minutes	Number of Students
0 to 19	10
20 to 39	21
40 to 59	9
60 to 79	3
80 to 99	3

121. The frequency distribution above summarizes the numbers of minutes, rounded to the nearest integer, that 46 students individually spent studying for an exam. Based on the information given, which of the following statement must be true? 【微信公众号：张巍 GRE】
Indicate all such statements.
A. The total number of minutes that all of the students spent studying is greater than 1,500.
B. The average(arithmetic mean) number of minutes spent studying per student is greater than 25.
C. The range of the number of minutes spent studying is greater than 60.

122. Tickets for a carnival are sold either individually or in packages of 10 tickets or 20 tickets at the costs shown in the table. A group of friends bought n tickets for the least possible total cost. A second group of friends bought more than n tickets for a total cost that was less than the first group's total cost. Which of the following could be the value of n?

Number of Tickets	Cost
1	\$1.25
10	\$11.00
20	\$20.00

Indicate all such values.
A. 8
B. 9
C. 11
D. 12
E. 16
F. 17
G. 18
H. 19

123. During archery practice, Kyle made 50 attempts to hit a target. After each attempt, his success rate was calculated as the number of successful attempts to hit the target as a percent of the number of attempts up to that time. After the 20th attempt, his success rate was 45 percent. After the 40th attempt, his success rate was 55 percent. Which of the following statements must be true?

Indicate all such statements.

A. Kyle's success rate was less than 45 percent at least once during the practice.

B. Kyle's success rate was between 48 percent and 52 percent at least once during the practice.

C. Kyle's success rate was greater than 55 percent at least once during the practice.

124. For each value x in a list of values with mean m, the absolute deviation of x from the mean is defined as |x-m|. 【微信公众号：张巍 GRE】

A certain online course is offered once a month at a university. The number of people who register for the course each month is at least 5 and at most 30. For the past 6 months, the mean number of people who registered for the course per month was 20. For the numbers of people who registered for the course monthly for the past 6 months, which of the following values could be the sum of the absolute deviations from the mean?

Indicate all such values.

A. 100

B. 90

C. 60

D. 30

E. 10

125. Tanya deposited money for one year in each of two different accounts, A and B. Account A paid x percent simple annual interest, and account B paid x/2 percent simple annual interest, where $x > 0$. The amount deposited in account A earned d dollars in interest, and the amount deposited in account B earn 2d dollars in interest.

Quantity A: The amount deposited in account A

Quantity B: The amount deposited in account B

126. A certain map is drawn to scale. The ratio of the actual distance between Cities A and B and the distance on the map between the points representing the two cities is 250,000 to 1. The distance on the map between Cities A and B is 20 centimeters.(1 centimeter = 0.01 meter, and 1 kilometer = 1,000 meters)

Quantity A: The actual distance between Cities A and B

Quantity B: 40 kilometers

127. The cube root of which of the following integers is equal to $\sqrt[3]{3}+\sqrt[3]{192}+\sqrt[3]{375}$?
   A. 570
   B. 3,000
   C. 9,000
   D. 27,000
   E. 216,000

128. A certain toy company manufactures tricycles at a costs of $750 for a batch of 50 small tricycles and at a cost of $1,000 for a batch of 50 large tricycles. The company sells small and large tricycles for $75 and $110, respectively. Which of the following represents the number of dollars profit that the company makes from the sale of 700 small and 600 large tricycles? (Profit is sales minus costs.)【微信公众号：张巍 GRE】
   A. 700(750-75) + 600 (1,000 - 110)
   B. 700(50(75) - 750) + 600(50(110) – 1,000)
   C. $700(75-\frac{750}{50})+600(110-\frac{1,000}{50})$
   D. 700(50)(750-75) + 600(50) (1,000 - 110)
   E. $700(\frac{750-75}{50})+600(110-\frac{1,000-110}{50})$

129. A certain charity is conducting a fund-raiser. For the first $9,000 raised by the charity, Company B will contribute $1 for every $3 collected by the charity. For any amount over $9,000 raised by the charity, Company B will contribute $2 for every $5 collected by the charity. How much money must the charity raise in order to reach a total of $68,000, including the contribution from Company B?
   A. $38,000
   B. $40,000
   C. $42,000
   D. $49,000
   E. $52,000

130. For which of the following values of x is the inequality $\frac{(x-4)^4(x+5)^6}{(x-2)^{10}}\le 0$ satisfied?

Indicate all such values.
   A. -9
   B. -5
   C. -2
   D. 0
   E. 1
   F. 4
   G. 6

131. On May 1, 2005, the population of Town Y was twice the population of Town X. On May 1 of the years 2006, 2007 and 2008, the population of Town X was 4 percent greater than it was the preceding May 1, and the population of Town Y was 1 percent greater than it was the preceding May 1. On May 1, 2008, the population of Town X was what fraction of the population of Town Y? 【微信公众号：张巍 GRE】

A. $(\frac{1.04}{1.01})^3$

B. $2(\frac{1.04}{1.01})^3$

C. $\frac{1}{2}(\frac{1.04}{1.01})^3$

D. $2(\frac{1+3(0.04)}{1+3(0.01)})^3$

E. $\frac{1}{2}(\frac{1+3(0.04)}{1+3(0.01)})^3$

132. A shipment of food consisted of boxes containing various items as shown in the table above. The items in the shipment were unpacked and arranged in groups of 6 cans of bean, 6 bottles of juice, and 6packages of flour. If after arranging the groups there were a total of 4 items left over, which of the following lists could be the number of boxes in the shipment?
Indicate all such lists.

Food	Contents
Box of beans	27 cans
Box of juice	32 bottles
Box of flour	40 packages

A. 1 box of cans of beans, 1 box of bottles of juice, 1 box of packages of flour
B. 2 box of cans of beans, 1 box of bottles of juice, 2 box of packages of flour
C. 2 box of cans of beans, 2 box of bottles of juice, 3 box of packages of flour

133.

16, 96, 108, 162, 200
Each of the five numbers shown can be expressed in the form $2^n \times k$, where n is a positive integer and k is a positive odd integer. When expressed in this form, for which of the numbers does n have the greatest value?
A. 16
B. 96
C. 108
D. 162
E. 200

134. Signs A, B, and C are positioned on a 1,000-meter-long airplane runway. Sign A is 100 meters from the beginning of the runway and it followed by sign B and C in that order. The distance between sign B and sign C is 120 meters more than twice the distance from sign C to the end of the runway. The distance between sign A and sign B is 1/2 the distance between sign B and sign C. What is the distance, in meters, from sign C to the end of the runway?
A. 150
B. 180
C. 200
D. 240
E. 360

135. In the decimal number 1,375.2648, which digit has a place value that is 1,000 times the place value of the digit 6?【微信公众号：张巍 GRE】
A. 1
B. 3
C. 4
D. 5
E. 7

136. A campaign director can divide the campaign workers into 10 or 11 teams. When the workers are divided into 11 teams, each of 10 of the teams has x workers and 1 team has 6 workers. When the workers are divided into 10 teams, each of 9 of the teams has x+1 workers and 1 team has 5 workers. How many campaign workers are there?
A. 72
B. 76
C. 82
D. 86
E. 94

137. The 1,200 seniors at a certain college, one-quarter live on campus and the rest live off campus. Three-quarters of the seniors at college own a car. Which of the following statements must be true?
A. No seniors live on campus and own a car.
B. At least 300 seniors live on campus and own a car.
C. At most 600 seniors live off campus and own a car.
D. At least 600 seniors live off campus and own a car.
E. At least 300 seniors live off campus and do not own a car.

138. The integer p is a prime number greater than 5, and 5 is a factor of $p+p^2$. what is the remainder when p is divided by 5?

A. 0
B. 1
C. 2
D. 3
E. 4

139. Each • in the table above represents an entry that is the product of the corresponding row and column values. What is the least positive entry of the 45 entries represented in the table?

	−30	−17	−13	−12	−5	2	3	8	22
−4	•	•	•	•	•	•	•	•	•
−1	•	•	•	•	•	•	•	•	•
0	•	•	•	•	•	•	•	•	•
3	•	•	•	•	•	•	•	•	•
6	•	•	•	•	•	•	•	•	•

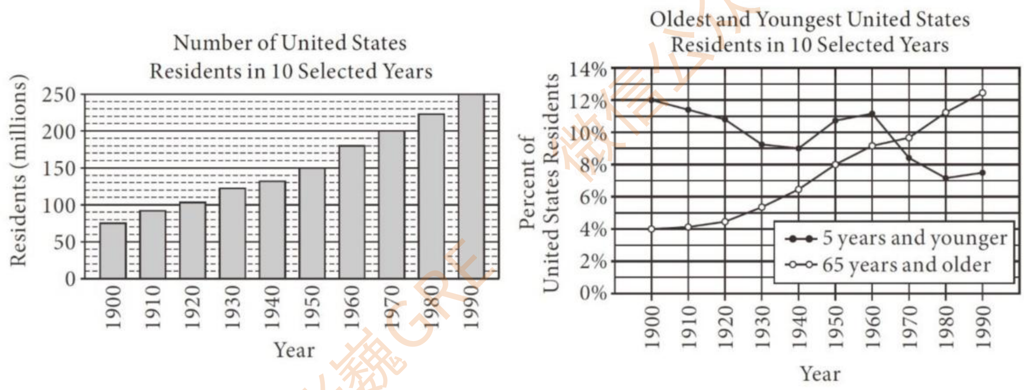
A. 2
B. 3
C. 5
D. 6
E. 9

140. In a certain university, 60 percent of all sophomores are liberal arts majors, 24 percent are education majors, and the rest are majoring in other areas or have not yet chosen a major. At the university, 55 percent of all sophomores are currently taking a psychology course. If x percent of all sophomores are liberal arts majors who are currently taking a psychology course, what is the least possible value of x?【微信公众号：张巍 GRE】

A. 15
B. 16
C. 31
D. 36
E. 40

Section 18-图表题

Questions 1 to 3 are based on the following data.



1. In 1970 approximately what percent of United States residents were older than 5 but younger than 65 years old? 【微信公众号：张巍 GRE】

A) 2%
B)18%
C) 64%
D) 82%
E) 98%

2. Of the 10 selected years, which had the greatest total number of United States residents in the combined age groups of "5 years and younger" and "65 years and older"?

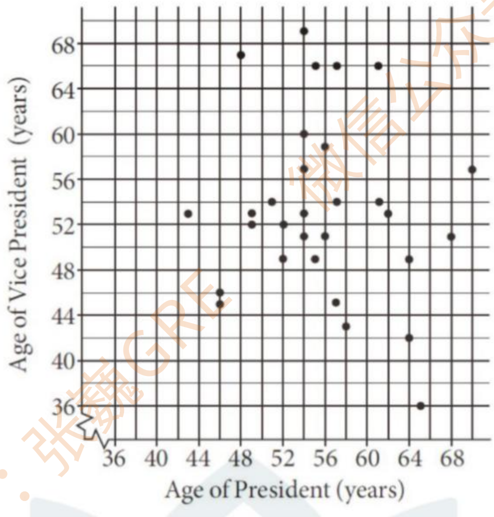
A)1900
B)1920
C)1960
D)1980
E)1990

3. Of the following which best describes how the number of United States residents that were in the age-group "5 years and younger" changed from 1900 to 1940?

A) It increased by more than 50 percent.
B) It increased by less than 50 percent but more than 5 percent.
C) It changed by less than 5 percent.
D) It decreased by less than 50 percent but more than 5 percent.
E) It decreased by more than 50 percent.

Questions 4 to 6 are based on the following data.

Ages* of Presidents and Corresponding
Vice Presidents of 30 Organizations



*as of most recent birthday

Select one answer choice.

4. What percent of all the presidents and vice presidents are less than 50 years old?
A)10%
B)15%
C)20%
D)25%
E)30%

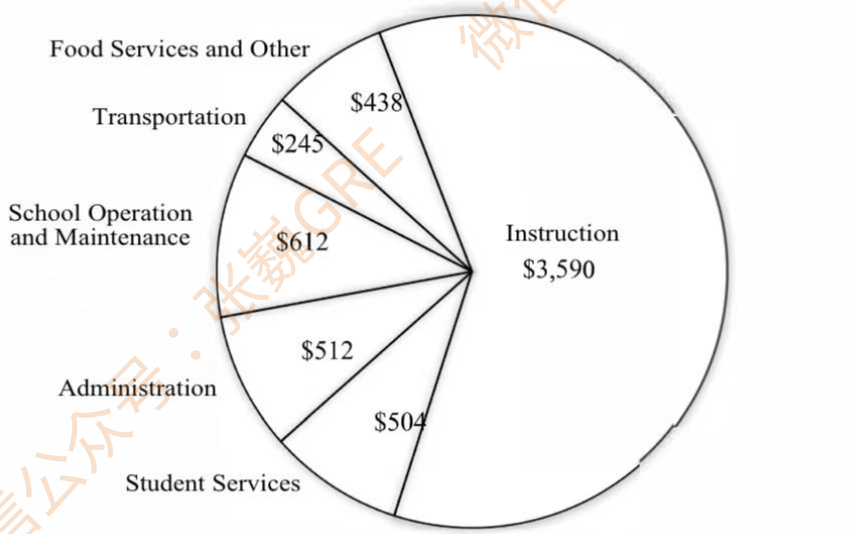
5. Two organizations will be selected at random, one at a time and without replacement, from the 30 organizations. Approximately what is the probability that the age of the vice president of at least one of the two organizations will be at least 1 year greater than the age of the corresponding president?【微信公众号：张巍 GRE】
A) 0.35
B) 0.45
C) 0.50
D) 0.55
E) 0.65

6. For how many of the organizations does the age of the president exceed the age of the corresponding vice president by 15 years?
A) None
B) One
C) Two

D) Three
E) Four

Questions 7-9 are based on the following data.

Per-Pupil Spending in United States
Public Schools for the School Year 1994-1995



Total Per-Pupil Spending: $5,901

7. Which of the following categories accounted for an amount closest to 10 percent of the total per-pupil spending? 【微信公众号：张巍 GRE】
   A. Administration
   B. Food Services and Other
   C. School Operation and Maintenance
   D. Student Services
   E. Transportation

8. If the total per-pupil spending for Transportation and Food Services and Other was 10 percent greater for the school year 1994-1995 than for the school year 1993-1994, approximately what was the total per-pupil spending for these two categories for the school year 1993-1994?
   A. $620
   B. $610
   C. $600
   D. $590
   E. $580

9. The sectors of the pie chart are to be reordered so that their sizes decrease when viewed consecutively in a clockwise direction starting with the largest. If the sectors are to be reordered by interchanging the positions of adjacent sectors, one interchange at a time, until the desired order is achieved, what is the minimum number of interchanges that will be required?
   A. Three
   B. Four
   C. Five
   D. Six
   E. Seven

Questions 10-12 are based on the following data.

Kim's Budget for June

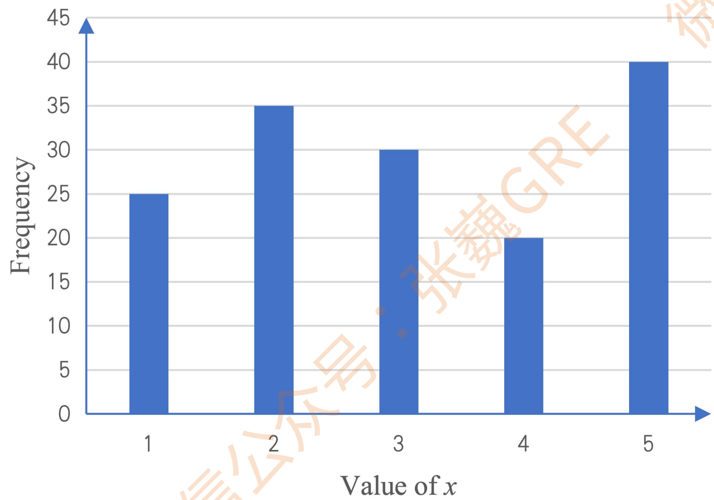
	Budgeted	Actual
Work-Related		
Expenditures		
Transportation	$120	$120
Tolls and Parking	100	100
Uniforms	75	40
Laundry/Dry Cleaning	55	40
Lunch and Snacks	50	80
Non-Work-Related		
Expenditures		
Rent	$800	$800
Automobile	550	500
Savings	300	300
Entertainment/Restaurants	300	380
Groceries	250	290
Heat/Light/Telephone	160	150
Insurance	100	100
Clothing	50	30
Other	90	70

10. Of the following fractions, which best approximates the ratio of total budgeted work-related expenditures to the total amount budgeted? 【微信公众号：张巍 GRE】
   A. 1/3
   B. 1/4
   C. 1/5
   D. 1/8
   E. 1/10

11. If 23.75 percent of actual work-related expenditures for lunch and snacks was for soft drinks, what percent of total actual work-related expenditures was for soft drinks?
   A. 5%
   B. 4%
   C. 3%
   D. 2%
   E. 1%

12. For how many of the 14 categories shown did actual expenditures differ from budgeted expenditures by more than 20 percent of budgeted expenditures? 【微信公众号：张巍 GRE】
   A. Three
   B. Five
   C. Six
   D. Seven
   E. Nine

13. The variable x takes on the values 1, 2, 3, 4, or 5. The graph shows the frequency distribution for 150 values of x. Which of the following is closest to the average (arithmetic mean) of the 150 values?
   A. 2.9
   B. 3.1
   C. 3.3
   D. 3.5
   E. 3.7



Questions 14-16 are based on the following data.

Weekly Rental Rates for Realtor M's Summer Houses on Island X

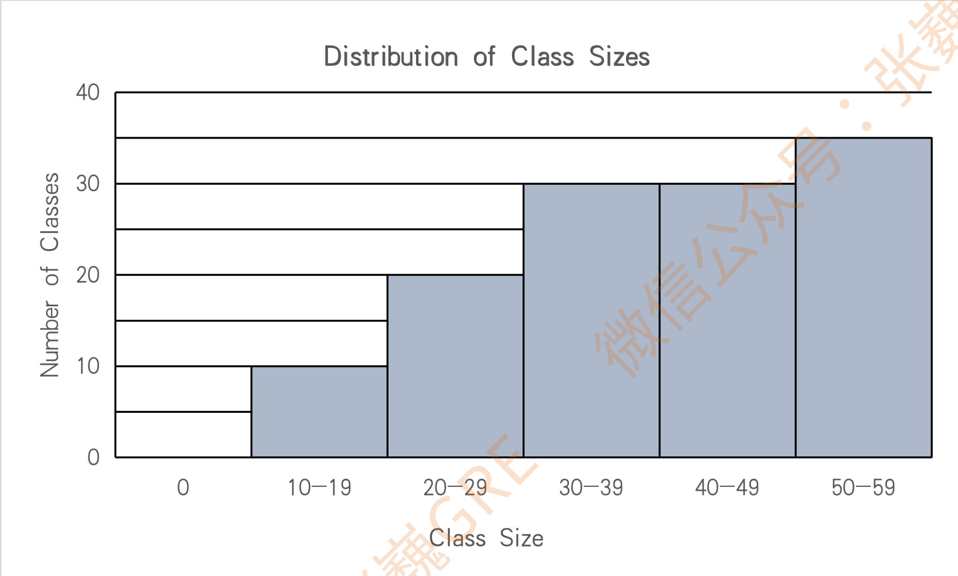
Location	Number of Bedrooms in House		
	2 Bedrooms	3 Bedrooms	4 Bedrooms
Oceanside	$800	$900	$1,000
Bayside	$600	$700	$800
Inland	$500	$600	$700

14. Of Realtor M's 3-bedroom summer houses on Island X, 5 are oceanside, 6 are bayside, and 1 is inland. What is the average (arithmetic mean) weekly rental rate for these 12 houses?
   A. $700
   B. $725
   C. $733
   D. $766
   E. $775

15. Realtor M has a special rate structure for an 8-week rental of a summer house: For each of the 1st and 2nd weeks, the rental charge is the listed weekly rate; for each of the 3rd and 4th weeks, the rental charge is x dollars less than the listed weekly rate; and for each of the remaining weeks, the rental charge is 2x dollars less than the listed weekly rate. In terms of x, what is the total rental charge, in dollars, for an 8-week rental of an oceanside 4-bedroom summer house? 【微信公众号：张巍 GRE】
   A. 8,000 – 16x
   B. 8,000 – 14x
   C. 8,000 – 12x
   D. 8,000 – 10x
   E. 8,000 – 8x

16. If Realtor M's inland summer houses on Island X generate a total weekly rental income of $8,800 when they are all rented, what is the greatest possible number of inland 4-bedroom summer houses that Realtor M can have?
   A. 8
   B. 9
   C. 10
   D. 11
   E. 12

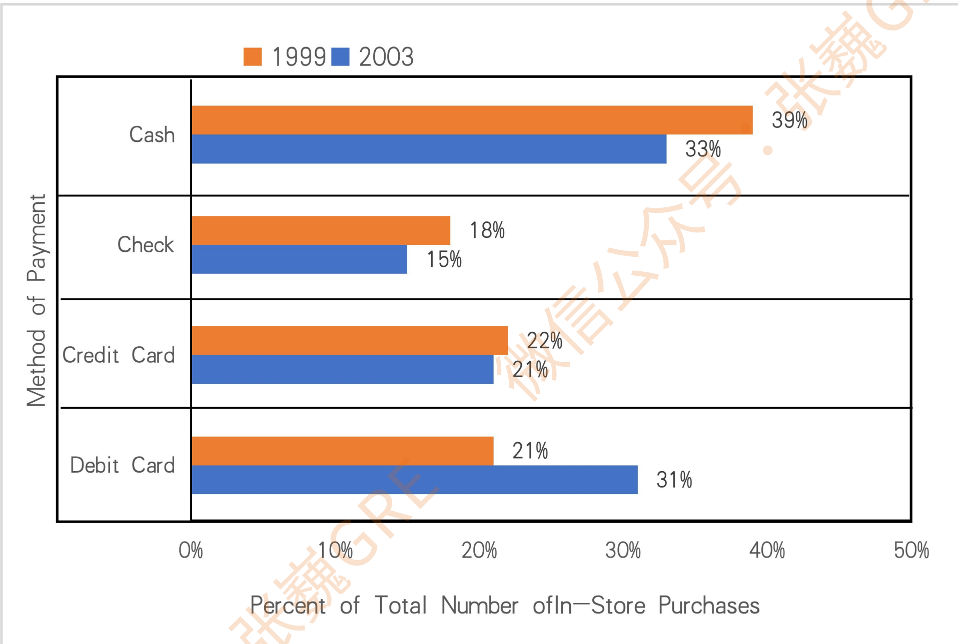
17. The figure shows the distribution of class sizes for 125 classes at a certain college. Based on the figure, which of the following values could be the median of the 125 class sizes?
   Indicate all such values.



A. 25
B. 34
C. 35
D. 40
E. 45

Questions 18 to 20 are based on the following data.

Methods of Payment for In-Store Purchases in 1999 and 2003



18. In 2003, what was the ratio of the number of in-store purchases paid for either by check or by credit card to the number of in-store purchases paid for in cash?
    A. 52 to 33

B. 36 to 31
C. 33 to 31
D. 31 to 21
E. 12 to 11

19. If the distribution of the number of in-store purchases in 2003 by method of payment is to be represented by a circle graph with a radius of 3.2 centimeters, approximately what will be the length, in centimeters, of the arc of the sector representing payment by credit card?
A. 4.2
B. 4.8
C. 5.4
D. 6.0
E. 6.4

20. In 2003, what percent of the in-store purchases that were <u>not</u> paid for in cash were paid for by check?【微信公众号：张巍 GRE】
Give your answer to the <u>nearest whole percent</u>.

Section 19-图表题

Questions 21 to 23 are based on the following data.

Measures Planned to be Used to Offset Rising Energy Costs:
Survey Results for 1,600 Companies

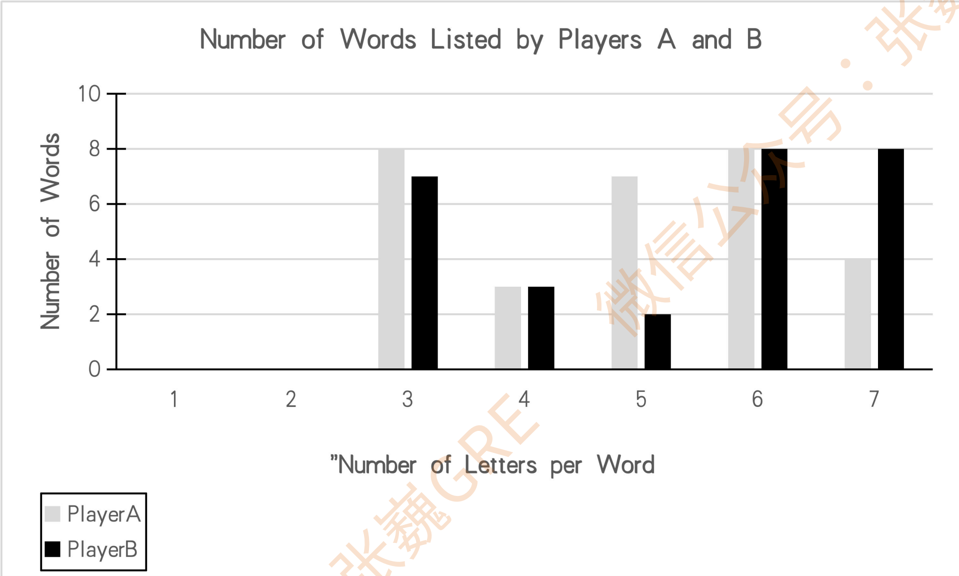
Measure	Percent of Companies
Absorb increased costs	48%
Consolidate debt	15%
Cut research and development	10%
Increase prices	41%
Negotiate raw-material costs	23%
Reduce workforce	18%
Trim advertising and marketing	12%

21. An analyst estimates that of the companies surveyed that plan to increase prices, 75 percent will actually increase prices. According to the analyst's estimate, what is the number of companies surveyed that plan to increase prices but will not actually increase prices?
   A. 164
   B. 262
   C. 381
   D. 547
   E. 640

22. For the seven measures shown, what is the median of the numbers of companies surveyed that plan to use the measures?【微信公众号：张巍 GRE】
   A. 656
   B. 548
   C. 402
   D. 368
   E. 288

23. The number of companies surveyed that plan to trim advertising and marketing is what percent less than the number of companies surveyed that plan to consolidate debt?
   A. 15%
   B. 20%
   C. 25%
   D. 30%
   E. 35%

Questions 24 to 26 are based on the following data.



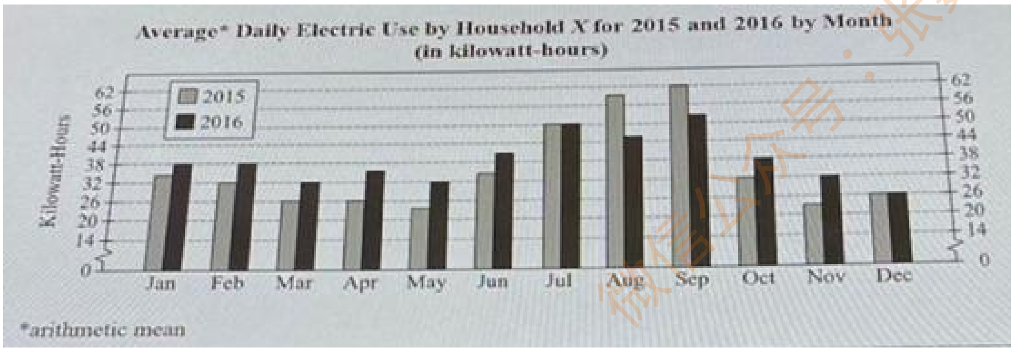
The graph shows the results of a word game in which two players were given identical sets of 12 different letters. Using only the 12 letters, each player listed as many words as the player could think of in three minutes, without repeating any letters within a word. Neither player listed a word with fewer than 3 or more than 7 letters. 【微信公众号：张巍 GRE】

24. In order to determine the scores for the players, each word listed with 3 or 4 letters was given 1 point, each word listed with 5 or 6letters was given 2 points, and each word listed with 7 letters was) given 3 points. The score for each player was the sum of the points given for the words that the player listed. What was the absolute value of the difference between the score for Player *A* and the score for Player *B*?

25. Of the 3-letter words listed by Player *A* or Player *B*, 4 words were listed by both players. How many of the 3-letter words listed were listed by only one of the two players?
    A. 7
    B. 8
    C. 11
    D. 15
    E. 19

26. What is the average (arithmetic mean) number of letters per word in the words listed by Player *B*?
    A. 4.95
    B. 5.00
    C. 5.25
    D. 5.50
    E. 6.00

Questions 27 to 29 are based on the following data.



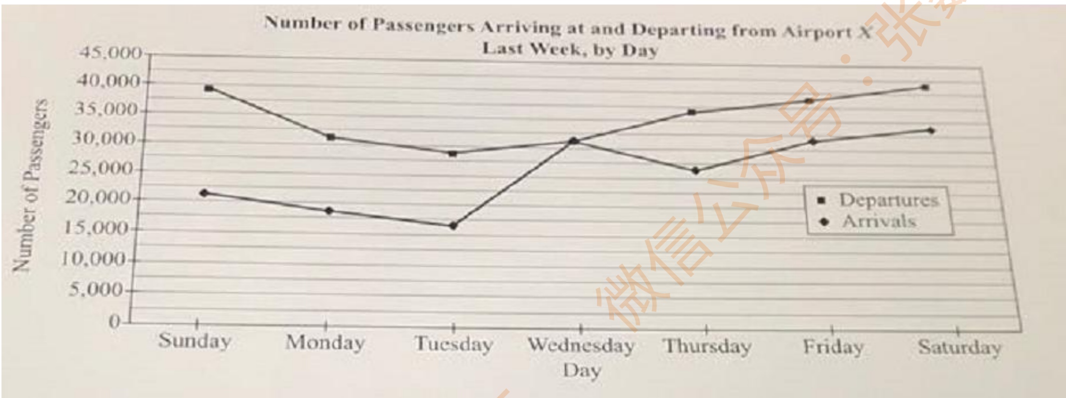
27. If the range of the average daily electric use for the months from January to June in 2015 was x kilowatt-hours less than that for the months from July to December in 2015 and if the range of the average daily electric use for the months from January to June in 2016 was y kilowatt-hours less than that for the months from July to December in 2016, approximately what is the value of $x-y$? 【微信公众号：张巍 GRE】
   A. 1
   B. 11
   C. 19
   D. 28
   E. 38

28. The total electric use, in kilowatt-hours, for the 31 days in March, the 30 days in April, and the 31 days in May was approximately hoy much greater for 2016 than for 2015?
   A. 30
   B. 90
   C. 480
   D. 600
   E. 730

29. What is the ratio of the number of months for which the percent increase from 2015 to 2016 in the average daily electric use was greater than 28 percent to the number of months for which the percent decrease from 2015 to 2016 in the average daily electric use was greater than 10 percent?

   Give your answer as a fraction.

Question 30-32 are based on the following data

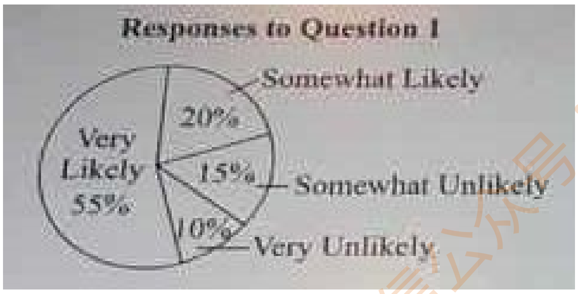


30. The number of passengers arriving on Sunday is approximately what fraction of the number of passengers arriving on Friday? 【微信公众号：张巍 GRE】
A 1/2
B 2/3
C 4/5
D 5/6
E 6/7

31. Which of the following is closest to the average (arithmetic mean) number of passengers departing per day from Airport X last week?
A 30,000
B 32,500
C 35,000
D 37,500
E 40,000

32. On how many of the days shown was the number of passengers departing at least 50 percent greater than the number of passengers arriving that day?
A Zero
B One
C Two
D Three
E Four

Questions 33 to 35 are based on the following data.
In a survey, 2,400 homeowners responded to the following question.
Question 1. How likely is it that you will continue to live in your current home for at least then more years?



The homeowners who responded “very likely” or “somewhat likely” to Question 1 were asked Question 2.

Question 2. What aspects of your current home make it likely that you will continue to live there? (Select all that apply.)

Responses to Question 2 as a Percent of Those Who Were Asked Question 2

Response	Percent
House is in a good location.	46%
House is attractive.	43%
House is comfortable.	38%
House is in good condition.	31%
House has no mortgage.	25%

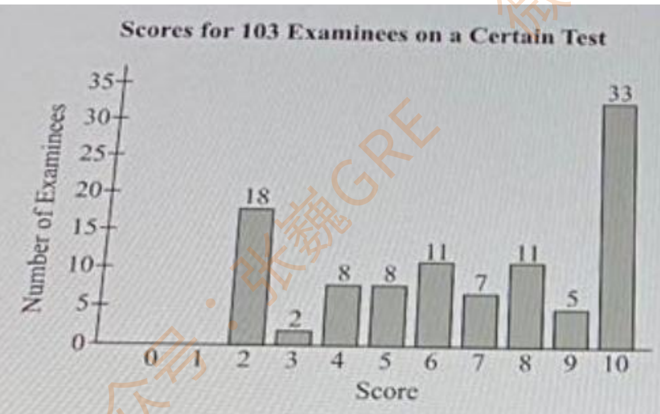
33. For how many of the five response categories in the table was the number of responses less than 750? 【微信公众号：张巍 GRE】
    A. None
    B. One
    C. Two
    D. Three
    E. Four

34. Based on the information given, which of the following statements are true?
Indicate all such statements.
    A. For Question 1, the ratio of the number of homeowners who responded “somewhat likely” to the number of homeowners who responded “somewhat unlikely” is 4 to 3.
    B. 75 percent of the homeowners in the survey were asked Question 2.
    C. More than 40 percent of the homeowners in the survey responded “house is attractive” to Question 2.

35. If 24 percent of the homeowners who were asked Question 2 responded both “house is comfortable” and “home is in good condition,” what percent of the homeowners who were asked Question 2 gave neither of these responses?
    A. 21%

B. 31%
C. 38%
D. 45%
E. 55%

Questions 36 to 38 are based on the following data.



The only possible scores on the test were integers between 0 and 10. No one received a score of 0 or 1. The mean score was 6.72 and the standard deviation was 2.98.

36. Examines who received a score of 6 or more passed the test. Which of the following is closest to the average(arithmetic mean) score of the examinees who passed the test?
A. 7.5
B. 8
C. 8.5
D. 9
E. 9.5

37. What was the median score?【微信公众号：张巍 GRE】
A. 6
B. 6.5
C. 7
D. 7.5
E. 8

38. Which of the following is closest to the percent of the examinees who received scores that were within 1.0 standard deviation of the mean?
A. 15%
B. 40%

C. 50%
D. 90%
E. 95%

Questions 39 to 41 are based on the following data.

Dakota Crossing Community Association
Landscape Maintenance Three-Year Bid Analysis, 1993–1995

Bidder	Bid (in dollars)			Three-Year Average		
	1993	1994	1995	Bid	Rank	Percent of Lowest Average Bid
Acme Lawn Care	30,874	30,874	32,418	31,388.67	7	138
Cott All-Season Landscaping	36,870	37,976	39,082	37,976.00	9	167
Doran Landscaping	30,496	30,496	31,106	30,699.33	6	135
Driscol and Son	47,246	48,664	50,610	48,840.00	11	215
King Farm	23,145	23,145	23,145	23,145.00	2	102
L & T Lawn Service	21,794	22,653	23,577	22,674.67	1	100
M. C. Keller	25,682	26,198	26,721	26,200.33	3	116
Pennsburg Landscaping	37,516	39,392	39,392	38,766.67	10	171
Raghavan Landscaping	29,139	29,650	30,603	29,797.33	4	131
Rockman Group	30,030	30,030	31,110	30,390.00	5	134
Sun Landscaping	34,553	34,553	34,553	34,553.00	8	152

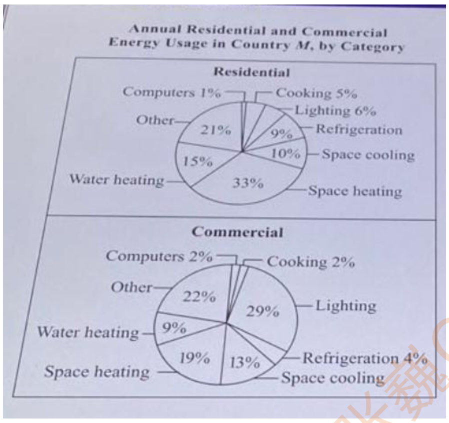
39. What was the median of the three-year average bids? 【微信公众号：张巍 GRE】
A. $31,110.00
B. $31,106.00
C. $30,699.33
D. $30,390.00
E. $29,797.33

40. The ratio of the three-year average bid of King Farm to that of Cott All-Season Landscaping was most nearly
A. 3/5
B. 3/4
C. 3/2
D. 5/3
E. 5/2

41. Approximately what percent of the highest total three-year bid is the lowest total three-year bid?
A. 45%
B. 55%
C. 70%
D. 85%
E. 115%

Section 20-图表题

Questions 42 to 44 are based on the following data.



42. If the annual commercial energy usage for lighting is w units of energy, which of the following is closest to the annual commercial energy usage for refrigeration, in the same units of energy? 【微信公众号：张巍 GRE】
    A. 0.03w
    B. 0.14w
    C. 0.26w
    D. 0.35w
    E. 0.42w

43. The ratio of the total annual residential energy usage to the total annual commercial energy usage is x to y. Which of the following represents the fraction of the total annual energy usage, both residential and commercial, that is for light?
    A. $\frac{x+y}{6x+29y}$
    B. $\frac{x+y}{(100)(6x+29y)}$
    C. $\frac{6x+29y}{(35)(x+y)}$
    D. $\frac{6x+29y}{(100)(x+y)}$
    E. $\frac{(100)(6x+29y)}{x+y}$

44. The annual commercial energy usage for space cooling is approximately what percent less than the annual commercial energy usage for space heating?
    A. 6%
    B. 22%

C. 32%
D. 39%
E. 46%

Questions 45 to 47 are based on the following data.

Annual Precipitation* and Evaporation for the Earth and its Continents According to a Certain Climate Model

Continent	Surface Area (millions of square kilometers)	Annual Precipitation (millimeters)	Annual Evaporation (millimeters)
Africa	29.8	696	582
Antarctica	14.1	169	28
Asia	44.1	696	420
Australia	8.9	803	534
Europe	10.0	657	375
North America	24.1	645	403
South America	17.9	1,564	946
Earth	510.0	973	973

* The annual precipitation for a region is the total volume of precipitation that falls on the region during a year divided by the total surface area of the region. The annual evaporation is defined similarly.

45. Which of the following is closest to the percent of the surface area of the Earth covered by Asia, Europe, and North America together? 【微信公众号：张巍 GRE】
A. 5%
B. 10%
C. 15%
D. 20%
E. 30%

46. In this climate model, the Earth's surface consists of either land or water regions. If all of the land regions and none of the water regions are accounted for in the surface areas of the seven continents, what is the total surface area, in millions of square kilometers, of the water regions on the Earth's surface?
A. 148.9
B. 361.1
C. 375.2
D. 403.1
E. 658.9

47. If the annual precipitation for a certain 10-million-square-kilometer region of Africa is 200 millimeters, which of the following is closest to the annual precipitation for the rest of Africa,

in millimeters?

A. 300
B. 500
C. 750
D. 950
E. 1,150

Questions 48 to 53 are based on the following data.

Population and Per Capita Carbon Dioxide Emissions for Selected Countries, 2001

Country	Population in 2001 (millions)	Per Capita Carbon Dioxide Emissions in 2001 (metric tons)	Percent Change in Per Capita Carbon Dioxide Emissions from 1990 to 2001
Canada	31	17.1	+10.3%
France	59	6.2	+1.6%
Germany	82	10.1	–16.6%
Japan	127	9.1	+10.3%
Mexico	99	3.7	+3.5%
United Kingdom	60	8.9	–8.6%
United States	284	20.6	+6.6%

48. In 2001, what was the range of the per capita carbon dioxide emissions, in metric tons, for the countries listed? 【微信公众号：张巍 GRE】

A. 3.5
B. 9.1
C. 12.1
D. 14.4
E. 16.9

49. For how many of the countries listed were the per capita carbon dioxide emissions lower in 1990 than in 2001?

A. 2
B. 3
C. 4
D. 5
E. 6

50. In Canada's population increased by 3 million from 1990 to 2001, which of the following is closest to the increase in total carbon dioxide emissions, in metric tons, for Canada from 1990 to 2001?

A. 100,000
B. 1,000,000
C. 10,000,000
D. 100,000,000
E. 1,000,000,000

51. In 1990, approximately how many metric tons of carbon dioxide were emitted per capita in Germany?
A. 8
B. 10
C. 12
D. 14
E. 16

52. Which of the following represents the per capita carbon dioxide emissions, in metric tons, for Canada and the United States combined in 2001? 【微信公众号：张巍 GRE】
A. $\frac{(31)(17.1)+(284)(20.6)}{315}$
B. $\frac{(31)(17.1)+(284)(20.6)}{37.7}$
C. $\frac{(31)(17.1)+(284)(20.6)}{2}$
D. $(\frac{284+31}{2})(37.7)$
E. $\frac{17.1+20.6}{2}$

53. Which of the following is closest to the total number of pounds of carbon dioxide emitted in the United States in 2001?(1 metric ton = 2,205 pounds, rounded to the nearest pound.)
A. 10^7
B. 10^{12}
C. 10^{13}
D. 10^{14}
E. 10^{15}

Questions 54 to 56 are based on the following data.

Nutrition Information on a Bag of Trail Mix
Serving Size: 28 grams　　Servings per Bag: 2.5

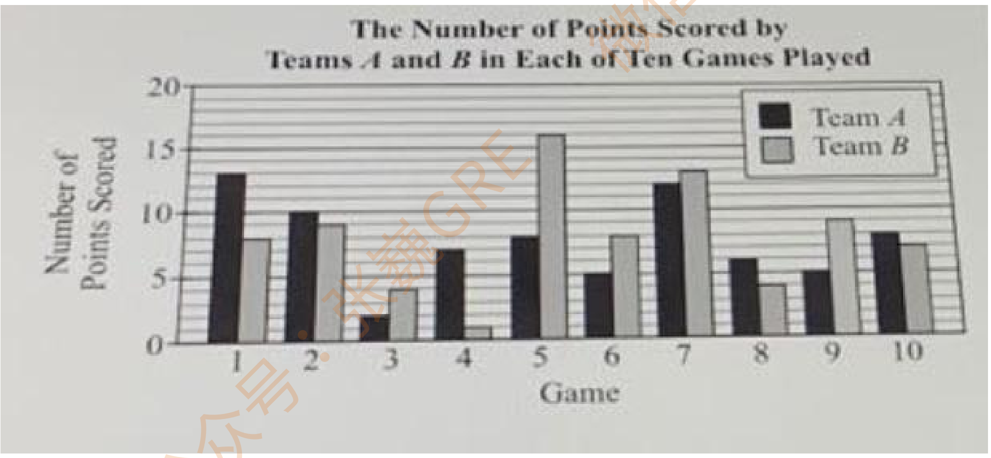
Amount per Serving		Calories per Serving	
Total fat	7 grams	Total Calories	139
Saturated fat	2 grams	Calories from fat	63
Total carbohydrates	15 grams	**Calories per Gram**	
Dietary fiber	2 grams		
Sugars	5 grams		
Protein	4 grams	Calories per gram of fat	9
Cholesterol	0 milligrams	Calories per gram of carbohydrates	4
Sodium	5 milligrams	Calories per gram of protein	4

54. The total number of calories from carbohydrates in a single serving of Trail Mix is approximately what percent of the total number of calories in the serving?
A. 10%
B. 25%
C. 30%
D. 45%
E. 60%

55. A certain candy bar contains a total of 28 grams of fat. The total number of grams of fat in the cady bar is what percent greater than the total number of grams of fat in one bag of Trail Mix? 【微信公众号：张巍 GRE】
A. 18%
B. 25%
C. 60%
D. 63%
E. 725

56. Which of the following expressions gives the number of grams of Trail Mix that contain N grams of saturated fat?
A. $\frac{9N}{139}$
B. $\frac{N}{14}$
C. $\frac{139N}{9}$
D. 14N
E. 9N

Questions 57 to 59 are based on the following data.



57. In what percent of the games for which the difference in the number of points scored by the two teams was equal to 1 point did team A score more points than team B?

A. 20%

B. 30%

C. $33\frac{1}{3}\%$

D. 50%

E. $66\frac{2}{3}\%$

58. Two of the ten games will be selected at random, without replacement. What is the probability that both of the games selected will be games in which team A scored more points than team B? 【微信公众号：张巍 GRE】

A. $\frac{2}{9}$

B. $\frac{1}{4}$

C. $\frac{5}{18}$

D. $\frac{1}{3}$

E. $\frac{1}{2}$

59. The number of games in which team B scored at least 50 percent more points than team A was what fraction of the number of games in which team B scored more points than team A?

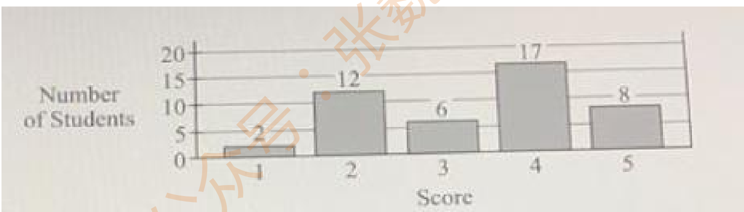
A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{3}{5}$

E. $\frac{4}{5}$



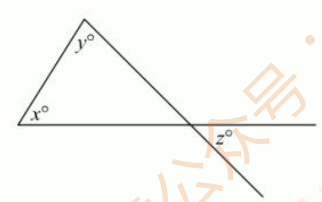
60. The chart shows the distribution of scores on a quiz in a geometry class.
Quantity A: The median of the quiz scores
Quantity B: The mode of quiz scores

新 Section 1 medium

1. a=$3^{11}+3^{16}+3^{19}+3^{23}$
Quantity A: The units digit of a
Quantity B: 3

2. On the number line, the points P, Q, R, and S have coordinates -3, -2, 4, and 6, respectively.
Quantity A: The coordinate of the midpoint of line segment PS
Quantity B: The coordinate of the midpoint of line segment QR

3. In the figure above, y+z =122【微信公众号：张巍 GRE】



Quantity A: x
Quantity B: 60

4. The operation ◇ is defined for all positive two-digit integers m and p as follows. m◇p is the sum of m and the integer obtained by interchanging the units digit and the tens digit of p. For example, 12◇53 = 12+35 = 47.
Quantity A: (17◇36)◇36
Quantity B: 140

5. n>1

Quantity A: $\frac{1}{n}+\frac{1}{n+1}-\frac{1}{n-2}$

Quantity B: $\frac{1}{n}-\frac{1}{n+1}+\frac{1}{n-2}$

6. The function h is defined above for all nonzero numbers x.

$$h(x)=\begin{cases}\frac{1}{x^2} & \text{if } x \text{ is a positive}\\ -\frac{1}{x^2} & \text{if } x \text{ is negative}\end{cases}$$

Quantity A: h(4) + h(-4)
Quantity B: 0

7. $ab<b<0$ and $at<t$

Quantity A: t

Quantity B: 0

8. A certain tutoring center offers only one-hour sessions. The costs of the sessions are as follows. A single session costs \$50, a package of 5 sessions costs \$225, and a package of 10 sessions costs \$400. At the tutoring center, Johann purchased 3 single sessions, 1 package of 5 sessions, and 2 packages of 10 sessions. What is the average (arithmetic mean) cost to the nearest dollar, of the sessions purchased by Johann?【微信公众号：张巍 GRE】

A. \$42
B. \$43
C. \$44
D. \$45
E. \$46

9. A student took two tests and received the same score on each test. On the first test, the students score was 0.5 standard deviation above the mean score of 20. On the second test, the student's score was 0.8 standard deviation below the mean score of 32. If r is the standard deviation of the scores on the first test and t is the standard deviation of the scores on the second test, what is the value of t in terms of r?

A. 2.4-1.6r
B. 1.5-0.625r
C. 12-r
D. 15-0.625r
E. 24-1.6r

10. If $x=\sqrt{10}$, what is the value of $(5\sqrt{10}+x)^2$?

Inventory of 500 Vehicles at a Car Dealership, Number of Vehicles by Type and Color

Vehicle Type		Black	Brown	Green	Red	Silver	White	Total
Sedan	4-door	25	34	42	33	30	36	200
	2-door	20	8	18	22	17	15	100
Specialty Vehicle	Minivan	12	6	10	10	8	14	60
	Sport-utility	12	16	22	9	3	18	80
	Station wagon	12	12	13	6	3	14	60
Total		81	76	105	80	61	97	500

11. For how many of the five vehicle types is the number of silver vehicles less than 20 percent of the total number of vehicles of that type?

A. One

B. Two

C. Three

D. Four

E. Five

12. By approximately what percent does the total number of green vehicles exceed the total number of brown vehicles?

A. 25%

B. 29%

C. 33%

D. 38%

E. 46%

13. For the 5 vehicle types and 6 vehicle colors, what is the average(arithmetic mean) number of vehicles per type per color, rounded to the nearest whole number?

A. 17

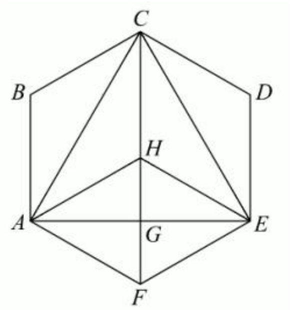
B. 33

C. 45

D. 83

E. 100

14. In the figure shown, ABCDEF is a regular hexagon. Point H is equidistant from each vertex of ABCDEF and angle EGF is a right angle. What is the ratio of the area of triangle EGH to the area of triangle ACE to the area of ABCDEF?【微信公众号：张巍 GRE】



A. 1 to 2 to 6

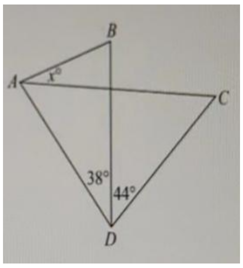
B. 1 to 3 to 12

C. 1 to 4 to 16

D. 1 to 5 to 10

E. 1 to 6 to 12

15. AD=BD=CD



Quantity A: x
Quantity B: 22

16. w = 26.956
Quantity A: The digit in the hundredths place of w
Quantity B: The digit in the hundreds place of 4w

17. -6 < r < -3, -7 < m < -5【微信公众号：张巍 GRE】
Quantity A: r^2
Quantity B: m^3

18. Last semester at School X, there was a total of 6 biology classes, and the numbers of students enrolled in these classes were 33, 34, 32, 32, 34, and 35. Of the total number of students enrolled in the biology classes, 20 percent were biology majors. Of the students enrolled in the biology classes who were not biology majors, 10 percent were enrolled in a chemistry class.
Quantity A: The number of students enrolled in the biology classes who were not biology majors and were enrolled in a chemistry class
Quantity B: 18

19. The least and greatest values in data set X are j and k, respectively, and the least and greatest values in data set Y are p and r, respectively, where j<k<p<r. The least value in data set Z is the average(arithmetic mean) of j and k, and the greatest value in data set Z is the average of p and r.
Quantity A: The average of the range of the values in data set X and the range of the values in data set Y
Quantity B: The range of the values in data set Z

20. Last month a recycling facility received 175 used automobile batteries, all of which arrived at the facility in crates. Each of these crates contained 19, 20, or 21 used automobile batteries.
Quantity A: The number of crates containing used automobile batteries that arrived at the facility last month.
Quantity B: 9

新 Section 2 medium

1. x and y are positive even integers.【微信公众号：张巍 GRE】

Quantity A: $(-1)^{\frac{x+y}{2}}$

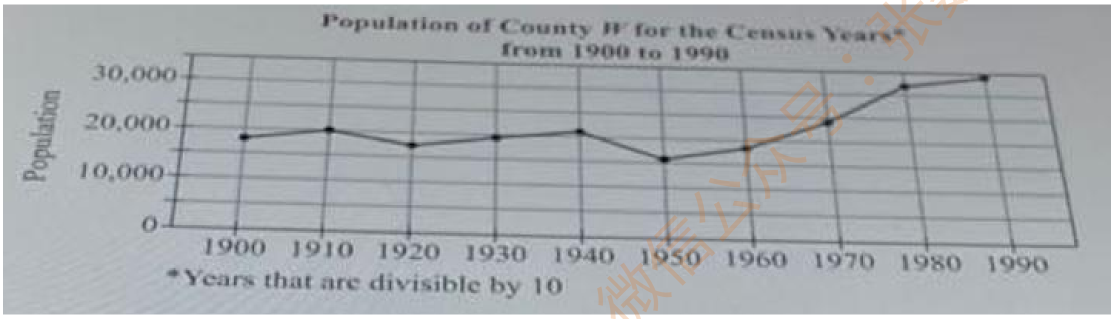
Quantity B: $(-1)^{x+y}$

2. $r = b^2 - 1$
$s = (b-1)^2$
$t = b^3 - 1$
If b is a negative number, which of the following must be true?
A. r<t<s
B. s<r<t
C. s<t<r
D. t<r<s
E. t<s<r

3. The average (arithmetic mean) of x_1, x_2, x_3, x_4, and x_5 is m. Which of the following is the average of $ax_1+b, ax_2+b, ax_3+b, ax_4+b$, and ax_5+b?
A. am
B. a+b
C. m+b
D. am+b
E. 5am+b

4. Thomas' daughter plans to begin college 6 years from now. Thomas plans to open a new college savings account that pays interest at the annual rate of 2.2 percent, compounded annually. Which of the following is closest to the amount of the initial deposit that Thomas should make if he plans to have $20,000 in the account at the end of 6 years, provided that there will be no transactions in the account other than the initial deposit and the interest payments?
A. $17,552
B. $17,668
C. $17,938
D. $18,054
E. $18,170

5. A={1,2,3,4,5,6}
B={1,2,5,6,9,10}
C={1,3,5,7,9,11}
For the sets A, B, and C above, how many numbers are in the set (A∪B) ∩ C?



6. For how many of the nine census years from 1910 to 1990 was the population of County W less than it was for the preceding census year?【微信公众号：张巍 GRE】

A. None
B. One
C. Two
D. Three
E. Four

7. Of the following groups of four consecutive census year, for which group is the range of the populations of County W for the four years in the group the least?

A. 1910-1940
B. 1920-1950
C. 1930-1960
D. 1940-1970
E. 1950-1980

8. Which of the following statements about the median population and the average (arithmetic mean) population of County W for the 10 census years shown is true?

A. The average is less than the median by an amount greater than 5,000.
B. The average is less than the median by an amount less than 5,000.
C. The average is equal to the median.
D. The average is greater than the median by an amount greater than 5,000.
E. The average is greater than the median by an amount less than 5,000.

9. The probability is 0.4 that none of the 3 college courses that Louise wants to take will be available at registration. What is the probability that at least one of the courses that Louise wants to take will be available at registration?

10. Let a, b, an c be integers greater than or equal to 2. When b is divided by c, the remainder is 0. When a is divided by b, the remainder is c. Which of the remainder when a is divided by c?

A. 0
B. 1
C. a
D. b

E. c

11. If x is 50 percent greater than y and if y is 30 percent greater than z, then x is what percent greater than z?【微信公众号：张巍 GRE】

A. 75%

B. 80%

C. 85%

D. 90%

E. 95%

12. Jim is 5 years younger than Mildred, and Mildred is 20 years older than Dinah.

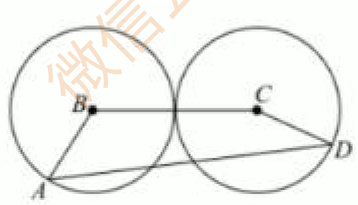
Quantity A: Jim’s age

Quantity B: Twice Dinah’s age

13. The two circles have centers at B and C, respectively, and are mutually tangent. Each circle has radius r.

Quantity A: The perimeter of quadrilateral ABCD

Quantity B: 8r

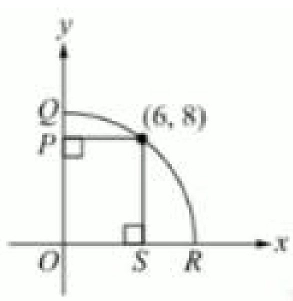


14. The two-digit positive integer m is equal to twice the sum of its digits.

Quantity A: m

Quantity B: 20

15. In the xy-plane, QR is an arc of the circle that has center O and passes through the point (6,8).



Quantity A: The length of SR

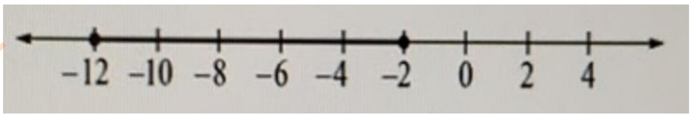
Quantity B: 3

16. \$325, \$418, \$722, \$818, \$955, \$1,092
The six amounts listed are the reported commissions earned by 6 salespeople during April. Two of the reported amount are in error; one is \$120 greater than the actual amount and one is \$120 less.
Quantity A: The standard deviation of the six reported commissions
Quantity B: The standard deviation of the six actual commissions

17. Quantity A: $(1.5+z)^2 - 0.25$
Quantity B: $(1.25+z)^2 + 0.75$

18. In a certain club of 65 members, 40 members enjoy skiing and 29 members enjoy hiking. If all but 4 of the members of the club enjoy skiing or hiking or both, how many members enjoy hiking but not skiing?【微信公众号：张巍 GRE】
A. 13
B. 21
C. 25
D. 32
E. 53

19. The solution set of which of the following inequalities is graphed on the number line shown?



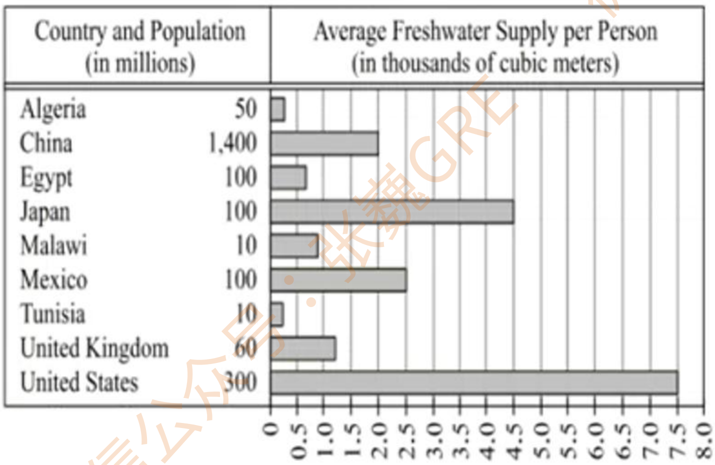
A. $|x|\le 12$
B. $2\le |x|\le 12$
C. $|x-2|\le 10$
D. $|x+7|\le 5$
E. $|x-7|\le 5$

20. $(2.82\times 10^{-51}) - (3.96\times 10^{-49}) =$
A. -3.9318×10^{-49}
B. -1.7804×10^{-51}
C. -1.14×10^{-100}
D. 1.7804×10^{-51}
E. 3.9318×10^{-49}

新 Section 3 medium

1. If n is an integer such that $12n - n^2 > 35$, what is the value of n?

Projected Information for Selected Countries in the Year 2025



2. The average freshwater supply per person for the United States is projected to decrease 20 percent from 1997 to 2025. Based on the data, which of the following was closest to the average freshwater supply per person for the United States in 1997, in cubic meters?

A. 6,000
B. 8,200
C. 8,800
D. 9,400
E. 9,900

3. The average amount of freshwater required per person per year is 1,700 cubic meters. According to the projections, in the year 2025 China will have a freshwater supply of how many millions of cubic meter above the amount required?【微信公众号：张巍 GRE】

A. 420,000
B. 300,000
C. 240,000
D. 180,000
E. 140,000

4. If the countries shown are ordered, from least to greatest, by the amount of their total projected freshwater supply for 2025, which of the following represents the country with the least amount following by the country with the greatest amount?

A. Algeria, United States

B. Tunisia, China
C. Algeria, Japan
D. Tunisia, Japan
E. Algeria, China

5. A list consists of six distinct positive integers less than or equal to 10. Which of the following CANNOT be the median of the six integers?
A. 3
B. 4
C. 5.5
D. 6
E. 7.5

6. The probability that event R will occur is 0.35, and the probability that events R and T will both occur is p. What is the least possible value of p?【微信公众号：张巍 GRE】
A. 0.00
B. 0.35
C. 0.50
D. 0.65
E. 0.70

7. For each of three departments of a certain business at the end of 2011, the table above shows the number of employees and the average(arithmetic mean) number of vacation days taken by the employees in 2011. The average number of vacation days taken by all of the employees in the three departments in 2011 was 10.0. If each employee worked in only one department, what is the value of x?

	Department A	Department B	Department C
Number of employees	25	15	19
Average number of vacation days	12.8	10.4	x

8. m is the least possible nonzero value of |15x+25y|, where x and y are integers.
Quantity A: m
Quantity B:5

9. Working at their own respective constant rate, copier A produces n copies in 30 minutes and copier B produces n copies in 20 minutes. The two copiers worked independently and continuously for 4 hours producing copies at their respective rates.

Quantity A: The number of copies produced by copier B during the 4 hours

Quantity B: 1.5 times the number of copies produced by copier A during the 4 hours

10. $y>|x|$ and $xy<0$

Quantity A: $x+y$

Quantity B: 0

11. Each beetle in a group of 350 beetles was weighted, and 203 of the beetles each weighted 0.62 grams or less. A beetle weight of 0.61 gram was at the kth percentile of the distribution of the 350 weights, where the higher percentile corresponds to the greater weight.

Quantity A: k

Quantity B: 60

12. The numbers in lists P and Q are integers. The range of the numbers in P is 50, and P contains that numbers 0 and -25. The range of the numbers in Q is 50, and Q contains that numbers 0 and -25. R consists of the numbers that P and Q have in common, and the range of the numbers in R is x. List S consists of the numbers in P and Q combined, and the range of the numbers in S is y.

Quantity A: The sum of the least possible value of x and the greatest possible value of y

Quantity B: The sum of the greatest possible value of x and the least possible value of y

13. List S consists of n numbers, where $n>1$, and the range of the numbers in S is r. List T consists of n number and is formed by raising each number in S to the third power.

Quantity A: The range of the numbers in T

Quantity B: r^3

14. In the xy-plane, a triangular region is enclosed by the x-axis, the y-axis, and the line with equation $2x-y+k=0$, where k is a positive constant. For which of the following values of k is the area of the triangular region greater than 1 and less than 4?【微信公众号：张巍 GRE】

A. 0

B. 1

C. 2

D. 3

E. 4

15. If q and r are prime numbers greater than 3 and $qr^2 < 450$, what is the greatest possible value of r?

A. 5

B. 7

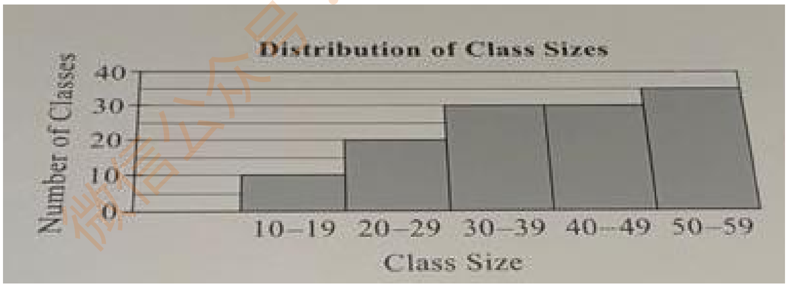
C. 11
D. 13
E. 17

16. What is the median of the number shown?

$$2\sqrt{2}-1,\quad \sqrt{2}-1,\quad 1+\sqrt{2},\quad 1-\sqrt{2},\quad \sqrt{2}$$

A. $2\sqrt{2}-1$
B. $\sqrt{2}-1$
C. $1+\sqrt{2}$
D. $1-\sqrt{2}$
E. $\sqrt{2}$

17. The figure shows the distribution of class sizes for 125 classes at a certain college. Based on the figure, which of the following values could be the median of the 125 class sizes?



Indicate all such values.
A. 25
B. 34
C. 35
D. 40
E. 45

18. At a college graduation ceremony, there will be 5 speakers seated in a row of 5 seats on a stage. One of the speakers is the president of the college. In how many of the 120 possible seating arrangements for the 5 speakers is the president seated in the third seat in the row?【微信公众号：张巍 GRE】

19. On a certain day, a fruit stand sold only two kinds of bananas: regular bananas and organic bananas. The fruit stand sold regular bananas at a price of \$0.60 per pound and organic bananas at a price of \$0.90 per pound. The average price per pound of all the bananas sold by the fruit stand that day was \$0.80 per pound. The total price of the regular bananas sold by the fruit stand that day was what fraction of the total price of all the bananas sold by the fruit stand that day?
Give your answer as a fraction.

20. If $\frac{1}{\sqrt{x}} > \frac{1}{24}$, where x>0, which of the following could be the value of x?【微信公众号：张巍 GRE】

Indicate all such values.

A. $\frac{1}{676}$

B. $\frac{1}{400}$

C. 324

D. 625

新 Section 4 medium

Sources of College Information-Survey of 2,400 High School Students

Source of Information	Percent Citing Source
Campus visit	55%
College alumni	15%
College catalog	28%
College ranking	20%
College Web site	25%
Current students	30%
Family members	43%
High school counselors	32%
High school teachers	17%

1. How many of the sources of information listed in the table were cited by more than 700 students?【微信公众号：张巍 GRE】
A. 2
B. 3
C. 4
D. 5
E. 6

2. How many of the students surveyed cited high school counselors as a source of information?
A. 408
B. 600
C. 628
D. 720
E. 768

3. Based on the information given, which of the following statements must be true?
Indicate all such statements.
A. The total number of students surveyed who cited either a college catalog or college alumni or both was less than the number of students who cited a campus visit.
B. Of the students surveyed, 73% cited either current students or family members or both.
C. Of the students surveyed, 8% cited both high school counselors and college ranking.
D. Of the students surveyed, 45% did not cite a campus visit.

4. The table shows the percent distribution of 1,200 respondents from a survey of adults, ages 55 to 64, who were asked to choose their favorite leisure activity. Based on the information in the table, which of the following statements are true?

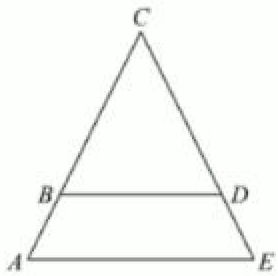
Indicate all such statements.

Response	Watching Television	Socializing	Reading	Exercising	Other
Percent	52%	13%	10%	5%	20%

A. Reading was chosen by 120 respondents.
B. Watching television was chosen by more than half the respondents.
C. The number of respondents who chose socializing was 30 percent greater than the number of respondents who chose reading.

5. $R = 1 - 2^{-1} + 4^{-1} - 8^{-1}$
$T = 1 - 2^{-1} + 4^{-1} - 8^{-1} + 16^{-1} - 32^{-1}$
Quantity A: R-T
Quantity B: -32^{-1}

6. In the figure above, triangle ACE is similar to triangle BCD. The height of ACE corresponding to the base AE is 7, and the height of BCD corresponding to the base BD is k. The area of BCD is equal to the area of trapezoid ABDE.【微信公众号：张巍 GRE】
Quantity A: k
Quantity B: 5



7. Two librarians, Robert and Patricia, cataloged a combined total of 180 new books, each of which was cataloged by only one of the two librarians. Although they spent the same amount of time cataloging their books, Robert spent an average of 15 minutes per book whereas Patricia spent an average of 12 minutes per book.
Quantity A: The number of books cataloged by Robert
Quantity B: 60

8. n is a positive integer and k=10,000n
Quantity A: The sum of the digits of n
Quantity B: The sum of the digits of k

9. The median of the numbers in a list is x, and the arithmetic mean of the numbers in the list is $\frac{x+2}{2}$

Quantity A: x

Quantity B: $\frac{x+2}{2}$

10. m is positive integer that is divisible by 8. q is a positive integer that is not divisible by 8 but is divisible by 4.

Quantity A: The remainder when m+q is divided by 8

Quantity B: 4

11. a, b, and c are positive integers【微信公众号：张巍 GRE】

Quantity A: $\frac{a+c}{b+c}$

Quantity B: $\frac{a}{b}$

12.

1 foot = 12 inches

1 yard = 3 feet

1 mile = 5,280 feet

Approximately how many cubic yards of asphalt are needed to cover a level road that is 1/2 mile long and 24 feet wide with a layer of asphalt that is 4 inches thick?

A. 800

B. 1,600

C. 2,400

D. 3,000

E. 9,000

13. The average(arithmetic mean) of 14 different positive integers in a set is 14. What is the greatest possible integer in any such set?

A. 14

B. 28

C. 60

D. 91

E. 105

14. How many positive integers less than or equal to 603 are multiples of 2 or multiples of 3 or both?
A. 101
B. 202
C. 303
D. 402
E. 501

15. In a data set, the value 79 is 1/2 standard deviation above the mean and the value 72 is 2/3 standard deviation below the mean. How many standard deviations above the mean is the value 87?
Give your answer as a fraction.

16. The function f is defined by f(x) = $\frac{x(x^2-4)(x^2-1)}{\sqrt{x^2-6x+9}}$ for all numbers x such that x≠3. For which of the following values of x is f(x) =0?
Indicate all such values.
A. -3
B. -2
C. -1
D. 0
E. 1
F. 2
G. 4
H. 6

17. If x and y are positive and x is 43 percent greater than y, then y must be what percent less than x?【微信公众号：张巍 GRE】
Give your answer to the nearest whole percent.

18. Which of the following expressions are equivalent to x+1?
Indicate all such expressions.
A. (2x+3)-(x+2)
B. 3(x+1) – 2(x-1)
C. (5x+1)-4x
D. $\frac{3(x+1)+4(x+1)}{7}$
E. $\frac{x^3+x^2+x+1}{x^2+1}$

19. If n is a multiple of 7 and $n=s^2t$, where s and t are each prime numbers, which of the following expressions must be a multiple of 49?【微信公众号：张巍 GRE】

A. s^2

B. t^2

C. st^2

D. s^2t

E. s^2t^2

20. List M:15, 25, 35, x, y, z

List M consists of 6 integers. The average (arithmetic mean) of the 6 integers is 40. None of the integers in list M is greater than 100.

Quantity A: The number of integers in list M that are greater than 40

Quantity B: 2

新 Section 5 hard

1. The standard deviation of n numbers $x_1, x_2, x_3, \ldots, x_n$ with mean $\overline{x}$ is equal to $\sqrt{\frac{S}{n}}$, where S is the sum of the squared differences $(x_i - \overline{x})^2$ for 0≤i≤n.

Each book that a bookstore sold last year was either new or used. The bookstore sold new books for $80 each and used books for $40 each. The standard deviation of the prices of the books sold last year was $16.

Quantity A: The fraction of the books sold last year that were new

Quantity B: 1/2

2. $(5^3)w+(5^2)x+5y+z=264$

In the equation shown, w, x, y, and z are nonnegative integers and each is less than 5. What is the value of w+x+y+z? 【微信公众号：张巍 GRE】

A. 6

B. 7

C. 8

D. 10

E. 12

3. In a group of 62 children, more of the children were born on a Friday than on any other day of the week. What is the least possible number of children in the group who were born on a Friday?

A. 11

B. 10

C. 9

D. 8

E. 7

4.

7, 9, 10, 12, 14, 15, 17, 18, 19, 21, 23

A car dealership sold n total of 165 cars for the first 11 months of a certain year, and the list shows the monthly numbers of cars sold, which are ordered from least to greatest. The dealership sold x cars for the last month of the years. If the average(arithmetic mean) of the 12 monthly numbers of cars sold is equal to the median of the 12 monthly numbers of cars sold, what is the greatest possible value of x?

5. For a positive integer n, when 2n+3 is divided by 11, the remainder is 3. What is the remainder when n+15 is divided by 11?

A. 0

B. 1

C. 2
D. 3
E. 4

6. All of the 80 science students at a certain school are enrolled in at least one of thee science courses: biology, chemistry, and physics. There are 60 students enrolled in biology, 50 students enrolled in chemistry, and 35 students enrolled in physics. None of the students are enrolled in all tree courses. Which of the following could be the number of students enrolled in both chemistry an physics?

Indicate all such numbers.

A. 0
B. 5
C. 10
D. 15
E. 20
F. 25
G. 30
H. 35

7. Let m = $(2^3)(3^2)(5)(7^2)$ and p= $(2^2)(3^5)(5^4)(11)$. What is the greatest common divisor of m and p?【微信公众号：张巍 GRE】

A. (2)(3)(5)
B. $(2^2)(3^2)(5)$
C. (2)(3)(5)(7)(11)
D. $(2^2)(3^2)(5)(7)(11)$
E. $(2^3)(3^5)(5^4)(7^2)(11)$

8. If a is an integer such that |a-b| > |a| - |b| for all negative integers b, which of the following could be the value of a?

Indicate all such values

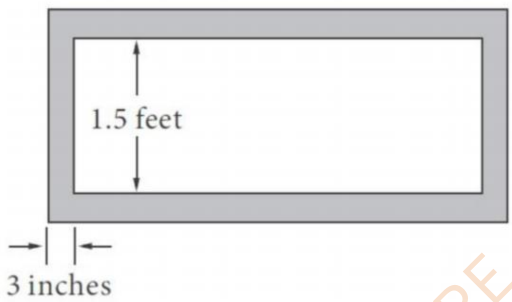
A. -1
B. 0
C. 3

9.

Quantity A: $\frac{9^{17}}{8^{17}}$

Quantity B: $\frac{9^{17}+5^{10}}{8^{17}+5^{10}}$

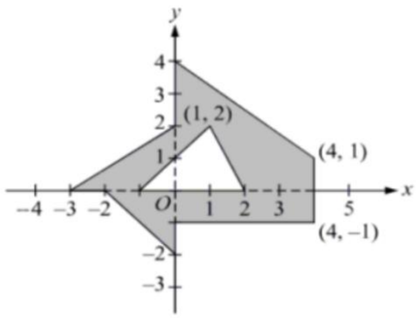
10. A certain rectangular painting is 1.5 feet high and 3 times as wide as it is high. The painting is to be framed by a wood frame that is 3 inches wide, as shown in the figure. Which of the following is closest to the minimum number of feet of 3-inch-wide wood framing that will be needed to frame the painting?(1 foot = 12 inches)



A. 9
B. 10
C. 12
D. 13
E. 16

11. x, y, z, u, v, w, and t are seven nonzero integers. Which of the following values can be the number of these seven integers that are negative if x×y×z=u×v×w×t?
Indicate all such values.
A. Three
B. Four
C. Five

12. What is the area of the shaded region shown in the xy-plane?【微信公众号：张巍 GRE】



13. The finite sets R, S, and T have the same number of elements. The number of elements in the set R∪T is less than the number of elements in the set S∪T.
Quantity A: The number of elements in the set R∩T
Quantity A: The number of elements in the set S∩T

14. Let S be the set of integers from 1 to 10. How many subsets of S contain at least one even integer and at least one odd integer?

A. 25

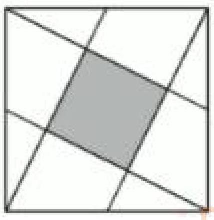
B. 31

C. 62

D. 252

E. 961

15. The figure shows a square in which each of four interior line segments connects the midpoint of a side to a vertex of the square. The area of the shaded region is what fraction of the area of the square?【微信公众号：张巍 GRE】



A. $\frac{1}{6}$

B. $\frac{1}{5}$

C. $\frac{\sqrt{5}}{10}$

D. $\frac{1}{4}$

E. $\frac{\sqrt{3}}{6}$

16. The figure shows a normal distribution with mean m and standard deviation d, including approximate percents of the distribution corresponding to the six regions shown.

The distribution of the random variable W is normal with mean 30,0. The value of 17.6 is at the 6th percentile of the distribution. Which of the following is the best estimate of the standard deviation of the distribution?

A. 2.0

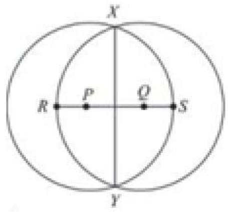
B. 4.0

C. 8.0

D. 12.0

E. 14.0

17. The figure shows two congruent circles with centers P and Q. If RP = QS = 1 and XY = 8, what is the distance between center P and Q?



A. 2
B. 10/3
C. 4
D. 20/3
E. 8

18. The table above consists of 10 rows and 10 columns. Each entry in the table is either a zero or a one. A zero is to be selected at random from the table. What is the probability that the zero selected will be in a row with an odd number of zeros and in a column with an odd number of zeros?

0	1	1	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1
0	0	0	0	1	1	1	1	1	1
0	0	0	0	0	1	1	1	1	1
0	0	0	0	0	0	1	1	1	1
0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0	0

A. 1/9
B. 3/20
C. 2/11
D. 1/4
E. 3/10

19. Kim, Megan, Nina, and Paula are planning to go to a movie theater, where three different movies--A, B and C--are being shown simultaneously in different auditoriums. Each of them will see only one of the movies but not necessarily the same movie as any of the other three people. One possible outcome is that Kim and Megan will see C while Nina and Paula will see A. How many possible outcomes are there?【微信公众号：张巍 GRE】
A. 12
B. 27
C. 48
D. 64
E. 81

20. The standard deviation of n values $x_1, x_2, x_3, \ldots, x_n$ with mean $\overline{x}$ is equal to $\sqrt{\frac{S}{n}}$, where S is the sum of the squared differences $(x_i - \overline{x})^2$ for 0≤i≤n.

List L consists of 80 values. List M consists of the values in L and the value 92, and the mean of the values in M is 12. If the standard deviation of the values in L is 0, what is the standard deviation of the values in M?【微信公众号：张巍 GRE】

A. 0

B. 6

C. $\frac{80}{9}$

D. $\frac{4\sqrt{5}}{3}$

E. $4\sqrt{5}$

新 Section 6 hard

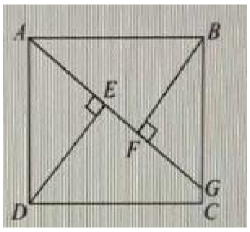
1. For positive integers a and b, a is a multiple of 3, and ab is a multiple of 18. When b is divided by 9, the remainder is R.
Quantity A: R
Quantity B: 1

2. Of an accounting firm's clients, 40 percent own a family business and 60 percent are at least 35 years old.
Quantity A: The percent of the accounting firm's clients who own a family business and are less than 35 years old
Quantity B: 16%

3. $y < \frac{x+z}{2}$ 【微信公众号：张巍 GRE】
Quantity A: y-x
Quantity B: z-y

4. The random variables X and Y are each normally distributed, and P(X>50) = P(Y>55).
Quantity A: The standard deviation of X
Quantity B: The standard deviation of Y

5. In rectangle ABCD shown, the length of side AB is greater than the length of side BC, the length of line segment DE is m, and the length of line segment BF is n.



Quantity A: The length of diagonal BD (not shown)
Quantity B: m+n

6. $h^2 > 1$
Quantity A: h
Quantity B: 1/h

7. x and y are positive numbers that differ by 1/2.
Quantity A: The number of positive integers that are less than the greater of x and y
Quantity B: The number of positive integers that are less than the lesser of x and y

8. Which of the following numbers is positive?【微信公众号：张巍 GRE】
A. $(-4)^{99}$
B. $(-5)^{66}$
C. $(-6)^{-77}$
D. $-(6^{-66})$
E. $-(6^{-77})$

9. A class of 20 students is divided into 4 groups, and each group has at least 3 students. Which of the following statements must be true?
Indicate all such statements.
A. The total number of students in any 3 of the groups is at least 9.
B. The total number of students in any 2 of the groups is at least 7.
C. At least 2 of the groups have the same number of students.

10. Three packages, A, B, and C, weigh a total of 60 pounds. Package A weighs twice as much as package B, and package C weighs 3 pounds less than twice as much as package A. How much does package B weigh, in pounds?
A. $8\frac{1}{7}$
B. 9
C. $10\frac{1}{2}$
D. 12
E. 15

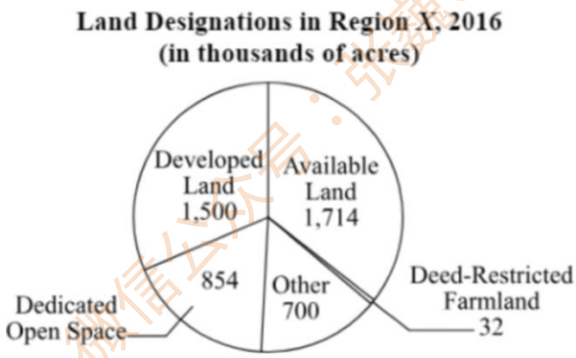
11. For what fraction of the integers from 301 to 1,000, inclusive, does the sum of the units digit and the tens digit equal 7?
A. 3/100
B. 1/25
C. 1/20
D. 1/14
E. 2/25

12. Two different integers are to be selected without replacement from the integers from 1 to 100, inclusive. What is the probability that both integers will be multiples of 6?

A. 4/165
B. 4/25
C. 7/330
D. 9/400
E. 16/625

13. If x=$\sqrt{5}$, what is the value of $(5\sqrt{5}+x)^2$?

Questions 14 to 16 are based on the following data.



14. In the circle graph, the measure of the central angle of the sector for Available Land exceed the corresponding measure for land designated as Other by approximately how many degrees?【微信公众号：张巍 GRE】
A. 26
B. 38
C. 53
D. 64
E. 76

15. In 2016, if y acres of land that were designated as Available Land had instead been designated as Dedicated Open Space, then the number of acres of Dedicated Open Space would have been 75 percent of the number of acres of Available Land. Approximately what is the value of y?
A. 247,000
B. 432,000
C. 1,063,000
D. 1,286,000
E. 1,726,000

16. Based on the information given, which of the following statements are true?
Indicate all such statements.【微信公众号：张巍 GRE】

A. The acreage of land designated as Developed Land was less than one-third of the total acreage of land in Region X.

B. The acreage of land designated as Dedicated Open Space was less than 50 percent of the total acreage of land designated as Developed Land.

C. The acreage of land designated as Available Land was more than 80 times the acreage of land designated as Deed-Restricted Farmland.

17. Line k in the xy-plane has y-intercept 8 and a negative x-intercept. The area of the triangle bounded by line k and the coordinate axes is 20. What is the slope of k?
Give your answer as a fraction.

18. A farmer has three flat circular gardens. The ratio of the radii of the gardens is 5 to 3 to 2. Which of the following statements are true about the areas of the gardens?
Indicate all such statements.

A. The area of the smallest garden is greater than 1/8 the total area of the three gardens.

B. The area of the second-largest garden is 64 percent less than the area of the largest garden.

C. The area of the largest garden is 2.5 times the area of the smallest garden.

19. If $(r+t\sqrt{2})^2 = x + y\sqrt{2}$, where r, t, x, and y are integers, then

A. $x = r^2 + t^2$ and y=2rt

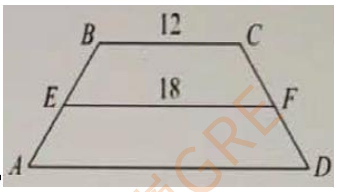
B. $x = r^2 + t^2$ and y=-2rt

C. $x = r^2 + 2t^2$ and y=2rt

D. $x = r^2 - 2t^2$ and y=2rt

E. $x = 2r^2 + t^2$ and y=-2rt

20. Quadrilateral ABCD shown is a trapezoid with bases AD and BC. Sides AB and CD are congruent, with midpoints E and F, respectively. If the height of ABCD is 8, what is the perimeter of ABCD?



A. 42

B. 46

C. 48

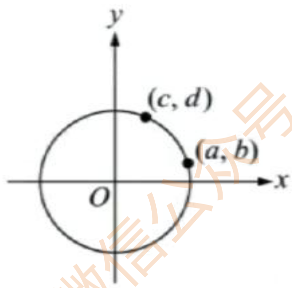
D. 52

E. 56

新 Section 7 hard

1. The total charge C that a customer pays to rent a certain piece of equipment is given by C=1.06 (20+5 (d-1)), where C is in dollars and d is the number of days for which the customer rents the piece of equipment. Customer Z paid a total charge of $106.00 to rent this piece of equipment.
Quantity A: The number of days for which Customer Z rented the piece of equipment
Quantity B: 18

2. In the rectangular coordinate system, O is the center of the circle and (a, b) and (c, d) are on the circle.【微信公众号：张巍 GRE】



Quantity A: $a^2 + b^2$
Quantity B: $c^2 + d^2$

3. x>0
Quantity A: $\frac{5+x}{\frac{15}{4}+x}$

Quantity B: $\frac{4}{3}$

4. k is a positive integer.
Quantity A: $(\frac{1}{5})^{-k}$

Quantity B: $(\frac{1}{2})^{-2k}$

5. x[(75+y)+(15-y)]=900
Quantity A: xy
Quantity B: 10

6. $|2x+1|=3$

Quantity A: x^2-1

Quantity B: 1

7. c is at least $\frac{1}{10}$ of d.

Quantity A: 10c+(d-c)

Quantity B: c

8. In the xy-plane, which of the following is an equation of the graph shown?

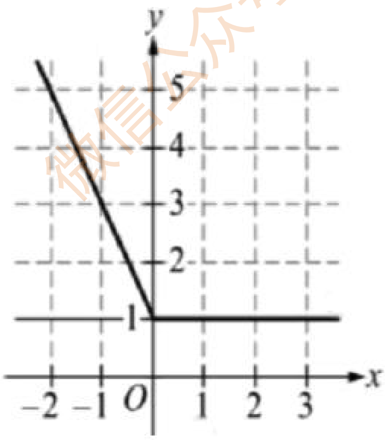
A. y=|x|+1

B. y=|x|+x+1

C. y=|x|-x+1

D. y=-|x|+x+1

E. y=2|x|+x+1



9. From a class of 8 students, of which 5 students are female, a president and a vice president are to be chosen at random. If a student cannot be both the president and the vice president, what is the probability that the president and the vice president will both be female? 【微信公众号：张巍 GRE】

A. 1/4

B. 5/16

C. 5/14

D. 1/2

E. 5/8

10. The table above shows the net annual incomes of a certain business from 2000 to 2006. Which of the following statements are true?

Year	2000	2001	2002	2003	2004	2005	2006
Net Income (in millions of dollars)	3.0	2.8	3.4	3.1	3.3	3.8	4.0

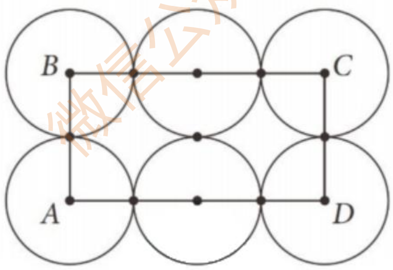
Indicate all such statements.

A. The range of the net annual incomes for the 7 years is $1.2 million.

B. The median of the net annual incomes for the 7 years occurred in 2004.

C. The average (arithmetic mean) change in the net annual income from 2000 to 2006 is an increase of 50.25 million per year.

11. Each of the thirteen marked points in either the center of one of the six circle or a point shared by two tangent circle. Each circle has a radius of r inches, and the rectangular region ABCD has a perimeter of x inches and an area of x square inches. What is the value of r?



A. 1.0

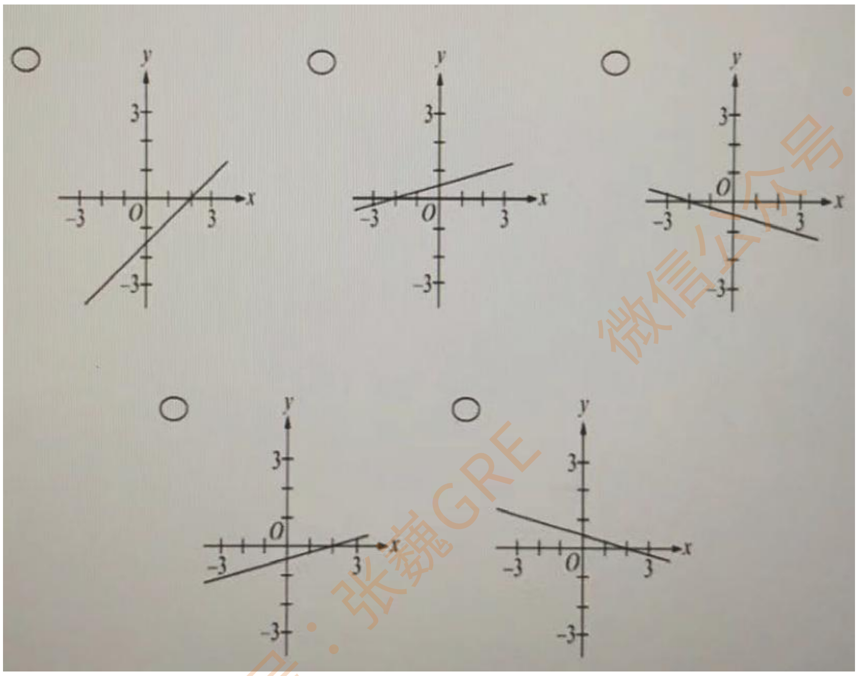
B. 1.5

C. 2.0

D. 2.5

E. 3.0

12. In the xy-plane, which of the following best represents the graph of the equation $2y - 2 = \frac{x}{2} - 3$? 【微信公众号：张巍 GRE】



13. x<-1 and 0<y<1

List L consists of the five numbers x, x, y, xy, and x^2y^2. Which of the following is the median of the five numbers in list L?

A. x

B. y

C. xy

D. x^2y^2

E. $\frac{x+xy}{2}$

14. If x and y are positive numbers, x is less than 6, and y is less than 8, which of the following statements must be true?【微信公众号：张巍 GRE】

Indicate all such statements.

A. x-y<2

B. x/y<2

C. x+y<14

15. If k is a positive integer and n is the ones digit of 7^k, what is the greatest possible value of |n-7|?

A. 0

B. 2

C. 4

D. 6

E. 8

Questions 16 to 18 are based on the following data.

Population of Nine States of the United States in 2000

State	Percent Increase in Population from 1990 to 2000	Population in 2000 (thousands)
Connecticut	3.6%	3,406
Maine	3.8%	1,275
Massachusetts	5.5%	6,349
New Hampshire	11.4%	1,236
New Jersey	8.9%	8,414
New York	5.5%	18,976
Pennsylvania	3.4%	12,281
Rhode Island	4.5%	1,048
Vermont	8.2%	609
		Total: 53,594

16. In which of the nine states did the population increase by the greatest number from 1990 to 2000?

A. New Hampshire
B. New Jersey
C. New York
D. Pennsylvania
E. Vermont

17. In 1990, the population of New Jersey was approximately what percent of the population of Pennsylvania?【微信公众号：张巍 GRE】

A. 45%
B. 50%
C. 55%
D. 60%
E. 65%

18. If the population of New York in 1990 was 7.1 percent less than the population in 2010, what was the population of New York in 2010?
Give your answer to the nearest 100,000

19. For the state that had the median population of the nine states in 1990, approximately what was the increase in the population from 1990 to 2000?

A. 118,000
B. 126,000
C. 218,000
D. 332,000
E. 404,000

20. The population of Connecticut and the population of New Hampshire increased by 5.0 percent and 6.7 percent, respectively, from 2000 to 2010. Which of the following is closest to the percent increase in the total population of the two states from 1990 to 2010?

A. 11.3%

B. 11.7%

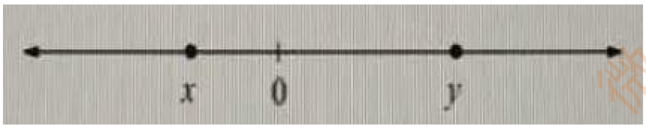
C. 12.8%

D. 13.2%

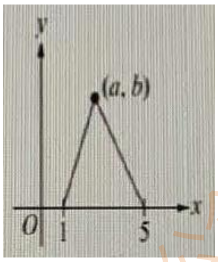
E. 13.8%

新 Section 8 hard

1. The numbers x and y are shown on the number line.
Quantity A: y-x
Quantity B: |x-y|



2. The area of the triangular region is 8.【微信公众号：张巍 GRE】



Quantity A: b
Quantity B: 4

3. x<y<-1<d<0
Quantity A: x(y+d)
Quantity B: y(d+x)

4. In the xy-plane, line k has equation 2x+y=2 and line m has equation x+2y=2. Lines k and m intersect at point (a, b).
Quantity A: a
Quantity B: 1

5. Classes X and Y have 8 and 10 students, respectively. On the day of an exam taken by both classes, one student was absent from each class. In each class, the average (arithmetic mean) score on the exam for the remaining students was 70 points. The students who were absent took the exam later, and both of them earned a score of r points.
Quantity A: The average score on the exam for the 8 students in class X
Quantity B: The average score on the exam for the 10 students in class Y

6. x is an integer.

Quantity A: $(8)^x\left(\frac{1}{2}\right)^{3x-4}$

Quantity B: 16

7. In the xy-plane, points (6,0) and (0,6) both lie on circle C.
Quantity A: The diameter of circle C
Quantity B: 12

8. If oranges are packed 24 to a bag and 8 bags to a box, how many dozen oranges are contained in 24 boxes? (1 dozen = 12)【微信公众号：张巍 GRE】
A. 192
B. 384
C. 576
D. 4,608
E. 6,144

9. A proposed rectangular athletic field will have a width between 150 and 175 feet and a length between 225 and 250 feet. Which of the following values could be the area, in square feet, of the proposed field?
Indicate all such values.
A. 33,250
B. 36,750
C. 39,500
D. 44,250
E. 46,500

10. A certain shirt sells for \$12.00 each or 2 for \$22.95. When buying 2 of these shirts at the same time, the savings is approximately what percent of the total cost of 2 shirts if purchased separately?
A. 1%
B. 4%
C. 6%
D. 8%
E. 10%

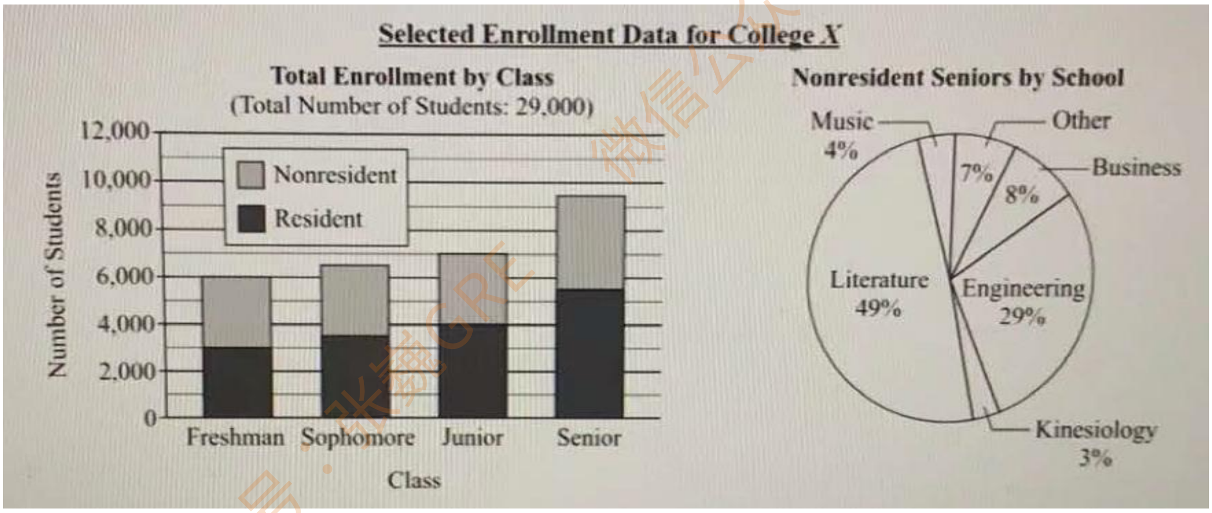
11. If the average (arithmetic mean) of the three numbers 7, k, and m is 14 and the median of these numbers is 12, what is the range of these numbers?
A. 16

B. 19
C. 24
D. 30
E. 35

12. The function f is defined by $f(x)=\frac{x}{x-2}$ for all numbers x such that $x \neq 2$. If $|f(a)| < 1$, which of the following could be the value of a ?
Indicate all such values.
A. -40
B. -4
C. 0
D. 4
E. 40

13. In a certain trivia game, each contestant answers 20 questions and earns or loses points as follows. The contestant earns 10 points for each correct answer and loses 2 points for each incorrect answer. If the contestant begins the game with 0 points, which of the following CANNOT be the total number of points that the contestant has after answering the 20 questions?【微信公众号：张巍 GRE】
A. 80
B. 104
C. 124
D. 140
E. 152

Questions 14 to 16 are based on the following data.



14. The fraction of all students at College X who are nonresident seniors in the kinesiology school is closest to which of the following?

A. 0.004
B. 0.006
C. 0.008
D. 0.012
E. 0.030

15. For the sophomores and juniors combined, the number of nonresident students is approximately what percent less than the number of resident students?

A. 5%
B. 10%
C. 20%
D. 25%
E. 40%

16. If two students are to be randomly selected, one at a time and with replacement, from the nonresident seniors, what is the probability that neither of them will be a student who is in the school of music?【微信公众号：张巍 GRE】

17. If x=$\sqrt{3}$, what is the value of $(4\sqrt{3}+2x)^2$

18. r=$\frac{18!}{15!}$, s= 18^3, t=(19)(17)(15)

Which of the following inequalities indicates the order of the integers r, s, and t, from least to greatest?

A. r<s<t
B. r<t<s
C. s<r<t
D. s<t<r
E. t<r<s

19. Which of the following is the total number of possible values for the units digit of the sum of a positive integer and its square?

A. Three
B. Four
C. Five
D. Six
E. Seven

20. The probability that event M occurs is 0.3, and the probability that event S occurs is 0.6. If M and S are independent, then the probability that neither of them occurs is

A. 0.10
B. 0.18
C. 0.28
D. 0.60
E. 0.82

新 Section 9 hard

1. Quantity A: x^{-3}

Quantity B: $\sqrt{x^{-5}}$

2. Liquids S is sold in 3-liter bottles for $6.33 each, and liquid T is sold in 5-liter bottles for $10.55 each.
Quantity A: The average (arithmetic mean) cost per liter for 3 of the bottles of liquid S and 2 of the bottles of liquid T
Quantity B: $2.11

3. A safe contains 2 bags of coins and 4 bags of paper money. The total value of the money in the 6 bags is $1,200.
Quantity A: The total value of the money in the bags that contain coins
Quantity B: $401

4. Sets R, S, and T are nonempty, and each has the same number of elements. The sets R∩S, R ∩T, and S∩T are nonempty, and each has the same number of elements. The set R∩S∩T is empty. Set V consists of all the elements in R that are not in S.
Quantity A: 3 times the number of elements in V
Quantity B: The number of elements in the set R∪S∪T

5. x is a positive odd integer.
y is a positive even integer.
Quantity A: The remainder when xy+1 is divided by 2
Quantity B: The remainder when $x^2y^2 - 1$ is divided by 2

6. m, n, p, r, and t are integers.
0<m<n<p<r<t
Quantity A: The median of m, n, p, r, and t

Quantity B: $\frac{m+n+p+r+t}{5}$

7. (x-1)(x-2)$(x+1)^2$>0【微信公众号：张巍 GRE】
Quantity A: x
Quantity B: 0

8. The tip of the minute hand of a clock is 5 centimeters from the center of the face of the clock.
Quantity A: The distance traveled by the tip of the minute hand of the clock in 38 minutes.
Quantity B: 21 centimeters

9. When the positive integer w is divided by 35, the remainder is 29. What is the remainder when w is divided by 7?【微信公众号：张巍 GRE】
A. 0
B. 1
C. 2
D. 3
E. 4

10. 0, 2, 4, 6, 12
If a and b are any two different numbers selected from the list shown, how many different values are possible for the product ab?
A. Five
B. Six
C. Seven
D. Eight
E. Nine

11. Based on the information in the table above, which of the following statements are true?

Value of Transatlantic Mergers for Selected Industries in 2006

Industry	Value in 2006 (in billions)	Percent Increase from 2005
Finance	\$15.8	487%
Real estate	\$11.1	281%
Technology	\$8.9	121%

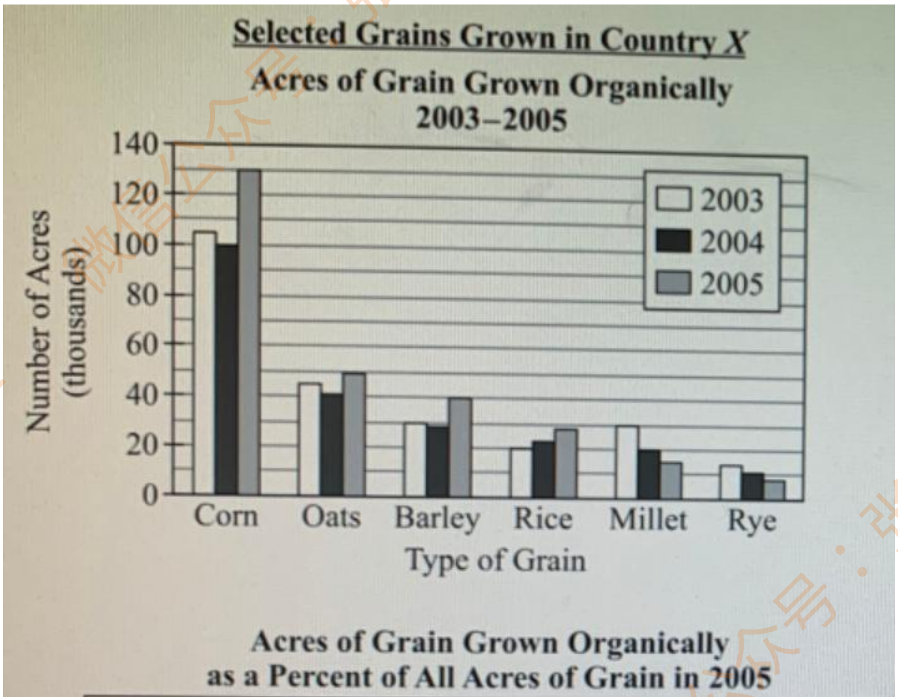
Indicate all such statements.
A. From 2005 to 2006, the value of transatlantic mergers in technology more than doubled.
B. In 2005 the value of transatlantic mergers in finance exceeded the value of transatlantic mergers in real estate.

C. In 2005 the value of transatlantic mergers in real estate was less than $3.5 billion.

12. How many distinct prime factors greater than 2 does the number 990 have?

A. One
B. Two
C. Three
D. Four
E. Five

13. A home aquarium in the shape of a rectangular box has an interior base that measures 20 inches by 10 inches. There are 2,400 cubic inches of water in the aquarium when it is filled to 75 percent of its capacity. What is the interior height, in inches, of the aquarium?



Acres of Grain Grown Organically as a Percent of All Acres of Grain in 2005

Type of Grain	Corn	Oats	Barley	Rice	Millet	Rye
Percent grown organically	0.2%	2.0%	1.0%	0.8%	2.5%	0.6%

14. In 2003, approximately what was the ratio of the number of acres of millet grown organically to the number of acres of rice grown organically?【微信公众号：张巍 GRE】

A. 6 to 5
B. 5 to 4
C. 4 to 3
D. 3 to 2

E. 2 to 1

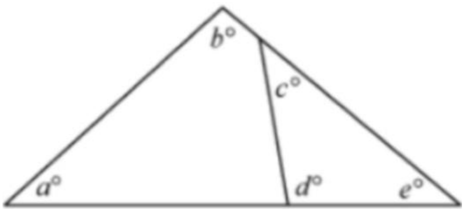
15. From 2005 to 2006, the number of acres of corn grown organically increased by half as much as it did from 2004 to 2005. From 2006 to 2007, the number of acres of corn grown organically increased by half as much as it did from 2005 to 2006. Which of the following is closest to the number of acres of corn grown organically in 2007?

A. 110,000
B. 130,000
C. 140,000
D. 150,000
E. 170,000

16. If the number of acres of barley grown was the same in 2004 as it was in 2005, approximately what percent of the number of acres of barley grown in 2004 was grown organically?【微信公众号：张巍 GRE】

A. 0.5%
B. 0.7%
C. 0.9%
D. 1.1%
E. 1.3%

17. In the figure above, if e=40, what is the value of a+b+c+d?

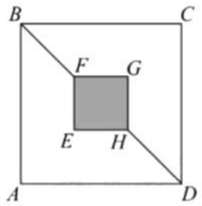


18. A beaker contained 6 grams of a solution that was 12 percent acid and 88 percent water, by weight. After some water was added to the beaker, the resulting solution was 10 percent acid, by weight. How many grams of water were added to the beaker?

A. 0.12
B. 0.2
C. 1
D. 1.2

E. 7.2

19. In the figure shown, diagonal FH of the small square EFGH lies on diagonal BD of the large square ABCD, and BF=FH=HD. If the area of square EFGH is 25, what is the area of the unshaded region of square ABCD?



20. The distance from Karen’s home to an office is 98.4 miles along a certain route. Along the route, a rest stop is located 83.8 miles from her home and a police car is always within 0.5 mile of the rest stop. Along the route, less than 0.5 mile before the office, Karen’s car broke down. Which of the following values could be the distance, in miles, along the route between Karen’s car and the police car?【微信公众号：张巍 GRE】

Indicate all such values.

A. 13.0
B. 13.5
C. 14.0
D. 14.5
E. 15.0
F. 15.5
G. 16.0
H. 16.5

新 Section 10 hard

1. 8x+24=56
Quantity A: 2x+6
Quantity B: 15

2. x is a positive integer.
Quantity A: 5^{-x}
Quantity B:1

3. The list price of a belt and the list price of a scarf were each reduced by $5. The percent decrease in the list price of the belt was less than the percent decrease in the list price of the scarf.
Quantity A: The list price of the belt
Quantity B: The list price of the scarf

4. 0<a<b<c<1

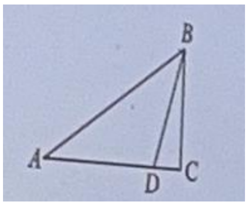
$x=\frac{\frac{a}{c}}{\frac{1}{b}}$ and $y=\frac{\frac{c}{b}}{a}$

Quantity A: x
Quantity B: y

5. x and y are positive numbers such that $\frac{1}{x}+\frac{1}{y}=\frac{1}{4}$ and x<12.【微信公众号：张巍 GRE】
Quantity A: y
Quantity B: 8

6. Quantity A: (1.98)(81)(99)(1,249)
Quantity B: 25,000,000

7. AD=4(DC)



Quantity A: The ratio of the area of triangle ABD to the area of triangle ABC

Quantity B: 3/4

8. The cost to rent a motorbike from a rental shop is x dollars per hour for the first 3 hours and y dollars for each hour thereafter. Which of the following represents the total cost, in dollars, of renting a motorbike from this shop for an 8-hour period?

A. x+3y

B. x+5y

C. x+8y

D. 3x+5y

E. 3x+8y

9. The standard deviation of the five values 2, 4, 6, 8, 10 is σ. For which of the following lists is the standard deviation of the values in the list equal to σ?

Indicate all such lists.

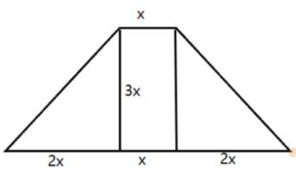
A. 1 ,3, 5, 7, 9

B. 2 ,4 ,8, 16, 32

C. 3, 5, 7, 9, 11

D. 4, 8, 12, 16, 20

10. A plot of land used to display flowers in a botanical garden was designed in the shape of a trapezoid in which the length of the two sides that are not parallel are equal. The length of the longer parallel side is 5 times the length of the shorter parallel side, and the distance between the two parallel sides is 3 times the length of the shorter parallel side. The perimeter of the plot is 135 meters. Approximately what is the length, in meters, of the shorter parallel side?【微信公众号：张巍 GRE】



A. 9

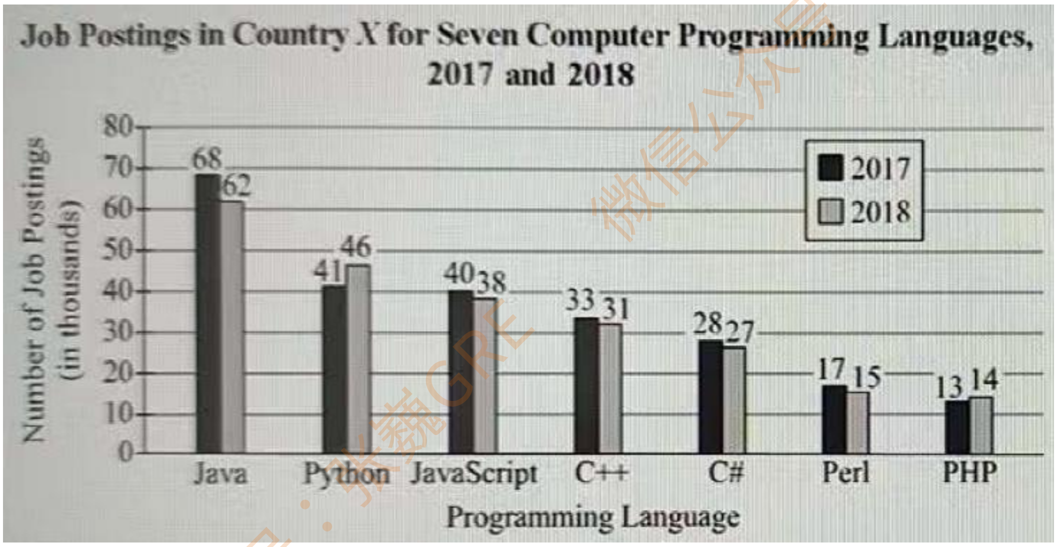
B. 10

C. 11
D. 12
E. 13

11. In a certain 10-sided polygon, 9 interior angles are congruent and the measure of the remaining interior angle is 108 degrees. What is the measure of an exterior angle at the vertex of one of the congruent angles? (Note: An exterior angle at a vertex of a polygon is the angle between one side of the polygon and a line extended form an adjacent side of the polygon.)【微信公众号：张巍 GRE】
A. 32
B. 34
C. 36
D. 38
E. 40

12.
n=12!
Which of the following is a factor of n?
A. 143
B. 250
C. 729
D. 2,048
E. 3,072

Questions 13 to 15 are based on the following data.



13. A storage facility rents units of two sizes, small and large. Each small unit has a

capacity of 800 cubic feet and a monthly rental fee of $95. Each large unit has a capacity of 2,000 cubic feet and a monthly rental fee of $200. The total of the capacities of all the units is 98,400 cubic feet. For a month in which all the units are rented, the total of the monthly rental fees for all the units is $10,560. What is the total number of units that the storage facility has?

14. For the programming languages shown, the ratio of the greatest number of job posting in 2018 to the least number of job postings in 2018 was closest to which of the following?
A. 4 to 1
B. 22 to 5
C. 9 to 2
D. 5 to 1
E. 31 to 5

15. For 2017, let M be the median number of job postings for the programming languages shown and let P be the median number of job postings for the programming languages shown except for PHP. What is the value of P-M?【微信公众号：张巍 GRE】
A. 3,500
B. 5,000
C. 7,000
D. 7,500
E. 8,000

16. If the numbers of job postings for each language and year except Java in 2018 were as shown, and if the range of the numbers of job posting for 2018 had been equal to that for 2017 instead of what it actually was, what would have been the number of job postings for Java in 2018?

17. Points A, B, and C are on a circle, and line segment AC is a diameter of the circle. If the radius of the circle is 6, by how much does the length of arc ABC exceed the length of diameter AC?
Give your answer to the nearest integer.

18. Each time a certain coin is tossed, it lands either heads up or tails up. The coin has the property that the probability that it will land heads up when tossed 1 time is 0.6. If the coin is to be tossed 3 times, what is the probability that it will land heads up at least 2 times?
A. 18/125
B. 27/125
C. 54/125
D. 75/125

E. 81/125

19. For any subset S of a universal set U, the set $\overline{S}$ consists of all the elements in U that are not in S. Of the elements in U, 50 percent are in set A, 30 percent are in set B, and 10 percent are in the set A∩B. If the set $A \cup \overline{B}$ contains 840 elements, how many elements does the set $\overline{A} \cup B$ contain?

A. 420
B. 504
C. 630
D. 1,120
E. 1,680

20. Let S be the sequence of consecutive odd integers from 1 to 99. Sequence T is obtained from S by removing the integer 3 and then removing every 3^{rd} integer after 3. Sequence V is obtained from T by removing the integer 7 and then removing every 7^{th} integer after 7. Which of the following integers is in V?【微信公众号：张巍 GRE】

A. 9
B. 15
C. 21
D. 23
E. 29

新 Section 11 hard

1. A set of 16 points are marked on a line. M and N are two of the points in the set, and 5 of the other 14 points are between M and N. The shortest of the distances between any two consecutive points in the set is d.
Quantity A：The distance between M and N
Quantity B：5d

2. A group of x girls and y boys took a test, and the average (arithmetic mean) score for the group was 74.4.The average score for the x girls was 75.6.【微信公众号：张巍 GRE】

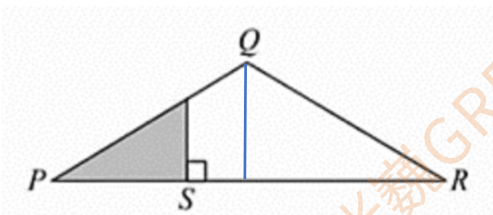
Quantity A: The average score for the y boys
Quantity B: 73.2

3. In the xy-plane, line k has a positive x-intercept and a slope of 2. Which of the following equations represents a line that intersects line k at a point above the x-axis?

A. y=-3x B. y=-2x-1 C. y=-x-1 D. y=x+1 E. y=2x+1

4. Last year the value of one share of a certain stock increased by 10 percent from January to June, and the value of one share of the stock increased by 50 percent from January to December. What was the percent increase in the value of one share of the stock from June to December of last year? Give your answer to the nearest whole percent.
【微信公众号：张巍 GRE】

5. PQ=QR
The area of the shaded triangular region is 1/3 the area of triangular region POR



Quantity A: PR/3
Quantity B: PS

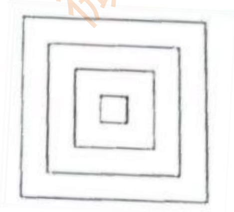
6. Quantity A: $11^5 13^5$
Quantity B: 12^{10}

$$12^{10}$$

7. A farmer has 4x meters of fencing to enclose a rectangular vegetable garden. Enclosing a garden with which of the following dimensions, in meters, would provide the greatest area for the garden？　【微信公众号：张巍 GRE】

A. x by x
B. (x-2) by (x+2)
C. 0.8x by 1.2x
D. 0.4x by 1.6x
E. It cannot be determined from the information given.

8. The design shown in the figure consists of squares and is created in several steps. In the first step, a square measuring 1 inch on each side is drawn. In the second step,a1-inch-wide border is drawn around the square, creating a larger square. in each subsequent step, a 1-inch-wide border is drawn around the largest existing square.What is the number of steps that will be needed in order to create a design in which the largest existing square measures 47 inches on each side?



9. n and q are positive integers such that n > q. When n is divided by q, the remainder is 5.
Quantity A：the least possible value of nq
Quantity B：60

10. In a certain school, 1/3 of the students are seniors and 5/6 of the students are studying French.

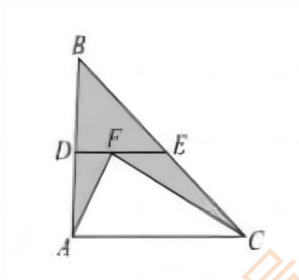
Quantity A: The fraction of the students at the school who are seniors and are not studying French
Quantity B: 1/12

11. At a fruit market, the price of each peach is x cents and the price of each plum is y

cents. If the total price of 4 peaches and 3 plums is equal to the total price of 7 peaches and 2 plums, which of the following statements must be true?
Indicate all such statements

A. The total price of 1 peach and 1 plum is less than the total price of 4 peaches.
B. The price of 1 plum is 3 times the price of 1 peach.
C. The total price of 14 peaches is greater than the total price of 5 plums.

12.



D is the midpoint of AB
E is the midpoint of BC
F is a point on DE

Quantity A: The area of triangular region ACF
Quantity B: The area of the shaded region

13. From a group of 100 people including Alice and Bob, 40 people are to be randomly selected at the same time to win movie tickets. What is the probability that both Alice and Bob will be selected to win movie tickets? 【微信公众号：张巍 GRE】

14. A bus traveled m miles in h hours at an average speed of r miles per hour. If the number of hours traveled by the bus had been 20 percent greater and the average speed had been 20 percent less, then the distance traveled by the bus would have been what percent less?

15. In City N, the sales tax is 8 percent of the selling price. For a certain week, the sales tax was suspended on items of clothing, but only on those with selling prices under$100. During that week, a customer bought 8 items of clothing with a total selling price of $292.

Quantity A: The total amount of sales tax that was suspended on the 8 items of clothing
Quantity B: 20
16. The inside of a rectangular aquarium has an inner base that measures 8 inches by

10inches, and has an inner height of 10 inches. The rectangular base of the aquarium is level, and the aquarium is filled with water so that the water level is 6 inches higher than the base. After a solid piece of steel is completely submerged in the water, the water level is 7 inches higher than the base.

Quantity A： The volume of the piece of steel
Quantity B： 100 cubic inches

17. A wooden board with a width of 2 inches is placed together with other boards, with an overlap of 0.5 inches. How many boards are needed to fill a photo frame with a width of 96 inch? (The default height is the same.)

18. A list consists of non-negative integers excluding the integers 1 and 2. The average (arithmetic mean) of the integers in the list is 1.
Quantity A： The fraction of the integers in the list that are equal to 0
Quantity B： $\frac{3}{5}$

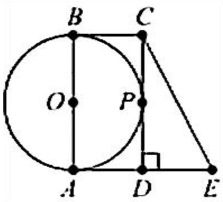
19. The closing price of a stock on a certain day was P dollars. For the next 8 consecutive trading days, the closing price changed as follows. For the 1st, 3rd, 5th, and 7th days, the closing price of the stock was10 percent less than its closing price on the preceding day. For the 2nd, 4th, 6th, and 8th days, the closing price of the stock was 10 percent greater than its closing price on the preceding day. Which of the following represents the closing price, in dollars, of the stock on the 8th day?【微信公众号：张巍 GRE】

A. 4(0.99)P B. 8(0.99)P C. (0.99^4)p D. (0.99^8)p E. P

20. When the positive integer p is divided by 7, the remainder is 3. When p is divided by 6, the remainder is 4. Which of the following could be the value of p ?
Indicate all such values.
A. 10 B. 52 C. 68 D. 80 E. 94

新 Section 12 hard

1. In the figure shown, the circle has center O and is tangent to line segment CD at point P. Quadrilateral ABCD is a rectangle, and AD = DE.



Quantity A：The area of quadrilateral ABCE

Quantity B：The area of the circle

2. If p is an even integer, which of the following must be an odd integer?

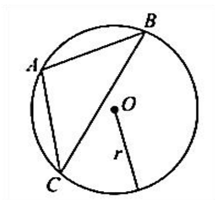
A. $\frac{3P}{2}$ B. $\frac{3P}{2}+1$ C. $\frac{3P^2}{2}$ D. $\frac{3P^2}{2}+1$ E. p^3

3. Theo owns $\frac{1}{3}$ of the number of pairs of shoes that Samantha owns, and Richard owns twice as many pairs as Samantha. If Richard owns fewer than 30 pairs of shoes, which of the following could be the number of pairs he owns?

Indicate all such numbers.

A. 29 B. 24 C. 18 D. 15 E. 12 F. 10 G. 9 H. 6

4. Point 0 is the center of the circle. 【微信公众号：张巍 GRE】



Quantity A: r^2

Quantity B：The area of triangular region ABC

5. Last month 200 students, including Jim and Joseph, took a certain test. Jim's score on the test was 96.which was at the 86th percentile of the 200 scores on the test. Joseph's score on the test was 89, which was at the nth percentile of the 200 scores on the test.

Quantity A：n

Quantity B：79

6. The average (arithmetic mean) and the median of a set of 5 numbers is 0, and the set contains at least one number that is not equal to 0, Which of the following statements must be true?

A. Exactly one of the numbers is not equal to 0.
B. Exactly one of the numbers is negative.
C. Exactly two of the numbers are negative.
D. The set has the same number of positive numbers as negative numbers.
E. The sum of the nonzero numbers in the set is 0.

7. Machines A and B each assembled 3,000 metal clips in 15 hours and 12 hours, respectively, working alone at their constant rates. On Monday, working simultaneously at their respective constant rates, the two machines assembled a total of n clips in x hours. The next day, machine A was modified so that its constant rate decreased by 25 percent, If the two machines worked simultaneously at their respective constant rates for x hours after the modification of machine A, then the total number of clips that they assembled in x hours was approximately what percent less than it was the day before?

A. 6% B. 11% C. 16% D. 20% E. 25%

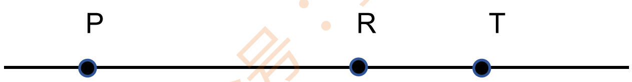
8. Of the 120 first-semester students entering a certain college, 3/5 registered for a statistics class and 1/2 registered for a calculus class. Which of the following could be the number of first-semester students entering the college who registered for a statistics class but did not register for a calculus class?
Indicate all such numbers.

A. 0 B. 6 C. 12 D. 30 E. 60 F. 72 G. 90

9. If n is a multiple of 7 and n $=s^2t$, where s and t are each prime numbers, which of the following expressions must be a multiple of 49 ?【微信公众号：张巍 GRE】

A. s^2 B. t^2 C. st^2 D. s^2t E. s^2t^2

10. On the number line, points P, Q, R, S, and T are ordered from left to right, PR=11. QS=7 and RT=5, points Q and S are not shown.



Quantity: PS
Quantity: QT

11. The function f is defined by $f(x) = 2^{-x^2}$ for all integers x. Which of the following could be the value of f(x)? Indicate all such values.

A. -2 B. -1 C. -0.5 D. 0 E. 0.5 F. 1 G. 2

12. Five gift cards will be distributed among 10 people so that no person receives more than one gift card. The gift cards consist of one \$100 gift card, one \$50 gift card, one \$25 gift card, and two \$10 gift cards. How many different distributions of the five gift cards among the 10 people are possible if the two \$10 gift cards are considered to be identical?
A. 720
B. 2,520
C. 5,040
D. 15,120
E. 30,240

13. If 1<r<s<t< 2 then, of the following, which is closest to $r + (s \times 10^6) + (t \times 10^{12})$?

A. $(r+s+t) \times 10^6$ B. $3s \times 10^9$ C. $t \times 10^{12}$ D. $(r+s+t) \times 10^{12}$
E. $(r+s+t) \times 10^{18}$

14. If |a|=b-c and c = a+ b, where a, b, and c are nonzero numbers, which of the following numbers must be negative?【微信公众号：张巍 GRE】
A. a only
B. b only
C. c only
D. a and c
E. b and c

15. A chair manufacturer' s weekly fixed cost is \$3,000, and the manufacturer s cost of producing each chair is \$100. If the manufacturer sells each chair for \$200, what is the least number of chairs the manufacturer would have to produce and sell in a week in order to make a profit of S1430 or more for that week?

16. Each employee in two separate groups of 15 employees, group I and group II, was

given a job-performance rating. The standard deviation of the ratings for group I was 8.5 and the standard deviation for group II was also 8.5.

Quantity A：The standard deviation of the 30 ratings
Quantity B：17

17. One month，George used 1/5 of his monthly salary for his car payment，For his rent that George used an amount equal to the amount of his car payment plus 1/3 of the amount of his car payment. What fraction of his monthly salary did George use that month for his car payment and rent combined?

A. $\frac{2}{15}$ B. $\frac{4}{15}$ C. $\frac{7}{15}$ D. $\frac{8}{15}$ E. $\frac{11}{15}$

18.
1,2,3...r
1,2,3....,r,r+1
The first sequence consists of r consecutive integers, where r is an even integer. The second sequence consists of r + 1 consecutive integers.
Quantity A:The percent of the integers in the first sequence that are even
Quantity B:The percent of the integers in the second sequence that are even

19. In a sequence of integers, the first two terms are odd integers and each term after the second term is 2 more than the sum of the two previous terms. How many of the first 57 terms of the sequence are odd integers? 【微信公众号：张巍 GRE】
A. 19 B. 28 C. 29 D. 38 E. 57

20. A large pump and a small pump are available to fill a public fountain with 7,500 gallons of water, The pumps can be used alone or simultaneously Working alone at their respective constant rates, the small pump would take 1.6 times as long as the large pump to fill the fountain. Working simultaneously at their respective constant rates the two pumps would take 3 hours to fill the fountain. How long would the small pump take to fill the fountain, working alone at its constant rate?

A. 1 hour 36 minutes
B. 4 hours 25 minutes
C. 4 hours 53 minutes
D. 5 hours 36 minutes
E. 7 hours 48 minutes

新 Section 13 hard

1. Of all the applicants for jobs at a certain company last year, 1/3 received a first interview. Of the applicants who received a first interview, 1/4 received a second interview. The applicants who received no interview or only a first interview were not hired. Of the applicants who received a second interview, 1/5 were hired and the rest were not hired. The applicants who received at least one interview and were not hired were what fraction of all the applicants for the jobs?

2. In a group of 10 people, exactly 4 are vegetarians. A committee of 3 will be formed by randomly selecting 3 people from the group. How many of the possible 3-member committees would include exactly 2 vegetarian members?

A. 3　　B. 6　　C. 24　　D. 36　　E.15

3. At the beginning of 2009, Kim bought 100 shares of stock in a company at a price of \$31 per share. During the year, the price of the stock changed daily; the highest price was \$35 per share and the lowest price was 25 per share. If Kim sold all 100 shares of stock on the same day in 2009, which of the following could be her percent profit or percent loss on the sale?【微信公众号：张巍 GRE】

Indicate all such percent profits or percent losses.

A. 15% loss

B. 10% loss

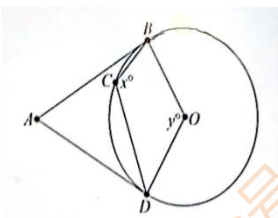
C. 10% profit

D. 15% profit

4. For which of the following values of x does the expression $\sqrt{24/(x+1)}$ represent an irrational number

A. 12.5　　B. 36.5　　C. 53　　D. 71　　E. 95

5. In the figure above, the circle has center O, and AB and AD are tangent to the circle. If the degree measures of angles ABC and ADC are 20 and 40, respectively, what is the value of x+y?



A. 140　　B. 160　　C. 190　　D. 210　　E. 240

6. If a sequence is defined by $a_n = n(-1)^n$ for all integers n from 1 through 499, what is the sum of all the terms in the sequence?

A. -499

B. -250

C. -1

D. 1

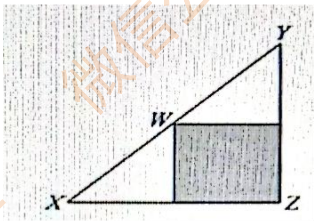
E. 499

7. Data set Q contains 10,000 values. The median of the values is equal to the third quartile of the values. 【微信公众号：张巍 GRE】

Quantity A：The 60th percentile of the values in Q

Quantity B：The 70th percentile of the values in Q

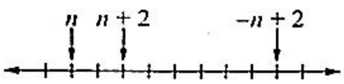
8. The shaded region is square, and the length of XW is greater than the length of WY.



Quantity A：The length of XZ

Quantity B：The length of YZ

9.



Quantity A	Quantity B
$-n+2$	4

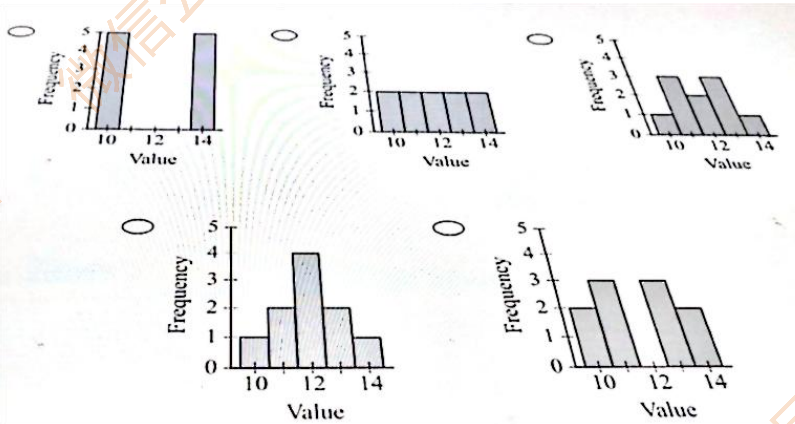
10. Data set T has.1.25 times as many values as data set S, and the average (arithmetic mean) of the values in each data set is 8, Data set U is obtained from data set S by including the additional value 320. Similarly, data set V is obtained from data set T by including the additional value 320, If the average of the values of U is 32, what is the average of the values of V?

11. The cost of producing 1 unit of a product at a certain manufacturing company is \$0.75.According to a sales model, if each unit has a selling price of \$1.00, the company expects to sell 1,000 units each month, For each \$0.05 increase in the unit selling price, the company expects to see a decrease of 100 units sold each month, According to the model, which of the following could be a unit selling price at which the company would expect a profit of at least \$250.00 each month? (Profit equals revenue minus cost.)

Indicate all such answers.

A. \$1.05
B. \$1.10
C. \$1.20
D. \$1.30
E. \$1.35

12. Each of five data sets consists of 10 values, all of which are integers, The distributions of the values in the five data sets are shown. The values in which distribution have the greatest standard deviation?【微信公众号：张巍 GRE】



13. A deposit of \$3,000 was made into an account that earned interest at the annual rate of r percent compounded annually, where r is a constant. The interest was paid into the account, and there were no other deposits to or withdrawals from the account, At the end of the first year, the amount in the account, including interest, was \$3,090. The amount of interest earned for the second year was what fraction of the total amount of interest earned for the 2 years combined?

14. n is an integer greater than 2.
List L: 2，n，n，n；　List M: 2，2，2，n
Quantity A：The standard deviation of the numbers in list L
Quantity B：The standard deviation of the numbers in list M

15. The number of feet in the perimeter of a rectangular game board is equal to the number of square feet in the area of the game board.

Quantity A：The width of the game board

Quantity B：2 feet

16. Polygon A has six equal sides and six equal angles, and polygon B has eight equal sides and eight equal angles. The longest diagonal of polygon A has the same length as the longest diagonal of polygon B.

Quantity A: The area of the six-sided polygonal region

Quantity B: The area of the eight-sided polygonal region

17. A certain monthly interest rate changed each month in 1999. For each month, the rate was either0.01 percentage point higher or 0.01 percentage point lower than that of the previous month. For January of 1999, the rate was 0.50 percentage point. How many of the five values 0.48, 0.49, 0.50, 0.51, and 0.52cannot be the rate, in percentage points, for April 1999 ?【微信公众号：张巍 GRE】

A. None

B. One

C. Two

D. Three

E. Four

18. At the beginning of 2009, Kim bought 100 shares of stock in a company at a price of \$31 per share. During the year, the price of the stock changed daily; the highest price was \$35 per share and the lowest price was\$25 per share. If Kim sold all 100 shares of stock on the same day in 2009, which of the following could beher percent profit or percent loss on the sale? Indicate all such percent profits or percent losses.

A. 15% loss

B. 10% loss

C. 10% profit

D. 15% profit

19. A sales representative’s monthly earnings consist of a base salary of \$3,500 plus a commission equal to 5 percent of monthly sales in excess of \$50,000. If monthly sales total s dollars and s> 50,000, what are the sales representative's monthly earnings in terms of s, in dollars?

20. Of the 29 students in a class, 15 students study French, fewer than 16 students study Spanish, and5 students study neither French nor Spanish, Which of the following could be the number of students in the class who study both French and Spanish?
Indicate all such numbers.【微信公众号：张巍 GRE】

A．0　　B. 3　　C. 5　　D. 6　　E. 8　　F. 9

新 Section 14 hard

1. A customer ordered 391 identical handbags and must choose a shipping plan. The shipper will use any combination of containers of sizes that each hold up to 20 handbags, up to 12 handbags, and up to 5 handbags. The shipping costs for these container sizes are $3, $2, and $1, respectively. At most one of the containers will be shipped holding less than the maximum number of handbags for that container, For the total shipping cost, the cost of using only the containers that hold up to 5 handbags is how much greater than the least possible cost?

2. A hexagon with sides of equal length and interior angles of equal measure is inscribed in a circle. If the perimeter of the hexagon is 12, what is the perimeter of an equilateral triangle inscribed in the same circle?【微信公众号：张巍 GRE】

3. List K consists of 9 positive integers. The median of the integers in K is 6, and the average (arithmetic mean) of the integers is 8. Which of the following could be the greatest integer in K?

A. 8　B. 11　C. 20　D. 32　E. 46

4. Jack and Marie each have a collection of United States postage stamps. Of all the various United States postage stamps issued from 1900 to 2000, 60 percent are in Jack's collection and 75 percent are in Marie's collection. What is the greatest possible percent of all the various stamps issued from 1900 to 2000 that are not in either of the two collections?

A. 10%　B. 15%　C. 25%　D. 35%　E. 40%

5. In a group of 50 students, 15 students enjoy board games and 20 students enjoy video games. Which of the following could be the number of students who enjoy both board games and video games? Indicate all such numbers.

A. 0　B. 5　C. 10　D. 15　E. 20　F. 25　G. 30

6. In the xy plane, the vertices of an equilateral triangle are (0, 1), (4, 3), and (a, b). Which of the following statements individually provide(s) sufficient additional information to determine the vertex(a.b)? Indicate all such statements.

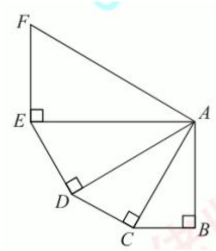
A. $2b-a>2$

B. $a<2$

C. $a^2+(b-1)^2=20$

20

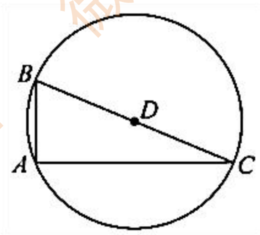
7. In the figure shown, the measure of angle BAC is 30 degrees. Triangles ABC, ACD, ADE, and AEF are similar. The area of triangle AEF is how many times the area of triangle ABC？



8. What is the number of positive integers n such that n < 165 and when 6^n is divided by 100, the remainder is 36 ?

A. 27 B. 32 C. 33 D. 40 E. 41

9. In the figure shown, D is the center of the circle whose area is 169 π . If the area of triangle ABC is 120， what is the perimeter of ABC ? 【微信公众号：张巍 GRE】



10. k is randomly selected from the set of integers from 1 to 100, inclusive.

Quantity A：The remainder when (k)(k+1)(k+2)(k+3)(k+4)is divided by 20

Quantity B：1

11. For all positive integers n , the function f is defined by the equation $f(n)=\frac{n(n+1)}{2}$ m is a positive integer.

Quantity A：$(-1)^{f(4m+1)}$

Quantity B：$(-1)^{f(4m+2)}$

12. If a triangle and a square have exactly n points in common, which of the following CANNOT be the value of n?

A. 2　B. 4　C. 5　D. 6　E. 8

13. In a list of 25 different numbers, the average (arithmetic mean) of all the numbers greater than the median is 200, and the average of all the numbers less than the median is 100. If the average of the 25 numbers is an integer, which of the following could be the value of the median?

Indicate all such values.

A. 120　B. 125　C. 140　D. 145　E. 160　F. 175　G. 180

14. Let a, b, c, and d be numbers such that $\frac{d}{a}=\frac{1}{2}$ and $\frac{c}{b}=\frac{2}{5}$. In the xy-plane, if the slope of the line with equation ax + by= 1 is $\frac{2}{3}$, what is the slope of the line with equation cx+dy=-1

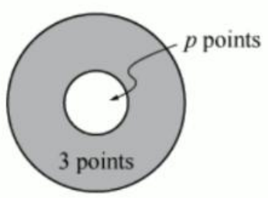
A.$-\frac{3}{2}$　B. $-\frac{2}{3}$　C. $\frac{3}{5}$　D.$\frac{3}{2}$　E. $\frac{6}{5}$

15. If -8≤h≤10 and h+ m = -4, what is the least possible value of mh ?【微信公众号：张巍 GRE】

A. 0　B. -32　C. -56　D. -112　E. -140

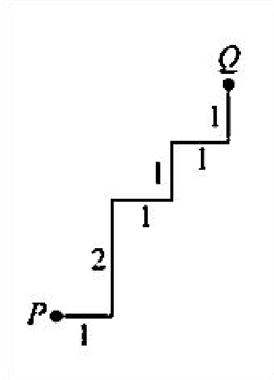
16. The figure above shows a circular dartboard at which players throw darts. A player earns 3 points each time a dart hits the shaded region and p points each time a dart hits the unshaded circular region. If a player ears a total of 47 points for a number of darts that hit the dartboard, which of the following could be the value of p ?

Indicate all such values.



A. 11　B. 12　C. 13

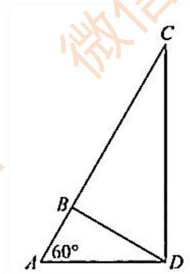
17. All angles in the figure above are right angles. What is the straight-line distance from point P to point Q?



A. $\sqrt{5}$ B. $\sqrt{5}+\sqrt{2}+1$ C. $\sqrt{5}+2\sqrt{2}$ D. 5 E. 7

18. Of the applicants for a job, 6 percent had a degree in computer science and spoke Russian. If 90 percent of the applicants who had a degree in computer science did not speak Russian, what fraction of all the applicants had a degree in computer science?

19. In the figure shown, angles ADC and ABD are right angles, If AB = 4, what is the length of AC?【微信公众号：张巍 GRE】



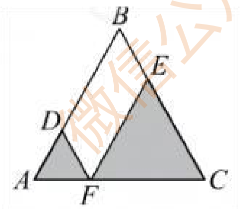
20. After taking a quiz, 25 students each received a score on the quiz that was an integer from I to 10inclusive. The average (arithmetic mean) of the 25 scores was 8.8. Only 1 student received the lowest score, and the sum of the 24 scores greater than the lowest score was 217. What was the range of the 25 scores?

新 Section 15 hard

1. In the xy -plane, three of the four vertices of a parallelogram are the points (6, 8), (12, 8), and (12, 4). Which of the following points could be the fourth vertex of the parallelogram?
Indicate all such points.

A. （6,4） B.（6,12） C.（5,3） D.（3,5） E.（12,12）

2. Three of the vertices of parallelogram RSTU in the xy-plane are R(-3,-2), S(1.-5), and T(9, 1).
Which of the following could be the x-coordinate of vertex U?
A. 3 B. 4 C. 5 D. 6 E. 7

3.


The sum of the areas of the shaded regions is what fraction of the area of triangle ABC above?

AD $=\frac{1}{3}AB, AF=\frac{1}{3}AC, BE=\frac{1}{3}BC$

4. A company has only two kinds of vehicles: cars and trucks. If 24 percent of the company's vehicles are cars that have automatic transmission and 60 percent of the company's cars have automatic transmission, what percent of the company's vehicles are cars?【微信公众号：张巍 GRE】

5. A scientist conducted an experiment and collected three measurements, Each measurement was an integer, The range of the three measurements was 2 and the least value was l. Which of the following values could be the average (arithmetic mean)of the measurements collected for the experiment?
Indicate all such values.

A. $\frac{4}{3}$ B. $\frac{5}{3}$ C. $\frac{7}{3}$ D. 2 E.3

6. In the ceiling of a room, an opening was cut in the shape of a square with sides that are

1 foot in length. A circular fixture will be placed over the opening. If the circular fixture covers the square opening completely which of the following could be the diameter of the circular fixture, in inches? (Note: 1 foot = 12 inches.)

A. 12　B. 13　C. 14　D. 15　E. 16　F. 17　G.18

7. The sequence a_1, a_2,a_3, ..., a_{200} is defined by a_n = n! for all integers n from 1 to 200. What is the units digit of the sum of the 200 integers in the sequence?

A. 1　B. 2　C. 3　D. 4　E.5

8. During each run of a computer simulation, either the letter X or the letter Y is displayed. For each run of the simulation, if the letter X is displayed, then the probability that X will be displayed in the next run is 0.3. Also for each run of the simulation, if the letter Y is displayed, then the probability that Y will be displayed in the next run is 0.4.

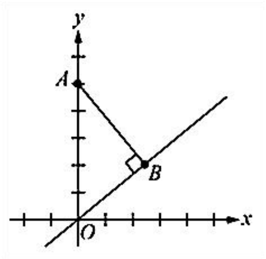
In 7 consecutive runs of the simulation, if X is displayed in the 5th run, what is the probability that X will be displayed in the 7th run?【微信公众号：张巍 GRE】

9. Last week a company manufactured N dolls that were to be packed into boxes for shipment, The company was able to pack a total of N-3 dolls into 5 boxes, each of which was packed with the same number of dolls.
Which of the following statements individually provide(s) sufficient additional information to determine the value of N?　Indicate all such statements.
A. N is a multiple of 3.
B. 6 dolls were packed into each box.
C. N is 27 greater than the number of dolls packed into each box.

10. If -1<x< 0< y< z< 1, which of the following must be true?
A. |x-z|>|y|
B. |x-z|<|y|
C. |x|<|y|+|z|
D. |x-y|>|z-y|
E. |x-y|<|z-y|

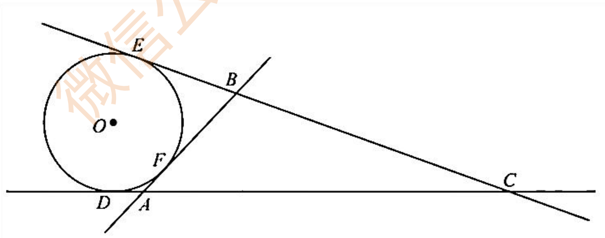
11. In the xy-plane shown, the x-axis represents an east-west road, the y-axis represents a north-south road, and the origin represents the intersection of the two roads. Point A and point B represent the centers of Town A and Town B , respectively, and the center of Town A is 5 kilometers north of the intersection. If the line through the origin and point B

has slope $\frac{3}{4}$ .which of the following is closest to the distance, in kilometers, between the centers of Town A and Town B ?【微信公众号：张巍 GRE】



A. 2 B. 2.5 C. 3 D. 3.5 E. 4

12. The circle shown has center 0 and a radius of 4. Lines AB, AC, and BC are tangent to the circle at points F, D, and E respectively. If the length of line segment CE is 23, what is the perimeter of triangle ABC ?



A. 38 B. 46 C. 49 D. 51 E. 54

13. A circle is inscribed in a regular octagon. If the perimeter of the octagon is 16, what is the circumference of the circle?

A. $2\sqrt{2}\pi$ B. $(2+\sqrt{2})\pi$ C.$3\sqrt{2}\pi$ D. $(2+2\sqrt{2})\pi$ E. $4\sqrt{2}\pi$

14. A bag contains only red marbles and blue marbles and contains at least two of each color.

Quantity A： The probability of drawing a red marble after an additional blue marble has been placed in the bag

Quantity B： The probability of drawing a red marble after one of the red marbles has been removed from the bag

15. List T consists of positive integers, and the range of the integers in T is 30. List Q consists of the integers in T and the integer 42, and the range of the integers in Q is 32. List R consists of the integers in T and the integer 45, and the range of the integers in R is

30, What is the greatest integer in T?

16. A telephone system has n telephone lines, For each of the n lines, the event that the line will fail during a certain reliability test has probability 0.3, and these n events are independent. If the probability that at least one of the n lines will not fail during the reliability test is greater than 0.99, what is the minimum value of n ?

A. 2 B. 3 C. 4 D. 5 E. 6

17. The average (arithmetic mean) and the median of the salaries of 5 employees in a certain company were equal until each of the 3 highest salaries was raised by $400 while the other 2 salaries remained the same. If X is the median and Y is the average(arithmetic mean) of the 5 salaries after the raises of the 3 salaries, then X-Y=

A. 100 B.160 C.200 D. 240 E.300

18. n is a positive integer.【微信公众号：张巍 GRE】

The remainder when 5n+2 is divided by 3 is 1.

The remainder when 5n+1 is divided by 4 is 1.

Quantity A: n

Quantity B: 3

19. n is the set of all numbers of the form $\frac{1}{n(n+1)}$, where n is a positive integer less than 4.

Quantity A: the sum of all numbers in S

Quantity B: $\frac{2}{3}$

20. List D：2，5，7，10，x，x+4

For which of the following values of x is the median of the six values in list D an integer? Indicate all such values.

A. 1 B. 2 C. 3 D. 6 E. 7 F. 8 G. 9

新 Section 16 medium

1. x and y are consecutive positive integers.
x<y

Quantity A：$(\frac{1}{x}-\frac{1}{y})^{-1}$

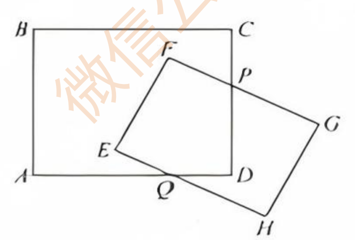
Quantity B：xy

2. -1<x<0

Quantity A：$(x+1)^2$

Quantity B：$\sqrt{x+1}$

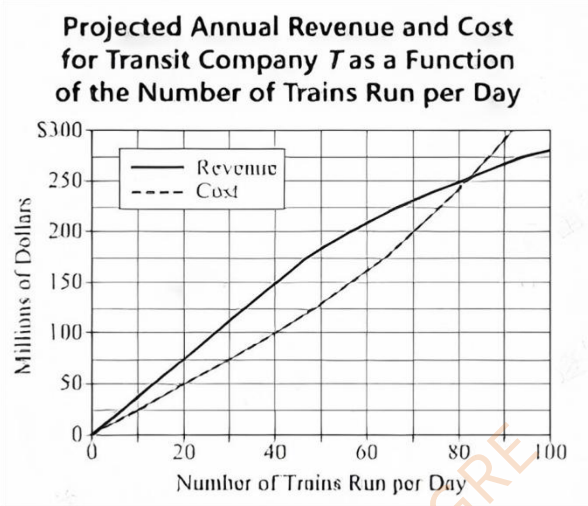
3.



In the figure shown, ABCD and EFGH are rectangles.
Quantity A：The sum of the measures of angle CPG and angle AQH
Quantity B：300°

4.In a certain sequence, each term after the 1st term is d less than the preceding term,where d is a positive number. If the 10th and 14th terms of the sequence are 37 and 31, respectively.

Quantity A：the 1st term
Quantity B：50



5. If Transit Company T will run 60 trains per day, then the projected annual revenue is approximately what percent greater than the projected annual cost?
A. 1%
B. 5%
C. 10%
D. 15%
E. 30%

6. If Transit Company T will run 40 trains per day for a year."approximately what is the projected average revenue per day?
A. $310,000
B. $360,000
C. $410,000
D. $460,000
E. $510,000

7. Transit Company T's projected annual cost consists of staffing costs, fuel costs, and equipment costs in the ratio of 1 to 2 to 5. If the company will run 50 trains per day.approximately what is the projected annual cost for fuel?【微信公众号：张巍 GRE】
A. $12 million
B. $17 million
C. $22 million
D. $27 million
E. $32 million

8. Theo owns $\frac{1}{3}$ of the number of pairs of shoes that Samantha owns, and Richard owns

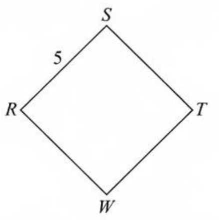
twice as many pairs as Samantha. If Richard owns fewer than 30 pairs of shoes, which of the following could be the number of pairs he owns?
Indicate all such numbers.
B. 29 B. 24 C. 18 D. 15 E. 12 F. 10 G. 9 H. 6

9. If x and y are integers such that x+y= 2k and x+ 2y= 2n +1 for some integers k and n,which of the following are possible values for x and y?
A. x=2 and y=4
B. x=3 and y=6
C. x=3 and y=7
D. x=4 and y=7
E. x=7 and y=8

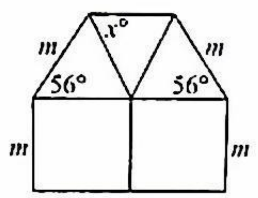
10. A farmer has 4x meters of fencing to enclose a rectangular vegetable garden. Enclosing a garden with which of the following dimensions, in meters, would provide the greatest area for the garden？ 【微信公众号：张巍 GRE】
F. x by x
G. (x-2) by (x+2)
H. 0.8x by 1.2x
I. 0.4x by 1.6x
J. It cannot be determined from the information given.

11. A box contains 100 purple balls, 100 red balls, and 100 white balls. What is the minimum number of balls that must be chosen to ensure that at least 4 of the balls chosen have the same color?
A. 5
B. 9
C. 10
D. 12
E. 13

12. Square RSTW is rotated 360 degrees around vertex S, where S is fixed. The path of vertex W is a circle,C,with radius r.
Quantity A： The area of the circular region enclosed by C
Quantity B： 50π



13.
The interior of the figure shown is divided into two squares and three triangles.



Quantity A： x
Quantity B： 56°

14.x=10^{10}
y=10^{-10}

Quantity A： $\frac{x+y}{x+2y}$

Quantity B： $\frac{1}{2}$

14. In the four quarters of 2013, denoted by Q1, Q2, Q3, and Q4, Company C hired the same number of employees in Q2 as in Q1 and twice as many employees in Q3 as in Q2. The number of employees hired by the company in Q4 was greater than the number hired in Q3; however, the number hired in Q4 was also less than 3 times the number hired in Q3. All of the employees were hired only once. If an employee is to be selected at random from all the employees hired during the four quarters, which of the following values could be the probability that the employee will be one who was hired in Q4 ?【微信公众号：张巍 GRE】

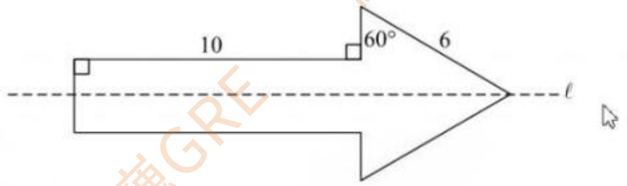
Indicate all such values.

A. 1/3 B. 3/8 C. 5/12 D. 1/2 E. 11/20 F. 3/5

16.0<a<b<c

Quantity A：The standard deviation of the seven numbers -c,-b,-a,0,a,b,c
Quantity B：The standard deviation of the seven numbers $-c^2,-b^2,-a^2,0,a^2,b^2,c^2$

17.



The arrow-shaped figure is symmetric with respect to line L .
Quantity A：The perimeter of the arrow-shaped figure
Quantity B：38

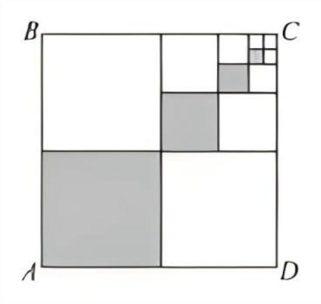
18.The median of the numbers in a list of consecutive positive multiples of 1.2 is 25.8. What
is the greatest possible number in the list?【微信公众号：张巍 GRE】
A. 50.4
B. 51.6
C. 52.8
D. 54.0
E. 55.2

19.

Machine	Minutes per Task
A	12
B	9
C	8

Three machines, A, B, and C, will work on three different kinds of tasks, respectively, over an 8-hour period. Each machine will work continuously for a constant time per task, as shown in the table, without stopping between consecutive tasks. If the three machines start working on their tasks simultaneously at the beginning of the 8-hour period, how many times during the 8-hour period will all three machines complete a task simultaneously?
A. 0　　B. 4　　C. 6　　D. 12　　E. 20

20.



The figure shown is the result of the following procedure. Square region ABCD is divided into 4 square regions, each with the same area, and the lower left square region of ABCD is shaded, The upper right square region of ABCD is then divided into 4 square regions, each with the same area, and the lower left square region of those 4 regions is shaded, These steps are repeated until a total of 4 square regions are shaded.What fraction of square region ABCD is shaded?

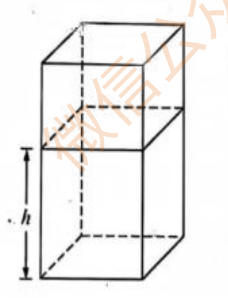
A. $\frac{1}{4}$ B. $\frac{5}{16}$ C. $\frac{5}{32}$ D. $\frac{21}{64}$ E. $\frac{85}{256}$

新 Section 17 hard

1. On a certain radio tower, a light flashes for 1 second once every 8 seconds, and a light on a certain television tower flashes for 1 second once every 6 seconds. if the radio tower light flashes at exactly 4:00:00 em, and the television tower light flashes exactly 4 seconds later, how many times will the two lights flash at the same time between 4:00:00 PM and 4:03:37 PM?

A. 0 B. 6 C. 9 D. 24 E. 36

2. The figure represents the interior of a rectangular tank with a volume of 175 cubic feet and a base area of 17.5 square feet, The tank contains 105 cubic feet of water, which fills the tank to a level of h feet above the bottom. The water level in the tank is to be raised 2.4 feet by adding water to the tank. The volume of additional water will be what fraction of the total volume of water in the tank after the water is added?【微信公众号：张巍 GRE】



A. 3/4
B. 3/5
C. 2/7
D. 2/5
E. 1/3

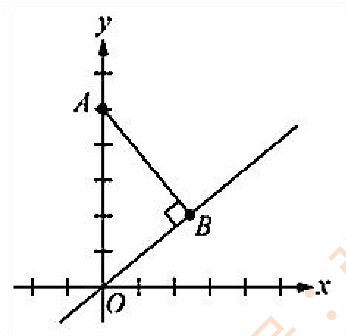
3. If twice the sum of the positive integers x, y, and z is divided by 7, the remainder is 1. What is the remainder if the sum of x, y, and z is divided by 7?

4. Each of the nine digits 1, 2, 3, 4, 5, 6, 7, 8, and 9 is marked on a separate slip of paper.and the nine slips are placed in a box, Three slips of paper will be randomly selected with replacement, and in the order selected the digits will be used to form a 3-digit number.

Quantity A:The probability that the 3-digit number will be greater than 600

Quantity B: 4/9

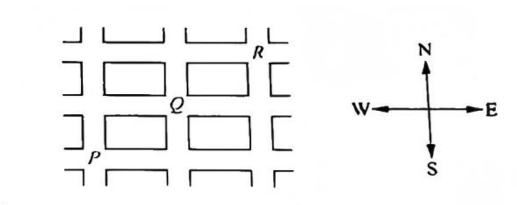
5.In the xy-plane shown, the x-axis represents an east-west road, the y-axis represents a north-south road, and the origin represents the intersection of the two roads. Point A and point B represent the centers of Town A and Town B , respectively, and the center of Town A is 5 kilometers north of the intersection. If the line through the origin and point B has slope $\frac{3}{4}$ .which of the following is closest to the distance, in kilometers, between the centers of Town A and Town B ?【微信公众号：张巍 GRE】



B. 2　　B.2.5　　C.3　　D.3.5　　E.4

6. j and k are integers such that 51 ≤ j< k ≤ 99.
Quantity A：The number of combinations of 100 objects taken j at a time
Quantity B：The number of combinations of 100 objects taken k at a time

7.



Each morning Betty walks from intersection P to intersection R, always walking along streets shown in the figure above and always going east or north. if she varies her walks as much as possible, taking every different path in turn before repeating the sequence of paths on how many mornings in 48 consecutive mornings will Betty walk through intersection Q?

A. 12
B. 16
C. 24
D. 32
E. 36

8.The length of each side of pentagon ABCDE is an integer, and the perimeter is 35. The length of diagonal AC is 10, the length of diagonal AD is 11, and the length of side AE is 12. What is the greatest possible length of side DE ?

A. 9　　B.10　　C.11　　D.12　　E.13

9.Five boxes contain 4, 7, 12, 14, and 20 building blocks, respectively. Each of three children will select one of the five boxes. If all the blocks in the three selected boxes can be distributed equally among the three children, what is the greatest possible number of blocks that each child could receive?

A. 10　　B.11　　C.13　　D.14　　E.15

10.N is a 3-digit integer with tens digit t and units digit u. If 621 < N < 685 and N is a multiple of 3，which of the following must also be a multiple of 3？

A. t+u　　B. t-u　　C. tu　　D. t^u　　E. u^t

11.n=1!+2!+3!+4!+5!+6!+7!+8!+9!+ 10!

What is the remainder when the integer n is divided by 20?

A. 19　　B.18　　C.13　　D.12　　E.9

12.In the xy -plane, if three of the vertices of a parallelogram have coordinates (2,-1), (1,2) and (5, 1), which of the following could be the coordinates of the fourth vertex?

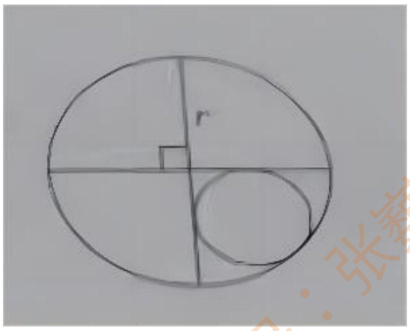
Indicate all such coordinates.【微信公众号：张巍 GRE】

A. (6,-2)

B. (-2,0)

C. (4,4)

13.In the figure, a circle with radius r is divided into four sectors by two diameters that are perpendicular. A circle with radius x is inscribed in one of the sectors.

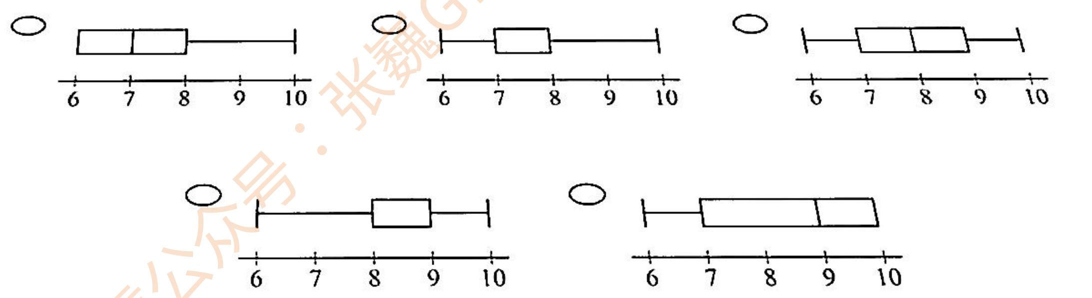


Quantity A: x

Quantity B： r/2

14.The table shows the relative frequency distribution of the numbers in data set B. Which of the following boxplots is a summary of the numbers in data set B ?

Number	Relative Frequency
6	0.1
7	0.2
8	0.1
9	0.2
10	0.4



15.A team of two students from an art school is to be selected to represent the school at a national event, The two students will be selected from the students in three classes that have no students in common. The three classes have 10 students, 8 students, and7 students. If the two students must be selected from different classes, how many teams are possible?

A. 168

B. 206

C. 210

D. 240

E. 560

16.There are 10 packages of candy on a shelf, 4 that are filled with blue candies and 6 that are filled with red candies, If 3 packages are to be chosen at random from the shelf without replacement, what is the probability that at least one of the 3 packages will be filled with blue candies? 【微信公众号：张巍 GRE】

A. $\frac{11}{12}$　　B. $\frac{5}{6}$　　C. $\frac{3}{4}$　　D. $\frac{2}{3}$　　E. $\frac{7}{12}$

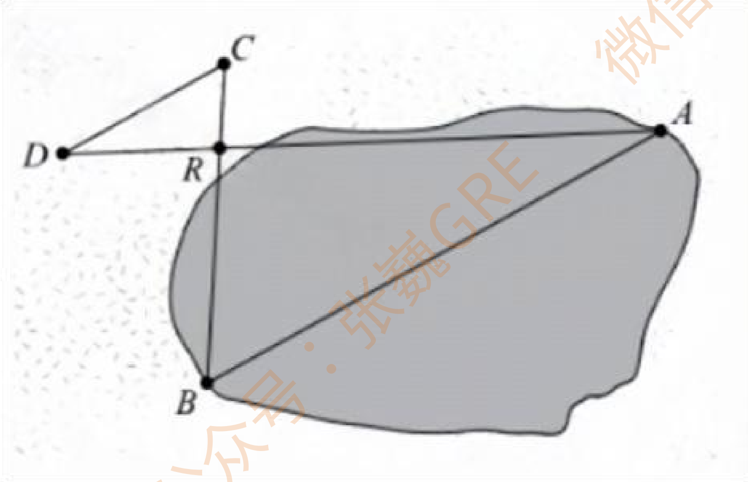
17.What is the ratio of the area of a square region with diagonal 10 to the area of a square region with diagonal 20?

A. 1:4 B. 1:2 C. 1:√2 D. 2:1 E. 4:1

18.Of the 70 students who took a typing test, 48 made 5 or more errors and 38 made 10 or fewer errors.

Quantity A：The number of students who made from 5 to 10 errors

Quantity B：16



19.The figure above represents a pond and the nearby land that surrounds it. Lucia plans to measure the distance across the pond from point A to point B , First, she will measure the distance from a rock on land at point R to point D on line AR, Next, she will measure the distance, along a line parallel to line AB , from point D to point C, which lies on line BR. Of the following, which additional measurement will be sufficient to determine the distance from A to 8 ?【微信公众号：张巍 GRE】

A. The distance from 4 to R

B. The distance from B to R

C. The distance from C to R

D. The measure of angle ARB

E. The measure of angle DAB

20.S= {1, 2,3,4,6}

T={1, 2,3, 6,8}

From set S , an integer is chosen and called s, and from set T an integer is chosen and called t. The product of the two integers s and t is called p. What is the total number of different values of p that can be determined in this way?

A.5 B.9 C.14 D.18 E.25

新 Section 18 hard

1.Statement: If a < x< b and a < y < b, where a and b are constants, then a < xy < b.
The statement shown is true for which of the following values of a and b?
A. a=-2 and b=-1
B. a=-1 and b=0
C. a=-0.5 and b=0.5
D. a=0.5 and b=1
E. a=1 and b=2

2.a b c d e f g h i j
Let S be the set of all 10-letter sequences consisting of the 10 letters above.
Let T be the set of all 8-letter sequences consisting of the first 8 letters above.
Quantity A：The number of sequences in S in which f, b, h, and g appear consecutively in that order
Quantity B：The number of sequences in T in which f and b appear consecutively in that order

3.For a group of 9 steel beams stored together, the average (arithmetic mean) length of the beams is 7.2 meters and the median length is 8.4 meters. Two additional steel beams-one that is longer than all 9 beams and one that is shorter than all 9 beams-will be stored with the 9 beams. For the combined group of 11 steel beams the average length is m meters and the median length is d meters. Which of the following statements must be true?【微信公众号：张巍 GRE】
Indicate all such statements.
A. m<7.2 B. d<8.4 C. m=7.2 D. d=8.4 E. m>7
F. d>8.4

4.If 0 < |x| - 2x < 3, which of the following is true?
A. x<-1 B. x=-1 C. -1<x<0 D. x=0 E. 0<x<1

5.A certain number of toys were packed into x boxes so that each box contained the same number of toys, with no toys left unpacked. If 3 fewer boxes had been used instead, then 12 toys would have been packed in each box, with 5 toys left unpacked. What is the value of x?
A. 11 B. 14 C. 28 D. 31 E. 34

6.In a local election there are 2 candidates for mayor, 4 candidates for sheriff, and 5 candidates for dogcatcher on the ballot, in each of the three categories a voter may vote for exactly one candidate or none. How many different ways can a voter fill out the ballot?

A. 11 B. 14 C. 40 D. 80 E. 90

7.In a distribution of the heights of 50,000 people, the 40th percentile is 63 inches and the 64th percentile is 69 inches.

Quantity A：The 52nd percentile in the distribution

Quantity B：66 inches

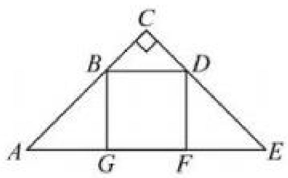
8.The width and the length of a rectangular piece of plywood are 4 feet and 8 feet,respectively. Along one edge of the plywood, a strip x inches wide and 8 feet long is removed. Then, along an edge perpendicular to the 8-foot edge, a strip x inches wide is removed. For what value of x will the remaining rectangular piece have width and length in the ratio of 2 to5?(1 foot=12 inches)

A. 4 B. 8 C. 12 D. 16 E. 20

9.Square GBDF is inscribed in the isosceles right triangle ACE.

Quantity A：The area of triangular region ABG

Quantity B：$\frac{1}{4}$ of the area of triangular region ACE



10.A certain firm has N female and male employees and 70 percent of them are lawyers. If 55 percent of the N employees are females and 60 percent of the females are lawyers,what percent of the N employees are male lawyers?【微信公众号：张巍 GRE】

A. 45%

B. 37%

C. 33%

D. 22%

E. 15%

11.Set S consists of 9 different integers whose sum is 116. The sum of the 5 least integers in S is 38, and the average (arithmetic mean) of the 5 greatest integers in S is 18.

Quantity A：The median of the integers in S

Quantity B：12

12.In the xy-plane, how many points P (x, y) have coordinates x and y that are integers and that satisfy the inequalities $2 < x < 8$ and $-2 < y < 7$?

13.A certain data set consists of n values, and the mean of the values is positive. If an additional value equal to 70 percent of the mean is included in the data set to create a new data set, then the mean of the values in the new data set will be 2 percent less than the mean of the values in the original data set. What is the value of n ?

A. 14
B. 15
C. 16
D. 34
E. 35

14.The positive integer c has a prime factor that is not a factor of 60, and $\frac{c}{450}$ is a prime number less than 100.

Quantity A：c
Quantity B：3000

15.In a poll of 200 people who answered either yes or no to each of two questions,170 people answered yes to the first question and 10 people answered yes to the second question. Which of the following could be the number of people polled who answered no to both questions?【微信公众号：张巍 GRE】

A. 15　B. 18　C. 20　D. 26　E. 32

16.The degree of a polynomial in x is the greatest exponent of x when the polynomial is written as a sum of terms such as a_1x^i, where i is a positive integer and a_1 is a number, For example, the degree of $9x^3 + 3x^5 - 100x$ is 5， What is the degree of the polynomial $(x+4x^3)^2(x^5-7)$?

A. 10　B. 11　C. 12　D. 13　E. 14

17.A rectangular piece of sheet metal has a width of w centimeters, where w≥30. The length of the sheet metal is 3.5 times its width. If squares with sides of length 10 centimeters are removed from each corner of the sheet metal and the sides are folded up to make an open box, what is the volume of the box?

A. (w-20)(35w-200)

B. (w-10)(35w-100)
C. $35w^2$
D. $35(w-10)^2$
E. $1,000w^3$

18.In the xy-plane, which of the following points are in the triangular region with vertices (0, 0), (3. 0), and (3, 2)？【微信公众号：张巍 GRE】
Indicate all such points.
A. (0.5, 0.2)
B. (1, 1)
C. (1.5, 0.5)
D. (2, 1)
E. (2, 3)
F. (2.5, 2)

19.There are 8 marbles in a box, and each marble is either green or yellow. One marble is to be chosen at random from the box, without replacement, and then another marbleis to be chosen at random from the box. If the first marble chosen is green, then the probability that the second marble chosen will be green is $\frac{2}{7}$

Quantity A：The probability that the second marble chosen will be yellow given that the first marble chosen is yellow

Quantity B：$\frac{4}{7}$

20.There are two overlapping circular regions on a level field, and the centers of the regions are 40 meters apart. One circular region has a radius of 25 meters, and within the region there is a marker located 5 meters away from its border, The other circular region has a radius of 30 meters, and within the region there is a marker located 7 meters away from its border. What is the greatest possible distance, in meters, between the two markers?

新 Section 19 hard

1.

	Data Set	
	X	*Y*
Mean	100	200
Range	100	100

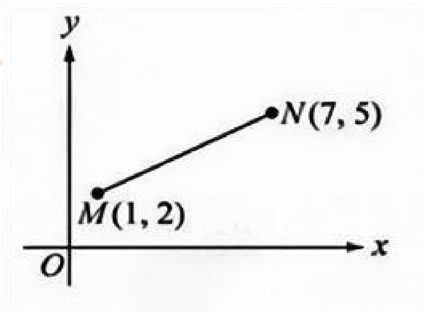
The table shows the means and ranges of two data sets, I and Y, each containing the same number of measurements.

Quantity A：The standard deviation of data set X

Quantity B：The standard deviation of data set Y

2.A cube has its faces numbered from 1 to 6. The cube is weighted in such a way that when it is rolled, the probability that X will appear on the top face is equal to k/X, where k is a constant. What is the value of k?【微信公众号：张巍 GRE】

3.



In the xy-plane above, MN is a line segment that contains point P (not shown). If the length of MP is twice the length of PN, what are the coordinates of P?

A. (3,3)　　B. (3,4)　　C. (4,2)　　D.(5,3)　　E. (5,4)

4.The average (arithmetic mean) of the four numbers a, b, c, and d is equal to m, and the average of the four numbers w, x,y, and z is equal to n, The standard deviation of the four numbers a + w, b +x,c +y, and d+z is equal to 0.

Quantity A：a+w

Quantity B：m+n

5.Set S={2, 4, 6) Set T={2, 4, 6, 8, 10, 12}

How many different sets M, other than S and T, can be formed such that S is a subset of

M and M is a subset of T?

A. Three B. Four C. Six D. Nine E. Twelve

6.For the values in data set A, the first quartile is 20 and the inter quartile range is 32. For the values in data set B, the third quartile is 40 and the inter quartile range is 8.

Quantity A：The median of the values in A

Quantity B：The median of the values in B

7.In a group of people, 30 people speak French, 40 speak Spanish, and 1/2 of the people who speak Spanish do not speak French. If each person in the group speaks French, Spanish, or both, which of the following statements are true?【微信公众号：张巍 GRE】

Indicate all such statements.

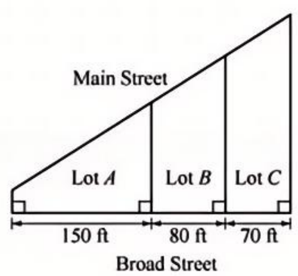
A. Of the people in the group, 20 speak both French and Spanish.

B. Of the people in the group, 10 speak French but do not speak Spanish.

C. of the people in the group, 1/5 speak French but do not speak Spanish.

8.The pieces of art in a certain collection are classified as either ancient or modern. All pieces in the collection are on display in one of two wings of a museum, C or D. Of the pieces in the collection on display in C, 1/3 are ancient pieces, and of the pieces in the collection on display in D, 1/4 are ancient pieces. The number of modern pieces on display in C is equal to the number of modern pieces on display in D. Of all the ancient pieces in the collection, what fraction are on display in wing C?

9.A plot of land is to be divided into three smaller lots, as shown in the figure above. If the total length of the sides of the three lots along Main Street is 360 feet, what is the length, in feet, of the side of lot A that is along Main Street?

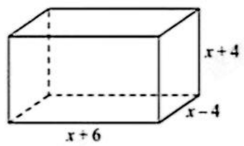


A. 160 B.170 C.180 D.190 E.200

10.List A contains 11 different positive integers that are less than or equal to 20. List B contains all the integers in A except for the greatest integer. The average (arithmetic mean) of the integers in A is x, and the average of the integers in B is y.

Quantity A：x
Quantity B：y+1

11.The figure shows the width, length, and height of a rectangular solid. Which of the following statements must be true?【微信公众号：张巍 GRE】
Indicate all such statements.



A. The polynomial representing the surface area of the solid has degree of 3.
B. The polynomial representing the volume of the solid has degree of 3.
C. The polynomial representing the perimeter of the base of the solid has degree of 2.

12.In the xy-plane, the point (2, 5) is a vertex of a square that has sides of length 3. Which of the following points could be the vertex of the square that is diagonally opposite the vertex (2, 5)
Indicate all such points.
A. (-1, 2) B. (-1, 8) C. (2, 8) D. (5, 8)

13.If an integer is chosen at random from the integers between 101 and 550, inclusive, what is the probability that the chosen integer will begin with the digit 1, 2, or 3, and end with the digit 4, 5 or 6?
A. 0.02
B. 0.05
C. 0.10
D. 0.15
E. 0.20

14.List D consists of 10 consecutive integers, including integers x and y. Of the integers in list D, 40 percent are greater than y, and 10 percent are less than x.
Quantity A：y-x
Quantity B：5

15.Carlos owns twice as many books as Surya and one-third as many books as Ian. Ian owns 10 more books than Emma.

Which of the following statements individually provide(s) sufficient additional information to determine how many books Carlos owns?

Indicate all such statements.

A. Emma owns 5 times as many books as Surya.

B. Emma owns 50 books.

C. Emma owns 30 more books than Carlos.

16.Sequence S:a_1, a_2, a_3, ...,a_n....

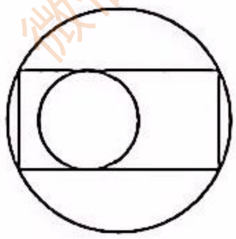
In sequence S, a_1 is an integer and $a_n = 2a_{n-1}$ for all integers n greater than 1. If no term of sequence s is a multiple of 100, which of the following could be the value of a_1?

Indicate all such values.

A. 5 B. 10 C. 20 D.25 E. 30 F. 40 G. 50

17.The figure below shows a rectangle inscribed in a large circle. inside the rectangle is a small circle of radius 2 that is tangent to two sides of the rectangle. If the length of the rectangle is twice its width, what is the area of the large circle?【微信公众号：张巍 GRE】

A. $12\pi\sqrt{3}$　B. $16\pi\sqrt{2}$　C. 16π　D. 18π　E. 20π



18.A child-care center has a supply of bicycles, each with 2 wheels, and tricycles, each with 3 wheels, for the children to use. The number of bicycles in the supply is odd, and the total number of wheels on all the bicycles and tricycles is 31. Which of the following could be the total number of bicycles and tricycles in the supply?

Indicate all such numbers.

A. 11

B. 12

C. 13

D. 14

E. 15

19.When positive integers k and n are each divided by 9, the remainders are 2 and 5, respectively. If k > n, what is the remainder when k-n is divided by 9?【微信公众号：张巍

GRE】

A. 2 B. 3 C. 4 D. 5 E. 6

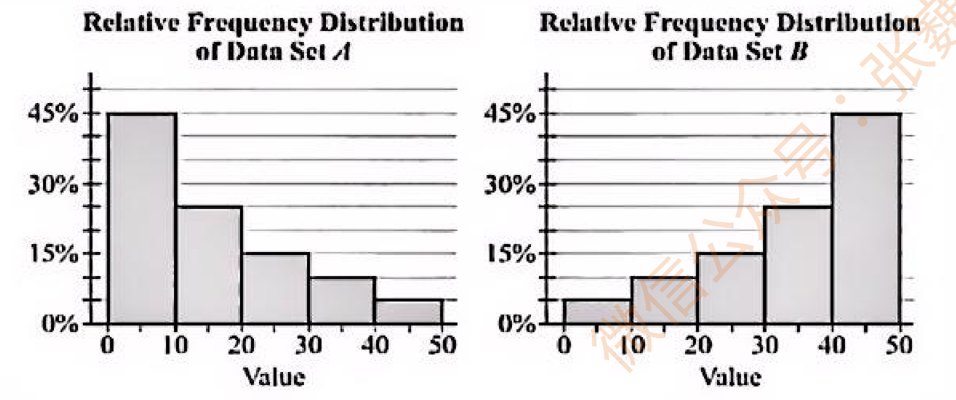
20.In order to use a certain automatic bank machine, customers are given a four-digit access number. A digit can have any integer value from 0 to 9, inclusive. How many different access numbers are there such that the first digit is 7,one of the other 3 digits is 2, and none of the digits is repeated?

A. 56 B. 72 C. 112 D. 168 E. 216

新 Section 20 hard

1. The relative frequency distributions of the values in data set A and data set B are shown in two histograms where each interval shown includes its left endpoint and excludes its right endpoint, The number of values in A is 2 times the number of values in B, Data set C consists of the values in A and the values in B. Which of the following could be the median of the values in ? 【微信公众号：张巍 GRE】

A. 5
B. 15
C. 25
D. 35
E. 45



2.

Number of colleges visited	0	1	2	3	4 or more
Number of students	3	x	24	32	7

The frequency distribution above shows the number of colleges visited by a sample of students. If the median of the numbers of colleges visited is 3, which of the following could be the value of x?

Indicate all such values.

A. 0 B. 6 C. 9 D. 15 E. 24

3.

1，2，10，74...

In the sequence shown, each term after the first term is 6 less than 8 times the preceding term.

Quantity A：The number of terms among the first 100 terms in the sequence that are multiples of 3

Quantity B：30

4. There are 25 students in Ms. Ware's class-15 of these students take French, 12 of these students take Spanish, and 8 of these students take Latin. If all 25 students in Ms. Ware's class take at least one of these three language courses, which of the following is the greatest possible number of students in the class who could be taking all three language courses?【微信公众号：张巍 GRE】

A. 13 B. 10 C. 7 D. 5 E. 3

5. A farmer will use fencing to enclose a rectangular pen that is located on a flat piece of

land. The pen will be divided into two adjacent rectangular regions, and an additional piece of fencing will be used as the common side of the two regions. If the areas of the two regions are l0 square meters and 20 square meters and the length of the common side is 5 meters, what is the total length of fencing, in meters, that will be used to enclose and divide the pen?

A. 32 B. 27 C. 22 D. 20 E. 17

6. Three people are in the process of reviewing 50 grant proposals. The numbers of proposals that each has reviewed so far are 32, 28, and 25, respectively.

Quantity A：The number of proposals that have been reviewed by at least 2 reviewers

Quantity B：32

7. The length of each side of regular pentagon P is twice the length of each side of regular pentagon Q.

Quantity A：One-third of the area of pentagon P

Quantity B：The area of pentagon Q

8. A survey of 500 households found that 450 households had a microwave oven, 300 had a food processor, and 150 had a trash compactor. Each of the 500 households had at least one of the appliances, and more than 100 households had all three appliances. Which of the following could be the number of households that had a food processor and a trash compactor but no microwave oven?

indicate all such numbers.

A. 5 B. 35 C. 50 D. 65 E. 100

9. The ratio of the number of males to the number of females in a mathematics class is 4 to 5. The corresponding ratio in an English class is 5 to 4. There are 50 percent more males in the English class than there are in the mathematics class. What is the ratio of the number of females in the mathematics class to the number of females in the English class?

A. 2 to 3

B. 24 to 25

C. 1 to 1

D. 25 to 24

E. 3 to 2

10.

Number of Books Read	Frequency
0	5
1	4
2	5
3	3
More than 3	3
Total	20

A town library held a reading drive, in which 20 elementary school students were asked to record the number of books they read over the summer. Only whole numbers of books were recorded. The table above shows the results of the reading drive. The average (arithmetic mean) number of books read by the students was 2 books per student.

Which of the following statements individually provide(s) sufficient additional information to conclude that only one student read only 5 books?
Indicate all such statements.
A. Two students read only 6 books each.
B. No student read more than 6 books.
C. No student read only 4 books.

11. The sequence of numbers $a_1, a_2, a_3 \ldots a_n \ldots$ is defined by $a_1 = 3$ and $a_n=1+2a_{n-1}$ for all integers $n \geq 2$. What is the ratio of a_5 to a_3?
Give your answer as a fraction.

12.
Quantity A: The number of ordered triples (x_1, x_2, x_3), where x_1, x_2, x_3 are non-negative integers such that $x_1+x_2+x_3=9$

Quantity B：55

13.
For each of 5 coins, the probability that the coin will land heads up when it is flipped is 1/2，The 5 coins will be flipped simultaneously. Which of the following statements are true? indicate all such statements.【微信公众号：张巍 GRE】
A. The probability that only 1 of the coins will land heads up is less than 1/2.
B. The probability that only 2 of the coins will land heads up is greater than the probability that only 1 of the coins will land heads up.

C. The probability that only 3 of the coins will land heads up is greater than the probability that only 2 of the coins will land heads up.

14.
$0<a<b<c$
Quantity A：The standard deviation of the seven numbers $-c, -b, -a, 0, a, b, c$
Quantity B：The standard deviation of the seven numbers $-c^2, -b^2, -a^2, 0, a^2, b^2, c^2$

15. a_1，a_2，a_3，a_4 ......
In the sequence of numbers shown, $a_1= 13^3$ and $a_n= a_{n-1}+ 13^3$ for each integer n greater than 1. If $a_k= 13^4$, what is the value of k ?

16. In the xy -plane, the graph of $y = ax+ b$, where $ab > 0$, has the y-intercept “s” and the x -intercept “t”.
Quantity A：s+t
Quantity B：b

17. Sets R, S, and T are nonempty, and each has the same number of elements. The sets R∩S, R∩T, and S∩T are nonempty, and each has the same number of elements. The set R∩S∩T is empty. Set V consists of all the elements in R that are not in S.
Quantity A：3 times the number of elements in V
Quantity B：The number of elements in the set R∪S∪T

18. A rectangular coordinate grid was created to represent the location of three campsites in a certain flat region.The centers of the three campsites are located at the points (0, 0), (0, 600), and (450, 300) on the arid, where 1unit on each axis represents 1 meter. The center of a water fountain in the region is equidistant from the centers of the three campsites. What point on the arid represents the location of the center of the water fountain?【微信公众号：张巍 GRE】
A. (125, 300)
B. (150, 300)
C. (225, 150)
D. (225, 300)
E. (300, 125)

19.

6			3
	p	x	
	y	q	
11			14

In the table above, if the sum of the numbers on each main diagonal is equal to the sum of the numbers in the four corners, what is the value of (p+ q) (x+y)?

A. 6 B. 0 C. -6 D. 16

F. It cannot be determined from the information given.

20. At a certain company, employees who earn \$20.00 per hour will be given an increase of \$1.00 per hour. For each of the other employees, either the employee will be given an increase of \$1.00 per hour or the employee will be given a percent increase equal to the percent increase that will be given to the employees who earn\$20.00 per hour, whichever results in a larger increase for that employee. Which of the following statements are true?

Indicate all such statements.

A. An employee who earns less than \$20.00 per hour will be given a percent increase that is greater than the percent increase that will be given to the employees who earn \$20.00 per hour.

B. An employee who earns \$22.00 per hour will be given an increase of \$1.10 per hour.

C. An employee who earns \$24.00 per hour will earn \$25.20 per hour after the increase.

新 section 21 hard

1. The manufacturing cost of a certain car model in 1995 was 10 percent greater than its manufacturing cost in 1990 and 10 percent less than its manufacturing cost in 2000.

Quantity A： The manufacturing cost of the car model in 2000 as a percent of its manufacturing cost in 1990

Quantity B： 120%

2. Five boxes each contain at least 1 ball. The mode of the numbers of balls in each of the boxes is 3. Which of the following statements must be true?【微信公众号：张巍 GRE】

Indicate all such statements.

A. The average (arithmetic mean) number of balls per box is 3.

B. The total number of balls in the five boxes is greater than 9.
C. At least one of the boxes contains fewer than 3 balls.

3. Which of the following numbers CANNOT be the value of $\frac{a}{|a|}+\frac{b}{|b|}+\frac{c}{|c|}$, where a, b, and c are nonzero

A. -3　　B.-1　　C.1　　D.2　　E.3

4. A furniture store purchased sofas for a wholesale price of $400 each and sold most of them for a retail price of $560 each. In a summer sale, the retail price of the last few sofas was reduced by x dollars, and they were sold at this reduced price. The store's profit from selling the last sofa was more than 15 percent of its wholesale price. Which of the following could be the value of x?

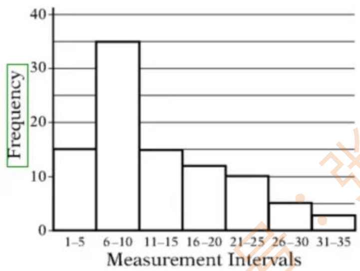
A. 50　　B. 75　　C.100　　D. 125

5. At a certain online bookstore, the price of each book is the same.The total price of a customer order is the sum of the prices of the books in the order plus a shipping fee that is constant regardless of the number of books purchased. Last year the average(arithmetic mean) number of books per customer order was 1.5, and the average of the total prices of the orders was $10.50.Which of the following amounts could be the book price and the shipping fee last year?

Indicate all such amounts.

A. $4.00 per book and $4.50 shipping fee
B. $4.50 per book and $3.50 shipping fee
C. $5.00 per book and $3.00 shipping fee.
D. $5.50 per book and $2.25 shipping fee

6. In the course of an experiment, 95 measurements were recorded, and all of the measurements were integers. The 95 measurements were then grouped into 7 measurement intervals. The graph above shows the frequency distribution of the 95 measurements by measurement interval.



Quantity A：The average (arithmetic mean) number of 95 measurements

Quantity B： The median number of 95 measurements

7. Seventy-five boxes of light bulbs were examined during a quality control check. The number of defective light bulbs per box and the corresponding frequency are given in the table.
Quantity A： The average (arithmetic mean) number of defective light bulbs per box for the75 boxes
Quantity B： The median number of defective light bulbs per box for the 75 boxes

Number of defective bulbs	0	1	2	3	4 or more
Frequency	35	20	12	2	6

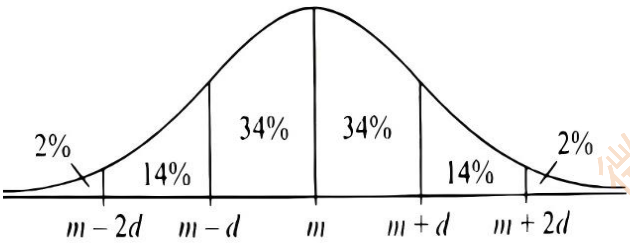
8. List S: 0， x， x^2
x is a positive number.
The median of the numbers in S is greater than the average (arithmetic mean) of the numbers in S.
Quantity A： x
Quantity B： 2

9. At a car dealership, Jason earned a total of N dollars in commissions for making 15 sales. If the commission on Jason's last sale had been \$210 greater, his average (arithmetic mean) commission on the 15 sales would have been x dollars greater.【微信公众号：张巍 GRE】
Quantity A： x
Quantity B： 16

10. A ball is dropped from a height of 3 meters and bounces repeatedly. After each bounce, it reaches a height that is 65 percent of its previous height, if the height of the ball after the nth bounce is less than 20 centimeters, which of the following could be the value of n?
Indicate all such values.
A. 4 B. 5 C. 6 D. 7 E. 8

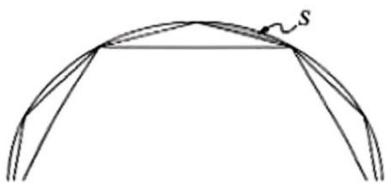
11. The figure shows a normal distribution with mean m and standard deviation d, including approximate percents of the distribution corresponding to the six regions shown.The continuous random variable D is normally distributed. If the value 75 is at the

98th percentile of the distribution of D and the value 66 is at the 16th percentile of the distribution of D, which of the following is the best estimate of the mean of the distribution?【微信公众号：张巍 GRE】



A. 68 B. 69 C. 70 D. 71 E. 72

12.

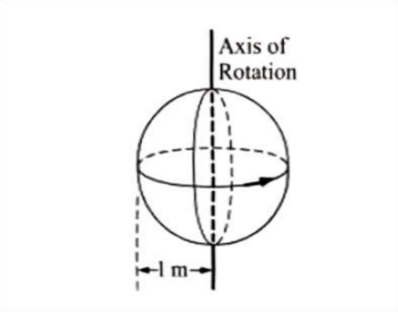


The figure shows part of a circle with two inscribed regular polygons—one with 6 sides and one with 12 sides. The two polygons have 6 vertices in common. The radius of the circle is r, and the area of the 6-sided polygon is x.

Quantity A	Quantity B
$\pi r^2 - x$	24 times the area of shaded region S

13. Two lists, each consisting of 6 consecutive integers, have exactly 2 integers in common. The sum of the integers in one list is x, and the sum of the integers in the other list is y.
Quantity A：|x-y|
Quantity B：24

14. Two seating arrangements of people around a circular table are called the same if, for each seated person, the same person is seated to that person's immediate right in both arrangements.
Quantity A：The number of different seating arrangements of 4 people around a circular table
Quantity B：8



15. A sphere, as shown above, with a radius of 1 meter is rotating 360 degrees per second about its vertical axis. The speed, in meters per second, of a point on the sphere due to this rotation is in the range from

A. 0 to1
B. 0 to π
C. 0 to 2π
D. 1 to π
E. 1to 2π

16. Which of the following is greatest?【微信公众号：张巍 GRE】

A. $3^{100}+2^{100}$
B. $3^{10}+2^{10}$
C. 30^2+10^2
D. $(3+2)^{100}$
E. $(30+2)^{10}$

17. The random variable W is normally distributed with P(W< k) < 0.6 and P(W > n) < 0.1,where k and n are constants.

Quantity A：P(k<W<n)
Quantity B：0.3

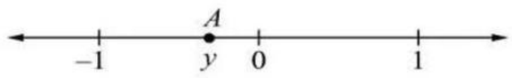
18. List A contains a certain number of consecutive integers including 2. The sum of all the
integers in List A is 11.

Quantity A：the number of integers in list A
Quantity B：10

19. On the number line, point A has coordinate y and point B (not shown) has coordinate x.The distance between A and B is equal to y^2.

Quantity A：x

Quantity B：0



20. The vehicles of Company W are numbered consecutively from 1 to 650. The vehicles with a number that ends with one of the digits 1, 2, 3, 4, or 5 are used by Division 1. Vehicles with a number between 130 and 389, inclusive, are trucks. What percent of the company vehicles are trucks used by Division 1? 【微信公众号：张巍 GRE】

A. 10% B. 20% C.30% D. 40% E. 50%

新 section 22 hard

1. Jim exercised last Monday at 6:00 in the evening and then exercised every third day after that. For example,after last Monday,he exercised the following Thursday and then the Sunday following that.How many days after last Monday will be the next Monday he exercises?【微信公众号：张巍 GRE】

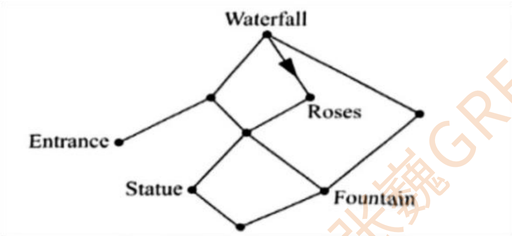
2. n is an integer greater than 1,000.
Quantity A：The number of prime numbers that are greater than n and less than n + 15
Quantity B：6

3. Pat has 7 identical pieces of paper, one or more of the pieces are cut into 7 parts each then one or more of the smaller pieces are cut into 7 parts each again. Which of the

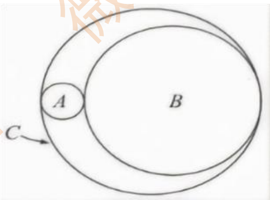
following could be the total number of pieces of paper after these cuts have been made?

A. 14　　B. 15　　C. 18　　D. 19　　E. 21

4. In the figure, the line segments represent paths in a garden and the dots represent benches, All of the paths allow two-way traffic except the path between the waterfall bench and the roses bench, which can be traveled only in the direction of the arrow. How many possible routes are there from the entrance bench to the fountain bench along these paths that a visitor can travel without passing the same bench more than once?



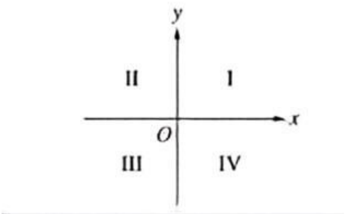
5. The figure shows three circles, A , B , and C, that are tangent to each other. The area of circle A is equal to 1/16 of the area of circle B. The circumference of circle C is equal to the sum of circumferences of circle A and circle B, If circle C has an area of 100π, what is the radius of circle A



6. In a survey of 100 teenagers, 71 of the teenagers reported using at least one of three social media sites A, B,and C and 30 reported using all three sites. At least 15 of the teenagers reported using only site A and at most 15 of the teenagers reported using only site B. Which of the following values could be the number of teenagers who reported using only site C?【微信公众号：张巍 GRE】
indicate all such values.

A. 0　　B. 10　　C. 20　　D. 30　　E. 40

7. The four quadrants of the xy-plane are shown. The graph of y = mx + b (not shown), where m and b are positive constants, must pass through which of the quadrants? Indicate all such quadrants.



A. Ⅰ　　B. Ⅱ　　C. Ⅲ　　D. Ⅳ

8. A company has assets worth $150,000 and liabilities worth $70,000, giving it an asset-to-liability ratio of approximately 2.1. The company will borrow x dollars, and the amount borrowed will be added to both the assets and the liabilities. if the asset-to-liability ratio is to be greater than 1.2 after the money is borrowed, which of the following could be the value of x?

Indicate all such values.

A. 300,000
B. 320,000
C. 340,000
D. 360,000

9.

Weekly Rental Rates for Realtor *M*'s Summer Houses on Island *X*

Location	Number of Bedrooms in House		
	2 Bedrooms	3 Bedrooms	4 Bedrooms
Oceanside	S800	S900	S1.000
Bayside	S600	S700	$800
Inland	S500	S600	$700

A group of 10 couples plans to rent bayside summerhouses on island X from Realtor M for a certain week.The total rental cost for the houses will be evenly distributed among the 10 couples, and each couple will have their own bedroom. If Realtor M has at least three bayside summer houses of each type available that week,what is the least possible rental cost per couple?【微信公众号：张巍 GRE】

A. S400　　B. S300　　C. S240　　D. $220　　E. S190

10. Last week, each employee at a small company, except for one employee, made a donation to a certain charity The average (arithmetic mean) amount of the donations was $39 per person. This week, after the remaining employee made a donation, the average amount of the donations increased to $40 per person. if the remaining employee' s donation was between $54 and $61, which of the following could be the total number of

employees at the company

A. 14　B. 17　C. 20　D. 23　E. 26

11. N points are labeled on a number line. The distance between any two adjacent labeled points is x, If N> 2 and x > 1, which of the following represents the distance between the labeled point with the least coordinate and the labeled point with the greatest coordinate?

A. xN
B. x(N-1)
C. x(N-2)
D. (-1)N
E. (- 1)(N-1)

12. a_1, a_2, a_3.....a_n, ...

In the sequence above, $a_1=2$, $a_2=5$, and $a_n=\frac{a_{n-1}}{a_{n-2}}$ for all integers n greater than 2. What is the value of a_{135}

A. 1/5　B. 2/5　C. 1/2　D. 2　E. 5/2

13. Of the people surveyed about the effectiveness of various treatments for insomnia,65 percent reported that prescription drugs were effective and 48 percent reported that exercise was effective. If 22 percent of those surveyed reported that prescription drugs were effective but did not report that exercise was effective, what percent of those surveyed reported that exercise was effective but did not report that prescription drugs were effective?

A. 5%　B. 9%　C. 17%　D. 26%　E. 39%

14. A box contains a total of 9 books consisting of 4 history books, 4 science books, and1 math book. Two books are to be selected at random from the box without replacement.Which of the following statements are true? Indicate all such statements

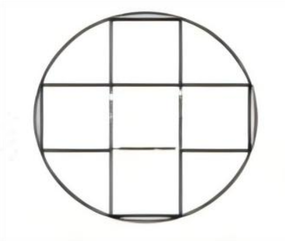
A. The probability that one of the two books selected will be a math book is 1/9
B. The probability that both books selected will be science books is 1/6
C. The probability that only one of the two books selected will be a history book is greater than 1/2

15. Telephone numbers that each consist of 7 digits are to be assigned to residents of a certain city. If the first 3 digits of each of these telephone numbers are 243 or 249, how many different telephone numbers are available to be assigned to the residents of this city?

A. 6561

B. 10000
C. 13122
D. 20000
E. 9000000

16.



In the figure, a “plus sign” is inscribed in a circle. The plus sign consists of five squares, each with perimeter 24
Quantity A: The radius of the circle
Quantity B: 9

17. A demographer studied the ages, at childbirth, of 14,000 first-time mothers in a certain region during 2012. The mothers’ ages were reported to the nearest 0.1 year. The first quartile and the median of the reported ages were23.5 years and 25.6 years, respectively, if the inter-quartile range of the reported ages was between 8.0 years and9.0 years, which of the following could be the number of reported ages that were less than 30.0 years?
A. 5,000 B. 6,500 C. 8,000 D. 9,500 E. 11,000

18. The mean and standard deviation of a list of data are 41 and 5, respectively. The maximum and minimum of the data are 60 and 30, respectively. The maximum is how many more standard deviations away from the mean than the minimum is?
A. 1.6 B. 2.0 C. 2.2 D. 3.8 E. 4.2

19. Vladimir invested $10,000 for one year. He invested some of the amount at 4 percent simple annual interest and the rest of the amount at 6 percent simple annual interest.If the total interest earned for the year was between $450 and $550, which of the following statements must be true?【微信公众号：张巍 GRE】
Indicate all such statements.
A. The amount invested at 6 percent simple annual interest was greater than $2,000.
B. The amount invested at 6 percent simple annual interest was less than $8,000.
C. The amount invested at 6 percent simple annual interest was more than 3 times the amount invested at 4 percent simple annual interest.

20. A bag contains 32 identically wrapped pieces of candy, of which 7 are mints, 11 are caramels, and the rest are chocolates, If Jill selects pieces of candy at random, what is the least number she must remove from the bag to be sure that she gets a caramel?

A. 11 B. 12 C. 19 D. 21 E. 22

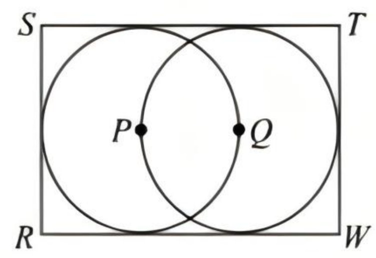
新 section 23 hard

1. Children at a summer camp are traveling on 5 buses to a museum, Each bus holds up to 48 passengers, consisting of children and adult chaperones. On each bus, the ratio of the number of children to the number of adult chaperones is no greater than 8 to 1. What is the greatest possible total number of children on the 5 buses?

2. A company announced that next year' s annual salary of each of its employees will be5 percent greater than this year's salary, plus $500 times the number of years of service that the employee has this year. If next year's annual salary of an employee who has 4 years of service this year will be $38,330, what is the employee's annual salary this year?

3. The figure shows two circles that are each tangent to three sides of rectangle RSTW and that have centers at P and Q, respectively. If each circle has a circumference of c,

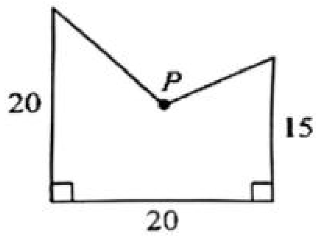
what is the area of rectangular region RSTW in terms of c?【微信公众号：张巍 GRE】



○ $3\left(\frac{c}{\pi}\right)^2$ ○ $\frac{3}{2}\left(\frac{c}{\pi}\right)^2$ ○ $\frac{3}{4}\left(\frac{c}{\pi}\right)^2$ ○ $\frac{1}{2}\left(\frac{c}{\pi}\right)^2$ ○ $\frac{1}{4}\left(\frac{c}{\pi}\right)^2$

4. In a certain town, the median daily noon temperature for a 7-day period was less than 60'F. in the town, which of the following could be the total number of days during the 7-day period on which the daily noon temperature was less than 60°F ?
Indicate all such numbers.

A. 1 B. 2 C. 3 D. 4 E. 5 F. 6 G. 7



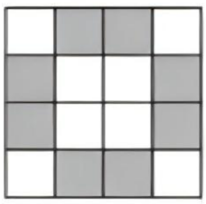
5. The figure above represents the surface of a wall with an irregular shape, where all measurements are in meters and point P is 10 meters from the bottom edge and 10 meters from the left edge. The surface is to be painted, and one bucket of paint will cover 170 square meters of the surface. If the bucket of paint will cover the part of the surface from the left edge to a vertical line that is x meters from the left edge, which of the following is true?

A. 8<x<9
B. 9<x< 10
C. 10<x< 11
D. 11<x<12
E. 12<x<13

6. Randy purchased 2 pens priced at x dollars each, and 5 pencils priced at y dollars

each, where x>y. The average (arithmetic mean) price of the 7 items was $1.75, Which of the following could be the value of x?【微信公众号：张巍 GRE】
Indicate all such values.
A. 1.40　　B. 1.65　　C. 2.50　　D. 5.50　　E. 6.25

7. Number $0.9\underline{54}$, with 54 repeating, can be expressed as k/n. What is the minimum value of k+n?

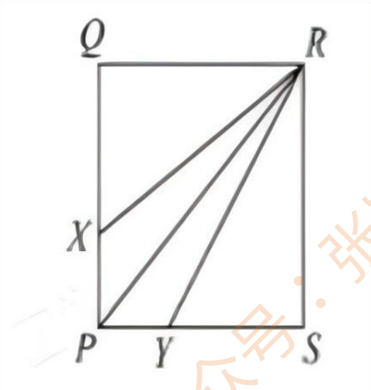


8. In the large square composed of the 16 small squares shown,certain small squares are shaded according to the following rule: only those small squares that have exactly one edge on the perimeter of the large square are shaded. If n is any integer greater than 4,then according to this rule,how many of the small squares would be shaded in a large square composed of n^2 small squares?
A. $2(n-2)^2$　　B. 2(n-1)　　C. 2(n-2)　　D. 4(n-1)　　E. 4(n-2)

9. x and y are positive integers, and x -y is an even integer.
Quantity A: The remainder when x^2+y is divided by 2
Quantity B: 1

10. 5, 8, 3, 7
How many different numbers k are there such that the sum of k and 3 of the 4 numbers shown equals 24?

11. PQRS is a rectangle, and the ratio of PQ to QR is 10 to 9. PX=1 and PY=0.9



Quantity A：The area of triangular region PXR

Quantity B：The area of triangular region PRY

12. A saltwater solution contains 25 percent salt by weight. To 1 kilogram of this solution,x grams of salt and 4x grams of water are added, where $x > 0$.

Quantity A：The percent of salt in the resulting solution

Quantity B：25%

13. A caterer has 20 liters of fruit punch that is 12 percent grape juice, by volume. How many liters of fruit punch that is 20 percent grape juice, by volume, must be added to the original 20 liters of fruit punch in order to create fruit punch that is 15 percent grape juice, by volume?【微信公众号：张巍 GRE】

A. 3

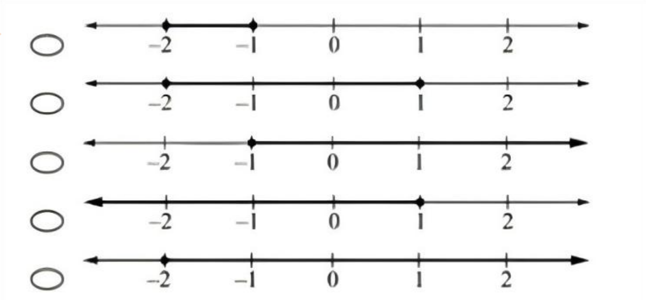
B. 10

C. 12

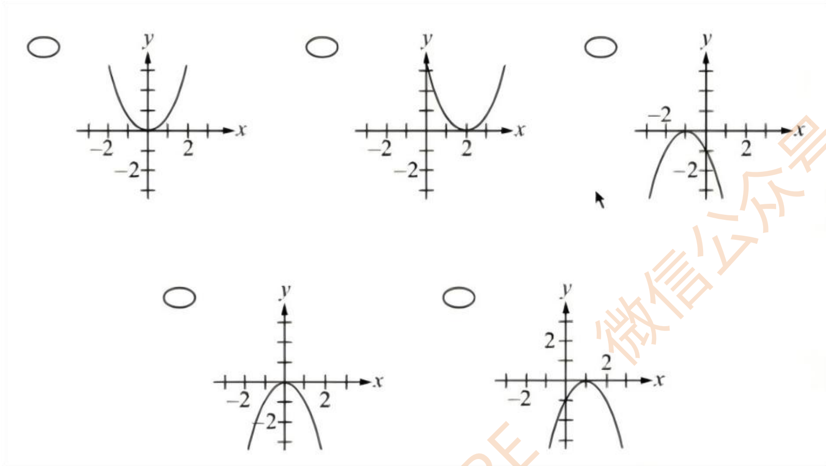
D. 15

E. 30

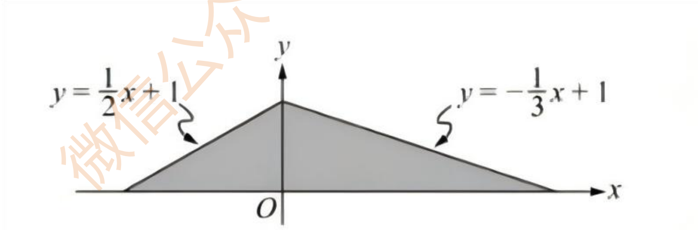
14. Which of the following number lines shows the graph of all values of p that satisfy both of the inequalities $p \geq -2$ and $-p \geq 1$



15. Which of the following is the graph of $y = -x^2 + 2x - 1$?



16. The shaded region shown is bounded by the lines $y=\frac{x}{2}+1$, $y=-\frac{x}{3}+1$, and the x-axis. What is the area of the shaded region?【微信公众号：张巍 GRE】



17. Eugene and Penny started a job in sales on the same day. Eugene's sales for the first month were r dollars. and each month after the first his sales for that month were twice his sales for the preceding month. Penny's sales for the first month were 10r dollars.and each month after the first her sales for that month were 10r dollars more than her sales for the preceding month. Which of the following statements are true?
Indicate all such statements.

A. The dollar amount of Penny's sales for the second month was 10 times that of Eugene's sales for that month.
B. The dollar amount of Penny's sales for the fourth month was 5 times that of Eugene's sales for that month.
C. The dollar amount of Eugene's sales for the eighth month was greater than that of Penny's sales for that month.

18. The midrange of a list of numbers is the average (arithmetic mean) of the least number and the greatest number in the list.
List D consists of positive numbers where the least number is x, and the greatest number is y. If the midrange and range of the numbers in D are equal, what is the value of $\frac{x}{y}$?

19. List R: 5，x，3，4，2
There are three different values of x for which the average (arithmetic mean) of the 5 numbers in R is equal to the median of the 5 numbers. What is the greatest of these three values?

20. If n, k, and r are positive integers such that $n^k = 10r + 3$, which of the following could be the value of n?【微信公众号：张巍 GRE】
A. 11 B. 12 C. 15 D. 17 E. 19

新 section 24 hard

1. Each of Rachelle's clients is a resident of one of four cities: R, S, T. or W. Of these clients, 20 percent are residents of City R.25 percent are residents of City S. and x percent are residents of City T. Which of the following expressions represent the number of Rachelle's clients who are residents of City W as a fraction of all her clients?
Indicate all such expressions.

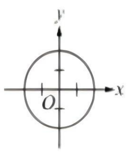
□ $1-\left(\frac{55-x}{100}\right)$ □ $1-\left(\frac{45+x}{100}\right)$ □ $\frac{45-x}{100}$ □ $\frac{55-x}{100}$ □ $\frac{55+x}{100}$

2. p is a prime number greater than 2.
Quantity A： The number of prime factors of p - 1
Quantity B： 2

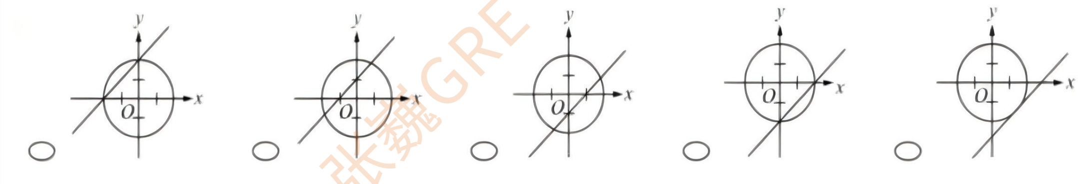
3. The line with equation y=3x-2 will be translated 4 units to the right and 5 units up in the xy -plane. Which of the following is an equation of the resulting line?
A. y=3x+15　　B. y=3x+5y　　C. y=3x-9　　D. y=3x-13　　E. y= 3x-21

4. The lengths of 5 fossils found at an excavation site were recorded to the nearest millimeter. The least length was 20 millimeters, and the average (arithmetic mean) of the lengths was 50 millimeters. Which of the following values could be the range, in millimeters, of the 5 lengths?
Indicate all such values.
A. 30
B. 50
C. 120
D. 170
E. 200

5. The diameter of wheel A is half the diameter of wheel B. Each wheel rolls along a flat surface without slippage.【微信公众号：张巍 GRE】
Quantity A: The number of revolutions that wheel A makes in traveling 5,000 feet
Quantity B: The number of revolutions that wheel B makes in traveling 2,500 feet

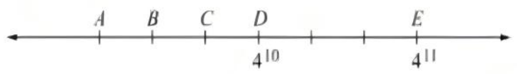


6. In the xy-plane, the circle shown is the graph of $x^2 + y^2 = 4$. Which of the following best represents the graph of the circle and the line y-x+2 =0?



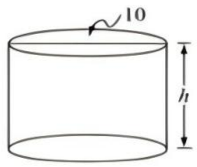
7. The tick marks shown on the number line are evenly spaced. Points D and E have

coordinates of $4^{10}$and 4^{11},respectively. The point that has a coordinate of 4^9 is



A. point A
B. between points A and B
C. between points B and C
D. point C
E. between points C and D

8.



The right circular cylinder shown has a diameter of 10, a height of h, and a total surface area of 130π, including the side, top, and bottom.【微信公众号：张巍 GRE】
Quantity A：h
Quantity B：10

9. The first and last terms in a certain sequence are -8 and 184, respectively. Each term in the sequence after the first term is 3 greater than the immediately preceding term.
Quantity A：The number of terms in the sequence
Quantity B：65

10. A wholesaler purchased a computer from a manufacturer and then sold it to a retailer,earning a profit of 20 percent of the wholesaler' s purchasing cost. The retailer then sold the computer to a customer, earning a profit of 30 percent of the retailer' s purchasing cost. If the retailer sold the computer for $975.00, what was the wholesaler' s purchasing cost? (Note; Profit equals selling price minus purchasing cost.)

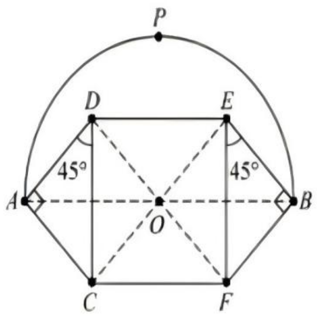
11. Ted sold two houses. He sold one of the houses for a profit of 20 percent of its purchase price, and he sold the other house for a loss of 20 percent of its purchase price. If the selling price of each house was $192,000, by what amount did the sum of the purchase prices of the two houses exceed $384,000?【微信公众号：张巍 GRE】
A. $0
B. $16,000
C. $24,000
D. $36,000

E. $40,000

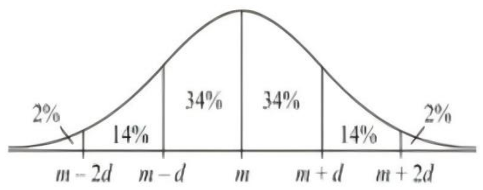
12. If x, y, and z are integers greater than 1 and xyz = 231, what is the value of x +y +z =?
A. 15
B. 19
C. 21
D. 23
E. It cannot be determined from the information given.

13. Two numbers are to be selected at random and without replacement from the set(1. 3, 5. 7.9, 11). What is the probability that the product of the two selected numbers will be greater than 50?【微信公众号：张巍 GRE】
Give your answer as a fraction

14. $1<a<b<c<d<e<f$
Quantity A: The median of a, c, and e
Quantity B: The average (arithmetic mean) of b, d,and f



15. In the figure, arc APB is a semicircle with center O and CDEF is a square. The area of the semicircular region is π .
Quantity A: The area of region ADEBFC
Quantity B: 3



16. The figure above shows a normal distribution with mean m and standard deviation d,including approximate percents of the distribution corresponding to the six regions shown.

The lengths of phone calls made on a certain weekend by students at High School H are approximately normally distributed with a mean of 30 minutes and a standard deviation of10 minutes. Which of the following statements must be true?【微信公众号：张巍 GRE】

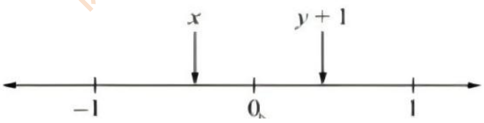
Indicate all such statements

A. The range of the lengths of the phone calls is less than 60minutes.

B. The lengths of half of the phone calls are each greater than 40minutes.

C. The length of a 35-minute phone call is 0.5 standard deviation from the mean of the lengths of the phone calls.

17.



Quantity A： xy

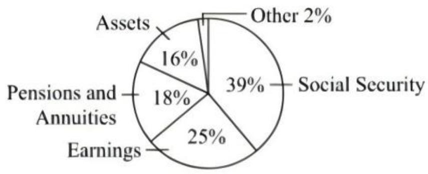
Quantity B： x+y

18. For all numbers a and b, the operation is defined by the following equation

$$a \otimes b = \begin{cases} 2 & \text{if } a > b \\ 0 & \text{if } a = b \\ b+1 & \text{if } a < b \end{cases}$$

What is the value of $((1 \otimes 2) \otimes 1) \otimes 2$?

A. 0　　B. 1　　C. 2　　D. 4　　E. 6



19. Greg' s income last year was $45,000. The graph above shows the distribution of his income, by income source. Based on the information given, which of the following statements about his income last year are true?【微信公众号：张巍 GRE】
Indicate all such statements.
A. His income from earnings was $11,250.
B. More than 2/5 of his income was from assets and earnings combined.
C. His income from social security was 56 percent greater than his income from earnings.

Mechanic	A	B	C	D	E
Time (seconds)	15	20	25	16	x

20. The table above shows the times, in seconds, that it took each of 5 mechanics to perform a simple task. For the 5 times shown, if the median is greater than the average (arithmetic mean), which of the following could be the value of x?
Indicate all such values
A. 17 B. 18 C. 20 D. 22 E. 26

新 section 25 hard

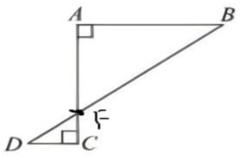
$$r = 4 + \frac{k}{d + 500}$$

1. When a certain lending business loans an amount of d dollars for one year, it charges interest equal to r percent of the amount of the loan so that the interest rate r and the amount d are related by the formula shown, where ks a constant. When the business loans 1,000 dollars for one year, the interest rate is 4.40 percent. When the business loaned an amount of money to Charles for one year, the interest rate was 4.12 percent. Approximately what was the total amount, including interest, that Charles had to repay the business for the loan?

A. $4,685 B. $5,365 C. $5,745 D. $6,355
E. $6,885

2. Telephone numbers that each consist of 7 digits are to be assigned to residents of a certain city. If the first 3 digits of each of these telephone numbers are 243 or 249, how many different telephone numbers are available to be assigned to the residents of this city?【微信公众号：张巍 GRE】

A. 6,561
B. 10,000
C. 13,122
D. 20,000
E. 9,000,000



3. In the figure above, AB and AC each have length 3, and DC has length 1. What is the length of DB?

A. 4 B. 5 C. 6 D. 7 E. 9

4. Working at their respective constant rates, machine A can complete 1 job in 2 hours and machine B can complete 2 jobs in 3 hours. Both machines can work at their respective constant rates continuously for n hours, where n > 10.

Quantity A：The number of jobs machine A,working at its constant rate, can complete in n hours

Quantity B：The number of jobs machine B,working at its constant rate, can complete in n-3 hours

5. Ina barrel there are n ping-pong balls labeled with consecutive integers from 1 to n, where n is an odd number greater than 50. If a ball is to be randomly selected from the barrel, what is the probability that the ball selected will be a ball that is labeled with an odd integer?

A. $\frac{1}{2}$ B. $\frac{2}{n}$ C. $\frac{n-1}{n}$ D. $\frac{n-1}{2n}$ E. $\frac{n+1}{2n}$

6. In a government' s current budget, 66 percent of the budget is allocated to the payment of entitlements and interest on the government' s debt If the dollar amount of the allocation for entitlements and interest on the government' s debt were to remain the same, but all the remaining allocations in the budget were to be reduced by 10 percent of their dollar amounts, by what percent would the entire government budget be reduced?

A. 1.5%
B. 3.4%

C. 5.0%
D. 6.0%
E. 6.6%

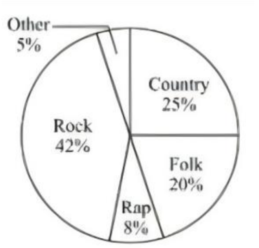
7. Faucets X and Y dispense water at their respective constant rates. Dispensing simultaneously, faucets X and Y fill an empty tub in 3 minutes. Faucet X, dispensing alone, takes 3/4 as long as faucet Y, dispensing alone, to fill the empty tub. Let x and y be the respective times, in minutes, that it takes faucets X and Y, each dispensing alone, to fill the empty tub. What are the values of x and y?【微信公众号：张巍 GRE】
A. x=3 and y=4
B. x=4.5 and y=6
C. x=5.25 and y=7
D. x=6 and y=8
E. x=7 and y=9.3

8. At a certain candy store, the price of a box of taffy is $6 and the price of a box of chocolates is $9. The store sells x boxes of taffy and y boxes of chocolates for a total price of $360. Which of the following statements must be true?
Indicate all such statements.
A. x≤30
B. y≤20
C. x+y≤60

9. x and y are positive integers, and y is odd.
Quantity A：The remainder when (x+y) (y+7) is divided by 2
Quantity B：1

10. A box contains yellow tennis balls and white tennis balls. The number of yellow tennis balls in the box is 10 more than the number of white tennis balls. If 2 of the yellow balls are replaced by 2 white balls, the ratio of the number of yellow balls to the number of white balls will be 4 to 3. What is the total number of tennis balls in the box?

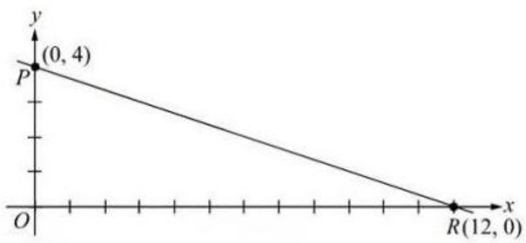
11. A sequence of numbers P_1，P_2，P_3...... P_n... is defined as follows: $P_1 = 1$， $P_2 = 2$， and $P_n = 4\frac{P_{n-1}}{P_{n-2}}$ for each integer n greater than 2. What is the value of P_4？【微信公众号：张巍 GRE】
A. 4 B. 6 C. 12 D. 16 E. 24



12. Rodger has a collection of 600 music CDs. The graph above shows the distribution of the CDs by music genre.
Which of the following statements about the CD collection are true?
Indicate all such statements.
A. There are more than 250 CDs in the rock genre.
B. The number of CDs in the country genre is 5 percent greater than the number of CDs in the folk genre.
C. The CDs in the country and rap genres together comprise 1/3 of the collection.

13. A robot processes orders in a warehouse, where the least and greatest possible processing times per order are 2 minutes and 5 minutes, respectively. Based on these times, which of the following could be the number of orders processed during a time period that is between110 minutes and 130 minutes in length?
Indicate all such numbers.
A. 20 B. 40 C. 60 D. 80 E. 100

14. In the rectangular coordinate system shown, how many points on line segment PRhave coordinates (x, y), such that both x and y are integers?



A. Four B. Five C. Six D. Seven E. Eight

Value of One Share of Stock

Company	A	B	C
Value of One Share of Stock	x	$2x$	$x+2$

15. The table above shows the values of one share of stock for companies A, B, and C at the close of the stock market yesterday, where x is in dollars and x > 2.
Which of the following statements individually provide(s) sufficient additional information

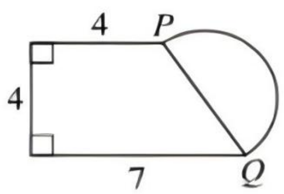
to determine the value of x?【微信公众号：张巍 GRE】
Indicate all such statements.
A. The average (arithmetic mean) of the three values of stock shown is $4.00.
B. The median of the three values of stock shown is $4.50.
C. The range of the three values of stock shown is $2.50.

16. Among 16 textbooks purchased by a bookstore, each textbook was either a mathematics textbook or a chemistry textbook, and each textbook was either new or used. Of the mathematics textbooks purchased, 1/2 were used. Of the chemistry textbooks purchased, 2/3 were used. Which of the following could be the total number of textbooks purchased that were used?
Indicate all such numbers.
A. 4 B. 5 C. 6 D. 7 E. 9
F.10 G.11 H. 12

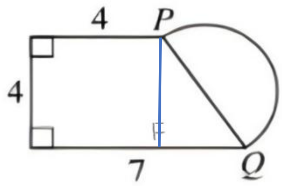
17. If k and n are distinct digits and k > n, which of the following numbers is greatest?【微信公众号：张巍 GRE】
- $1-0.\overline{kn}$
- $1-0.\overline{nk}$
- $1-0.\overline{knk}$
- $1-0.\overline{knn}$
- $1-0.\overline{nkn}$

18. Three children and one adult played a game in which each player scored a number of points. The average (arithmetic mean) number of points scored by the 4 players was1,200. If the average number of points scored by the 3 children was 1,300, how many points did the adult score?【微信公众号：张巍 GRE】

19.



Quantity A：The length of the semicircular arc PQ
Quantity B：15



20. Stations A, M, and B are located along a certain train route, and Station M is between Stations A and B. At noon, a train engine passed Station A traveling at a constant speed of 80 kilometers per hour toward Station B. Also at noon, another train engine passed Station B traveling at a constant speed of 60 kilometers per hour toward Station A. Both train engines passed Station M at the same time. What is the ratio of the distance along the route between Stations A and M to the distance along the route between Stations A and B?

A. $\frac{1}{4}$ B. $\frac{3}{7}$ C. $\frac{1}{2}$ D. $\frac{4}{7}$ E. $\frac{3}{4}$